

Chapter 1: Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for **ProcessMind AI**, an AI-powered Autonomous Process Intelligence Platform.

The purpose of this document is to provide a complete and unambiguous specification for the design, development, testing, deployment, and maintenance of the platform. It serves as the primary reference for software developers, system architects, UI/UX designers, testers, project managers, stakeholders, and future contributors throughout the software development lifecycle.

This SRS documents the system's objectives, architecture, business requirements, functional modules, interfaces, constraints, security requirements, performance expectations, and quality attributes.

Furthermore, this document establishes a common understanding between all stakeholders regarding the platform's intended behavior, ensuring consistency throughout implementation while minimizing ambiguity and requirement changes.

The specification follows software engineering best practices and is intended to support future product evolution from a Minimum Viable Product (MVP) into a scalable enterprise Software-as-a-Service (SaaS) platform.

1.2 Scope

ProcessMind AI is an AI-powered Process Intelligence Platform designed to transform organizational Standard Operating Procedures (SOPs) into intelligent, executable, and continuously optimized business processes.

Unlike conventional workflow automation systems that require users to manually design workflows, ProcessMind AI leverages Artificial Intelligence to understand natural-language SOPs, automatically generate workflow models, validate process logic, simulate execution outcomes, orchestrate workflow execution, analyze operational performance, and recommend process improvements.

The primary objective of the platform is to bridge the gap between organizational knowledge and operational execution by enabling organizations to convert static documentation into intelligent operational systems.

The initial release (MVP) focuses on demonstrating these capabilities using a Leave Management workflow. However, the platform is designed with a domain-independent architecture, allowing it to support various enterprise business processes, including:

- Human Resource Management
- Procurement Management
- Invoice Processing
- Employee Onboarding
- Compliance Management

- IT Incident Management
- Vendor Approval
- Risk Assessment
- Operations Management

The platform aims to reduce manual effort, improve operational efficiency, enhance process visibility, minimize compliance risks, and support continuous business process optimization through AI-driven intelligence.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
AI	Artificial Intelligence
API	Application Programming Interface
SOP	Standard Operating Procedure
SaaS	Software as a Service
JWT	JSON Web Token
RBAC	Role-Based Access Control
CRUD	Create, Read, Update, Delete
BPM	Business Process Management
BPMN	Business Process Model and Notation
UI	User Interface
UX	User Experience
REST	Representational State Transfer
JSON	JavaScript Object Notation
SQL	Structured Query Language
ORM	Object Relational Mapping
MVP	Minimum Viable Product
LLM	Large Language Model
AI Agent	An autonomous software component capable of reasoning, decision-making, and performing tasks using Artificial Intelligence

Workflow	A structured sequence of business activities executed to complete an organizational process
Process Intelligence	The capability of understanding, analyzing, validating, monitoring, optimizing, and improving business processes using AI and operational data
Organizational Knowledge	Business knowledge stored within SOPs, policies, procedures, documentation, and organizational rules
Organizational Digital Twin	A virtual representation of organizational processes used for simulation, prediction, and optimization before implementing operational changes

1.4 References

The design and implementation of ProcessMind AI are guided by established software engineering principles, enterprise architecture practices, workflow management concepts, and modern Artificial Intelligence technologies.

The following standards, methodologies, and technologies are referenced throughout the development of the platform:

Software Engineering Standards

- IEEE 29148 – Systems and Software Engineering Requirements Engineering
- IEEE 1016 – Software Design Description
- IEEE 830 (Legacy Software Requirements Specification Standard)
- ISO/IEC 25010 – Software Product Quality Model

Architectural Principles

- RESTful API Architecture
- Layered Software Architecture
- Clean Architecture
- Domain-Driven Design (DDD)
- SOLID Design Principles

Workflow and Business Process Standards

- Business Process Management (BPM)
- Business Process Model and Notation (BPMN)
- Workflow Orchestration Concepts
- State Machine Design Patterns

Artificial Intelligence Technologies

- Google Gemini Large Language Models
- Prompt Engineering
- AI-assisted Workflow Generation
- AI-powered Process Intelligence

Development Technologies

- Next.js
- React
- Spring Boot
- PostgreSQL
- Docker
- JWT Authentication

1.5 Document Overview

This Software Requirements Specification is organized into multiple chapters to provide a structured description of the ProcessMind AI platform.

The document begins with an overview of the product vision, objectives, and system context before defining detailed functional and non-functional requirements.

Subsequent chapters describe the platform architecture, user roles, business requirements, software modules, database design, AI components, workflow engine, security mechanisms, deployment architecture, and operational constraints.

The specification further includes comprehensive use case descriptions, workflow diagrams, sequence diagrams, state diagrams, database models, API specifications, and acceptance criteria to ensure complete traceability from business objectives to implementation.

The overall structure of this SRS is as follows:

- **Chapter 1:** Introduction
- **Chapter 2:** Overall Description
- **Chapter 3:** System Architecture
- **Chapter 4:** Business Requirements
- **Chapter 5:** Functional Requirements
- **Chapter 6:** AI Functional Requirements
- **Chapter 7:** Database Requirements
- **Chapter 8:** REST API Specifications

- **Chapter 9:** Security Requirements
- **Chapter 10:** Non-Functional Requirements
- **Chapter 11:** Use Case Specifications
- **Chapter 12:** Activity Diagrams
- **Chapter 13:** Sequence Diagrams
- **Chapter 14:** State Machine Diagrams
- **Chapter 15:** Entity Relationship Diagram
- **Chapter 16:** Deployment Architecture
- **Chapter 17:** Acceptance Criteria
- **Chapter 18:** Future Enhancements

This document is intended to evolve alongside the product, providing a comprehensive blueprint for the successful development, deployment, maintenance, and future expansion of ProcessMind AI into a full-scale enterprise Process Intelligence Platform.

Chapter 2: Overall Description

2.1 Product Perspective

ProcessMind AI is an AI-powered Process Intelligence Platform designed to transform organizational knowledge into intelligent, executable, and continuously optimized business processes.

The platform bridges the gap between business documentation and operational execution by enabling organizations to convert Standard Operating Procedures (SOPs) into automated workflows. Unlike traditional workflow automation platforms that require users to manually design process flows, ProcessMind AI leverages Artificial Intelligence to understand SOPs, generate workflows, validate business logic, simulate execution outcomes, orchestrate workflow execution, and recommend process improvements.

ProcessMind AI is designed as a cloud-based Software-as-a-Service (SaaS) platform with a modular and scalable architecture. The system follows a layered architecture consisting of a web-based user interface, backend services, AI intelligence layer, workflow orchestration engine, and data management layer.

The initial release of the platform focuses on Leave Management as the Minimum Viable Product (MVP). However, the architecture is domain-independent and is designed to support multiple enterprise domains, including Human Resources, Procurement, Finance, Compliance, IT Operations, Customer Support, and other workflow-driven business processes.

In the long term, ProcessMind AI aims to evolve into an AI Organizational Operating System capable of understanding, managing, and continuously optimizing organizational operations through intelligent process analysis and automation.

2.2 Product Functions

The primary functions of ProcessMind AI include:

SOP Management

- Upload SOPs in multiple formats (Text, PDF, DOCX)
- Store and manage organizational SOPs
- Maintain version history of SOPs

SOP Intelligence

- Analyze natural-language SOPs using Artificial Intelligence
- Extract business rules, conditions, approvals, and actions
- Convert unstructured procedures into structured process definitions

Workflow Generation

- Automatically generate workflow models from SOPs
- Create decision trees and approval chains
- Visualize generated workflows

Process Validation

- Detect missing approvals
- Detect conflicting business rules
- Identify undefined roles
- Detect circular dependencies
- Validate workflow completeness

Process Simulation

- Simulate workflows using different business scenarios
- Predict workflow execution paths
- Estimate completion times
- Analyze process outcomes before deployment

Workflow Execution

- Execute workflows dynamically
- Manage workflow states

- Route tasks automatically
- Pause and resume workflow execution
- Track workflow progress

Human Approval Management

- Request approvals
- Escalate pending approvals
- Track approval history
- Enforce approval deadlines

Process Analytics

- Monitor workflow performance
- Analyze workflow completion rates
- Measure approval delays
- Identify process bottlenecks

Process Optimization

- Recommend workflow improvements
- Identify redundant process steps
- Suggest approval optimizations
- Improve organizational efficiency

Administration

- Manage users and organizations
- Configure workflows
- Manage roles and permissions
- Configure AI settings
- Monitor platform health

Notification Management

- Email notifications
- In-app notifications
- Workflow status notifications
- Approval reminders

- Escalation notifications

2.3 User Classes and Characteristics

ProcessMind AI supports multiple user roles, each with different responsibilities and system privileges.

System Administrator

The System Administrator manages the overall platform configuration and operational environment.

Responsibilities include:

- Manage organizations
- Configure system settings
- Manage user accounts
- Assign user roles
- Monitor system health
- View audit logs
- Configure AI services

Required Skills:

- Basic system administration knowledge
- Organizational process understanding

Organization Administrator

The Organization Administrator manages business processes within a specific organization.

Responsibilities include:

- Upload SOPs
- Generate workflows
- Deploy workflows
- Manage departments
- Configure approvals
- Monitor organizational workflows

Required Skills:

- Business process management
- Organizational policy knowledge

Process Analyst

The Process Analyst evaluates and improves business processes.

Responsibilities include:

- Review SOPs
- Validate workflows
- Run process simulations
- Analyze workflow performance
- Review optimization recommendations

Required Skills:

- Business analysis
- Process improvement methodologies

Manager / Approver

Managers participate in approval workflows.

Responsibilities include:

- Review approval requests
- Approve or reject business processes
- Track assigned workflows
- Monitor approval history

Required Skills:

- Departmental knowledge
- Organizational decision-making authority

Employee

Employees interact with workflows generated by the platform.

Responsibilities include:

- Submit requests
- Track workflow progress
- Respond to assigned tasks
- View notifications

Required Skills:

- Basic computer literacy

AI Services (External Actor)

Artificial Intelligence services assist the platform by performing reasoning-based operations.

Responsibilities include:

- SOP understanding
- Workflow generation
- Validation reasoning
- Process optimization
- Simulation support

2.4 Operating Environment

ProcessMind AI is designed as a cloud-native web application.

Client Environment

Supported Web Browsers:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Supported Devices:

- Desktop Computers
- Laptops
- Tablets

Server Environment

Backend:

- Java 21
- Spring Boot

Frontend:

- Next.js
- React
- TypeScript

Database:

- PostgreSQL

AI Services:

- Google Gemini API

Containerization:

- Docker
- Docker Compose

Deployment:

- Vercel (Frontend)
- Render (Backend)
- Neon PostgreSQL (Database)

Network Requirements

- Stable internet connection
- HTTPS-enabled communication
- REST API communication
- Secure authentication

2.5 Assumptions and Dependencies

The development and operation of ProcessMind AI assume the following:

Assumptions

- Users have internet connectivity.
- Organizations possess documented SOPs or business procedures.
- Uploaded SOPs are written in understandable natural language.
- Users have valid system accounts before accessing protected resources.

- AI services are available during workflow generation and validation.
- Users possess basic computer literacy.
- Organizations are willing to digitize business processes.
- System clocks remain synchronized across infrastructure.

Dependencies

The platform depends on:

- Google Gemini API for AI reasoning
- PostgreSQL database availability
- Cloud hosting infrastructure
- Email service provider for notifications
- Internet connectivity
- Modern web browsers

Future versions may additionally depend on:

- Temporal.io
- Neo4j Graph Database
- Enterprise Identity Providers
- Cloud Object Storage

2.6 Constraints

The following constraints apply to ProcessMind AI.

Technical Constraints

- The frontend shall be developed using Next.js and React.
- The backend shall be implemented using Java Spring Boot.
- PostgreSQL shall be used as the primary relational database.
- Artificial Intelligence capabilities shall initially rely on Google Gemini APIs.
- Communication between frontend and backend shall use RESTful APIs.
- The system shall support Docker-based deployment.

Business Constraints

- The initial release shall focus on Leave Management as the primary demonstration domain.

- Workflow execution shall require administrator deployment before becoming active.
- Human approvals shall remain mandatory for configured approval stages.
- AI-generated workflows shall be reviewed by users before deployment.

Operational Constraints

- AI-based features depend on external AI service availability.
- Workflow execution requires continuous backend availability.
- Email notifications require a configured email service.
- Cloud deployment requires internet access.

Security Constraints

- User authentication shall be mandatory.
- Role-Based Access Control (RBAC) shall be enforced.
- Passwords shall never be stored in plain text.
- All communication shall occur over HTTPS.
- Sensitive credentials shall be stored as secure environment variables.

Scalability Constraints

- The MVP is designed for small to medium-sized organizations.
- Future releases shall support multi-tenant enterprise deployments.
- The architecture shall allow horizontal scaling of backend services.
- AI components shall be replaceable without significant changes to the overall system architecture.

This overall description establishes the context, scope, user environment, assumptions, and operational boundaries of ProcessMind AI and provides the foundation for the detailed functional and non-functional requirements described in subsequent chapters.

Chapter 3: System Architecture

3.1 System Architecture Overview

ProcessMind AI follows a modular, cloud-native, layered architecture that separates presentation, business logic, artificial intelligence, workflow orchestration, and data management into independent components.

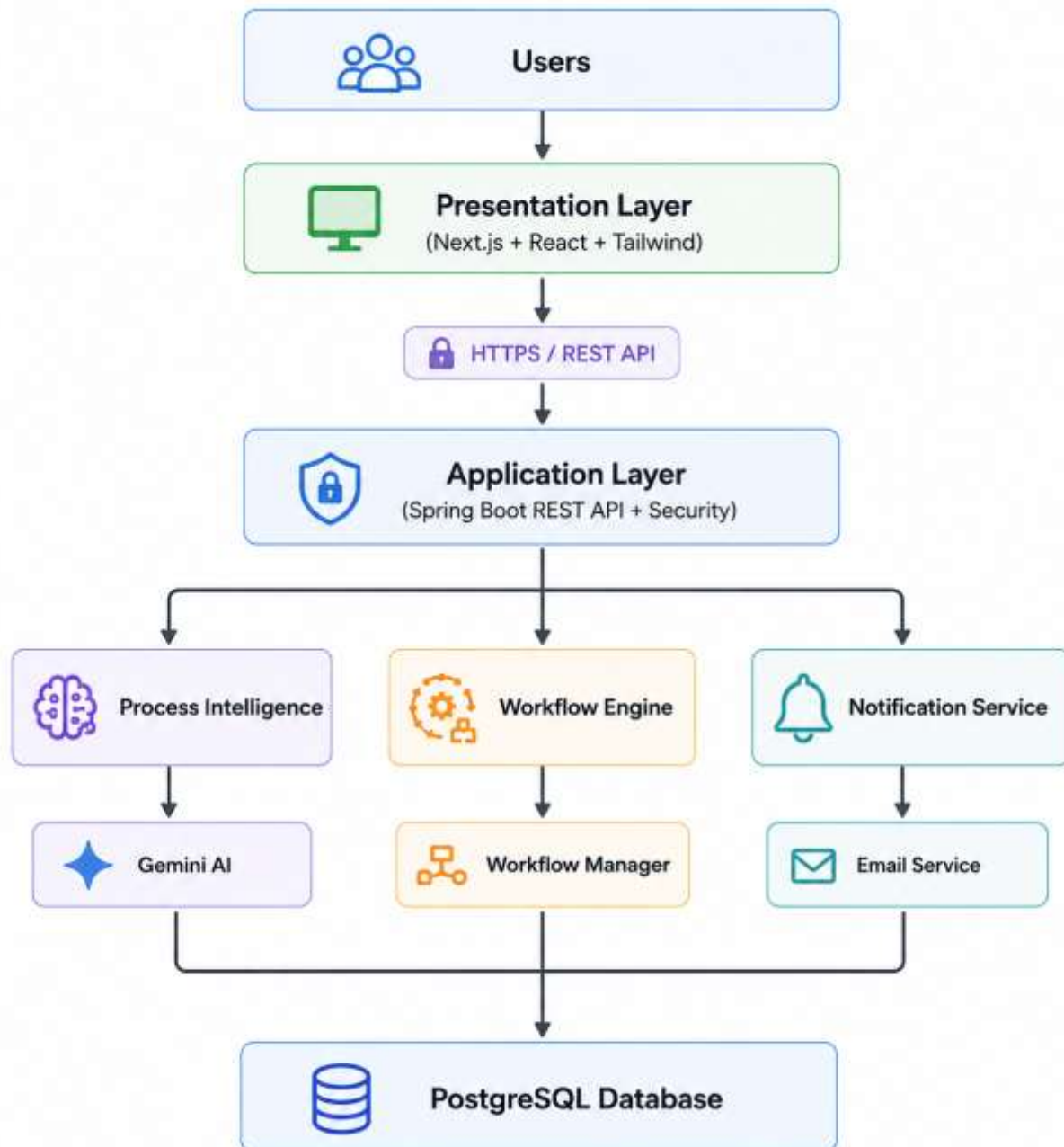
This architecture promotes scalability, maintainability, security, extensibility, and high system availability while allowing future integration of additional AI services, workflow engines, enterprise systems, and organizational intelligence capabilities.

The platform consists of six primary architectural layers:

1. Presentation Layer
2. Application Layer
3. Process Intelligence Layer
4. Workflow Orchestration Layer
5. Data Layer
6. External Services Layer

Each layer has clearly defined responsibilities and communicates through secure RESTful APIs and internal service interfaces.

3.2 High-Level Architecture



3.3 Architectural Layers

3.3.1 Presentation Layer

Purpose

Provides the graphical user interface through which users interact with the system.

Technologies

- Next.js
- React
- TypeScript
- Tailwind CSS

- React Flow

Responsibilities

- User Authentication
- Dashboard
- SOP Upload
- Workflow Visualization
- Process Simulation Interface
- Approval Dashboard
- Analytics Dashboard
- Administration Panel
- Notification Center

Users

- Employee
- Manager
- Process Analyst
- Organization Administrator
- System Administrator

3.3.2 Application Layer

Purpose

Acts as the central backend responsible for handling business logic and coordinating communication between all system components.

Technologies

- Java 21
- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate

Responsibilities

- REST API Management
- Authentication

- Authorization
- Request Validation
- Business Logic
- Workflow Coordination
- Database Communication
- External AI Communication

3.3.3 Process Intelligence Layer

Purpose

Provides Artificial Intelligence capabilities for understanding, analyzing, validating, and optimizing business processes.

This layer differentiates ProcessMind AI from traditional workflow automation platforms.

Components

SOP Intelligence Engine

Responsibilities:

- Understand SOPs
- Extract business rules
- Identify approvals
- Identify actions
- Detect dependencies

Workflow Generation Engine

Responsibilities:

- Generate workflow structures
- Create decision trees
- Build workflow graphs
- Generate workflow JSON

Process Validation Engine

Responsibilities:

- Detect logical conflicts
- Detect missing approvals
- Detect undefined roles

- Detect circular dependencies
- Evaluate workflow quality

Process Simulation Engine

Responsibilities:

- Execute virtual workflow scenarios
- Predict workflow outcomes
- Estimate completion time
- Identify execution paths

Process Optimization Engine

Responsibilities:

- Detect bottlenecks
- Recommend workflow improvements
- Reduce approval delays
- Simplify workflows

3.3.4 Workflow Orchestration Layer

Purpose

Executes generated workflows and manages workflow lifecycle.

Components

Workflow Engine

Responsibilities:

- Start workflows
- Execute workflow steps
- Route tasks
- Pause workflows
- Resume workflows
- Complete workflows

Approval Engine

Responsibilities:

- Approval routing
- Approval notifications

- Escalations
- Approval history

State Manager

Responsibilities:

Maintain workflow states:

- Draft
- Active
- Running
- Waiting Approval
- Completed
- Failed

Notification Manager

Responsibilities:

- Email notifications
- In-app notifications
- Approval reminders
- Escalation alerts

3.3.5 Data Layer

Purpose

Store all business and operational data.

Database

PostgreSQL

Stored Data

- Users
- Organizations
- Departments
- SOP Documents
- Workflow Definitions

- Workflow Instances
- Workflow Logs
- Approval Records
- Notifications
- Analytics Data
- Audit Logs

3.3.6 External Services Layer

Artificial Intelligence

Google Gemini API

Responsibilities

- SOP reasoning
- Workflow generation
- Validation reasoning
- Simulation reasoning
- Optimization recommendations

Email Service

Responsibilities

- Send notifications
- Approval requests
- Reminder emails

Cloud Storage (Future)

Responsibilities

- Store uploaded documents
- Archive SOP versions

3.4 Component Interaction

The following sequence describes the interaction among the major components.

1. User uploads an SOP.
2. Frontend sends the document to the backend.
3. Backend forwards the SOP to the SOP Intelligence Engine.

4. Gemini AI analyzes the SOP.
5. Structured workflow information is returned.
6. Workflow Generation Engine creates the workflow.
7. Process Validation Engine verifies the workflow.
8. Process Simulation Engine performs simulations.
9. Administrator reviews the generated workflow.
10. Workflow is deployed.
11. Workflow Engine executes workflow instances.
12. Analytics Engine collects execution data.
13. Process Optimization Engine generates recommendations.

3.5 Data Flow



3.6 Deployment Architecture

The platform follows a cloud-native deployment model.

Frontend

- Next.js
- Hosted on Vercel

Backend

- Spring Boot
- Hosted on Render

Database

- PostgreSQL
- Hosted on Neon

AI Services

- Google Gemini API

Containerization

- Docker
- Docker Compose

3.7 Security Architecture

The architecture incorporates multiple security layers.

Authentication

- JWT Authentication
- Spring Security

Authorization

Role-Based Access Control (RBAC)

Password Protection

- BCrypt Hashing

Communication Security

- HTTPS
- Secure REST APIs

Secrets Management

- Environment Variables
- API Keys
- Database Credentials

Audit Logging

All critical operations are logged for traceability.

3.8 Scalability Considerations

The architecture is designed to support future growth.

Horizontal Scaling

Backend services can be replicated.

Database Scaling

PostgreSQL supports replication and future sharding strategies.

AI Provider Abstraction

The AI layer is designed with provider abstraction, allowing Gemini to be replaced or complemented by other AI providers such as OpenAI, Anthropic, or self-hosted models without significant architectural changes.

Future Enterprise Extensions

The architecture supports future integration of:

- Multi-Agent Systems
- Organizational Knowledge Graph
- Organizational Digital Twin
- Process Mining
- Enterprise Identity Providers
- Temporal.io Workflow Orchestration
- Predictive Process Analytics

This modular architecture ensures that ProcessMind AI can evolve from a startup Minimum Viable Product (MVP) into a scalable enterprise Process Intelligence Platform capable of supporting organizations of varying sizes and operational complexity.

Chapter 4: Business Requirements

4.1 Business Overview

Organizations rely on Standard Operating Procedures (SOPs) to define how business processes should be executed. These SOPs are commonly maintained as documents such as PDFs, Word files, spreadsheets, or internal knowledge-base articles.

Although these documents define organizational operations, they are passive and require employees to manually interpret procedures, make decisions, request approvals, and execute tasks. This manual approach often results in inconsistent execution, process delays, compliance risks, operational inefficiencies, and limited visibility into business performance.

ProcessMind AI addresses these challenges by transforming static SOPs into intelligent, executable, and continuously optimized workflows. The platform combines Artificial Intelligence with workflow orchestration to automate process understanding, workflow generation, validation, simulation, execution, monitoring, and optimization.

The platform is designed to reduce operational costs, improve business efficiency, accelerate decision-making, and provide organizations with greater visibility into their operational processes.

4.2 Business Goals

The primary business goals of ProcessMind AI are:

- Enable organizations to digitize business processes from existing SOPs.
- Reduce manual effort involved in interpreting and executing procedures.
- Automate workflow generation using Artificial Intelligence.
- Improve workflow accuracy through automated validation.
- Minimize approval delays and operational bottlenecks.
- Enhance compliance with organizational policies.
- Improve visibility into workflow execution and performance.
- Support continuous process improvement through AI-driven recommendations.
- Provide a scalable Software-as-a-Service (SaaS) platform suitable for organizations of different sizes.

4.3 Business Objectives

The platform shall achieve the following objectives:

BO-001

Automatically convert natural-language SOPs into structured workflow definitions.

Priority: High

BO-002

Reduce the time required to design business workflows compared to manual workflow creation.

Priority: High

BO-003

Detect process design issues before workflow deployment.

Priority: High

BO-004

Allow organizations to simulate workflow behavior prior to deployment.

Priority: High

BO-005

Execute approved workflows using an automated workflow engine.

Priority: High

BO-006

Provide real-time visibility into workflow execution and operational performance.

Priority: High

BO-007

Identify workflow bottlenecks and recommend optimization opportunities using Artificial Intelligence.

Priority: Medium

BO-008

Support multiple business domains through reusable workflow architecture.

Priority: Medium

BO-009

Provide secure, role-based access to organizational resources.

Priority: High

BO-010

Enable future integration with enterprise systems and third-party services.

Priority: Medium

4.4 Business Requirements

BR-001: SOP Digitization

Priority: High

The system shall enable organizations to upload and manage Standard Operating Procedures (SOPs) in digital formats and convert them into structured business process definitions.

BR-002: AI-Powered Process Understanding

Priority: High

The platform shall analyze uploaded SOPs using Artificial Intelligence to identify business rules, conditions, actions, approval chains, roles, and dependencies.

BR-003: Automatic Workflow Generation

Priority: High

The platform shall automatically generate executable workflow models from AI-generated process definitions without requiring manual workflow design.

BR-004: Workflow Validation

Priority: High

The platform shall validate generated workflows to identify logical conflicts, missing approvals, undefined roles, circular dependencies, and incomplete business rules before deployment.

BR-005: Workflow Simulation

Priority: High

The platform shall simulate business workflows using different input scenarios to predict execution outcomes before workflows are deployed into production.

BR-006: Workflow Execution

Priority: High

The platform shall execute approved workflows while maintaining workflow state, routing tasks, and tracking execution progress.

BR-007: Human Approval Management

Priority: High

The platform shall support configurable approval workflows involving managers, department heads, administrators, and other authorized personnel.

BR-008: Organizational Analytics

Priority: High

The platform shall provide dashboards that visualize workflow performance, completion rates, approval delays, execution statistics, and operational metrics.

BR-009: Process Optimization

Priority: Medium

The platform shall analyze workflow execution history and recommend improvements that reduce delays, eliminate redundant activities, and improve operational efficiency.

BR-010: Notification Management

Priority: Medium

The platform shall notify users regarding workflow status changes, approval requests, reminders, escalations, and completed tasks.

BR-011: User and Organization Management

Priority: High

The platform shall provide facilities to manage organizations, departments, users, roles, permissions, and system configurations.

BR-012: Security and Access Control

Priority: High

The platform shall ensure that organizational data is protected using authentication, authorization, encrypted communication, audit logging, and role-based access control mechanisms.

BR-013: Multi-Tenant Architecture

Priority: Medium

The platform shall support multiple organizations while ensuring complete isolation of organizational data and business processes.

BR-014: Scalability

Priority: Medium

The platform shall support increasing numbers of organizations, users, workflows, and process executions without significant degradation in performance.

BR-015: AI Extensibility

Priority: Medium

The platform shall be designed with an abstract AI service layer to allow future integration of multiple AI providers without significant architectural modifications.

4.5 Business Value

The implementation of ProcessMind AI provides measurable business value to organizations.

Expected benefits include:

- Reduced workflow design time.
- Faster process execution.
- Improved operational efficiency.

- Increased process consistency.
- Reduced manual errors.
- Improved compliance with organizational policies.
- Enhanced process transparency.
- Reduced dependency on external business process consultants.
- Continuous AI-driven process optimization.
- Improved decision-making through operational analytics.

4.6 Stakeholders

The primary stakeholders of ProcessMind AI include:

Stakeholder	Responsibility
Organization Owners	Adopt and manage the platform within the organization
Organization Administrators	Configure workflows, users, and system settings
Process Analysts	Analyze, validate, and optimize business processes
Managers	Review and approve workflow requests
Employees	Submit requests and participate in workflows
System Administrators	Maintain the platform infrastructure
Software Developers	Design, implement, and maintain the application
Quality Assurance Engineers	Verify system quality and compliance
Business Analysts	Gather and validate business requirements
Future Investors	Evaluate the business value and scalability of the platform

4.7 Business Success Criteria

ProcessMind AI shall be considered successful when it demonstrates the following outcomes:

- Organizations can successfully upload SOPs and generate workflows automatically.
- Workflow generation significantly reduces manual process design effort.
- Workflow validation identifies logical issues before deployment.
- Simulations accurately predict workflow execution paths.
- Workflow execution operates reliably in production.

- Approval processes are completed efficiently.
- Analytics provide actionable operational insights.
- AI-generated optimization recommendations contribute to measurable process improvements.
- The platform supports multiple organizations securely.
- The system provides a scalable foundation for future enterprise expansion and advanced AI capabilities.

The business requirements defined in this chapter establish the strategic objectives and expected business outcomes of ProcessMind AI. These requirements provide the foundation for the detailed functional requirements described in the following chapters.

Chapter 5: Functional Requirements

5.1 Overview

This chapter specifies the functional requirements of **ProcessMind AI**. Functional requirements describe the services, behaviors, and operations that the system must perform to satisfy business objectives.

Each functional requirement is assigned a unique identifier (FR-XXX) to support requirement traceability throughout the software development life cycle.

The functional requirements are organized according to the major software modules of the platform.

Functional Modules

The ProcessMind AI platform consists of the following functional modules:

1. Authentication and Authorization Module
2. User Management Module
3. Organization Management Module
4. SOP Intelligence Engine
5. Workflow Generation Engine
6. Process Validation Engine
7. Process Simulation Engine
8. Workflow Execution Engine
9. Human Approval Engine
10. Analytics Dashboard
11. Process Optimization Engine

12. Notification Module

13. Administration Module

Each module is described in detail in the following sections.

5.2 Authentication and Authorization Module

5.2.1 Module Description

The Authentication and Authorization Module is responsible for securely identifying users, authenticating their credentials, authorizing access to system resources, and enforcing Role-Based Access Control (RBAC). This module ensures that only authorized users can access protected functionalities while maintaining the confidentiality, integrity, and security of organizational data.

5.2.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-001	The system shall allow Organization Administrators to register new users within their organization.	High
FR-002	The system shall allow registered users to log in using their email address and password.	High
FR-003	The system shall validate user credentials before granting access to the system.	High
FR-004	The system shall generate a secure JWT access token after successful authentication.	High
FR-005	The system shall verify the validity of the JWT token before processing every protected API request.	High
FR-006	The system shall deny access to protected resources when authentication fails.	High
FR-007	The system shall enforce Role-Based Access Control (RBAC) based on assigned user roles.	High
FR-008	The system shall allow authenticated users to securely log out of the system.	Medium

FR-009	The system shall securely hash user passwords using BCrypt before storing them in the database.	High
FR-010	The system shall never store passwords in plain text.	High
FR-011	The system shall allow users to change their passwords after successful authentication.	Medium
FR-012	The system shall provide a secure password reset mechanism for users who forget their passwords.	Medium
FR-013	The system shall enforce configurable password complexity requirements during password creation and updates.	Medium
FR-014	The system shall automatically invalidate expired JWT tokens.	High
FR-015	The system shall terminate user sessions when users log out or when authentication tokens expire.	High
FR-016	The system shall restrict users from accessing modules outside their assigned permissions.	High
FR-017	The system shall prevent disabled or inactive user accounts from authenticating.	High
FR-018	The system shall record all authentication activities, including successful logins, failed login attempts, password changes, and logout events in the audit log.	Medium
FR-019	The system shall allow administrators to activate, deactivate, suspend, or reactivate user accounts.	Medium
FR-020	The system shall return appropriate HTTP status codes (401 Unauthorized, 403 Forbidden) for unauthorized access attempts.	High

5.2.3 Supported User Roles

Role	Description
System Administrator	Full access to the entire platform and system configuration.
Organization Administrator	Manages users, SOPs, workflows, departments, and organizational settings.
Process Analyst	Reviews SOPs, validates workflows, performs simulations, and analyzes business processes.
Manager	Reviews and approves workflow requests assigned to them.

Employee	Submits requests, tracks workflow status, and performs assigned workflow tasks.
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5.2.4 Preconditions

- Users must have a registered account.
- User account must belong to an active organization.
- Authentication service must be available.
- Database connection must be active.

5.2.5 Postconditions

- Authenticated users receive a valid JWT access token.
- User session is established.
- User role and permissions are loaded.
- Authentication events are recorded in the audit log.

5.2.6 Business Rules

Rule ID	Business Rule
BR-AUTH-001	Every protected resource shall require authentication.
BR-AUTH-002	Each user shall belong to exactly one organization.
BR-AUTH-003	Passwords shall never be stored in plain text.
BR-AUTH-004	JWT tokens shall have a configurable expiration period.
BR-AUTH-005	Only authorized administrators may create, activate, suspend, or deactivate user accounts.
BR-AUTH-006	Access permissions shall be determined solely by assigned user roles.

5.2.7 Exception Handling

Exception	Expected System Response
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Invalid email address	Reject authentication and display an appropriate error message.
Incorrect password	Reject authentication without revealing which credential is incorrect.
Expired JWT token	Reject the request and require the user to authenticate again.
Disabled account	Prevent login and notify the user that the account is inactive.
Unauthorized access	Return HTTP 403 (Forbidden).
Missing authentication token	Return HTTP 401 (Unauthorized).
Database unavailable	Return an internal server error and log the incident.

5.2.8 Acceptance Criteria

The Authentication and Authorization Module shall be considered complete when:

- Users can securely authenticate using valid credentials.
- JWT-based authentication functions correctly.
- Role-Based Access Control (RBAC) is enforced across all protected resources.
- Passwords are securely hashed using BCrypt.
- Unauthorized users cannot access restricted modules.
- Authentication events are successfully recorded in the audit log.
- Users can securely log out and terminate active sessions.

5.3 User Management Module

5.3.1 Module Description

The User Management Module enables administrators to manage user accounts within their organizations while allowing users to maintain their personal profiles. The module supports user creation, profile updates, role assignment, account status management, and user search operations.

The module ensures that only authorized administrators can perform administrative user management functions, while individual users are limited to managing their own profile information.

5.3.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-021	The system shall allow Organization Administrators to create new user accounts.	High
FR-022	The system shall allow administrators to assign roles during user creation.	High
FR-023	The system shall validate that each email address is unique within the organization.	High
FR-024	The system shall automatically generate a unique User ID for every registered user.	High
FR-025	The system shall allow administrators to update user information.	High
FR-026	The system shall allow users to update their own profile information.	Medium
FR-027	The system shall prevent users from modifying their assigned roles.	High
FR-028	The system shall allow administrators to change user roles.	High
FR-029	The system shall allow administrators to activate or deactivate user accounts.	High
FR-030	The system shall allow administrators to temporarily suspend user accounts.	Medium
FR-031	The system shall allow administrators to permanently delete inactive user accounts.	Medium
FR-032	The system shall prevent deletion of users who own active workflows unless ownership is transferred.	High
FR-033	The system shall maintain complete user profile information, including name, email, department, role, and contact information.	Medium
FR-034	The system shall allow administrators to search users using name, email, department, or role.	Medium
FR-035	The system shall support filtering users based on account status.	Medium
FR-036	The system shall display the user's account status (Active, Inactive, Suspended).	Medium
FR-037	The system shall display each user's assigned role and department.	Medium
FR-038	The system shall record the account creation date and last login date for every user.	Medium

FR-039	The system shall maintain an audit history of user account modifications.	High
FR-040	The system shall prevent duplicate user records from being created.	High

5.3.3 User Information

Each user account shall maintain the following information:

Field	Description
User ID	Unique system-generated identifier
First Name	User's first name
Last Name	User's last name
Email Address	Login email address
Password	Securely hashed password
Role	Assigned system role
Department	Associated department
Contact Number	Optional contact information
Account Status	Active, Inactive, Suspended
Organization	Organization to which the user belongs
Created Date	Account creation timestamp
Last Login	Most recent successful login
Updated Date	Last profile modification timestamp

5.3.4 Preconditions

- Organization exists.
- Administrator is authenticated.
- Administrator has permission to manage users.
- Database connection is available.

5.3.5 Postconditions

- User account is successfully created, updated, suspended, activated, or deleted.
- Audit logs are updated.
- Notifications are generated where applicable.

- Database records are synchronized.

5.3.6 Business Rules

Rule ID	Business Rule
BR-USER-001	Every user shall belong to exactly one organization.
BR-USER-002	Every user shall have exactly one primary system role.
BR-USER-003	Email addresses shall be unique within an organization.
BR-USER-004	Users shall only modify their own personal profile information.
BR-USER-005	Only administrators may modify user roles.
BR-USER-006	Only administrators may deactivate or delete user accounts.
BR-USER-007	Deleted accounts shall be retained in audit logs for traceability.

5.3.7 Exception Handling

Exception	Expected System Response
Duplicate email address	Reject account creation and display an appropriate validation message.
Invalid user information	Reject submission and display validation errors.
Unauthorized role modification	Return HTTP 403 Forbidden.
Attempt to delete active workflow owner	Prevent deletion and request workflow reassignment.
Database unavailable	Return internal server error and log the incident.

5.3.8 Acceptance Criteria

The User Management Module shall be considered complete when:

- Administrators can successfully create, update, suspend, activate, and delete user accounts.
- Users can update their personal profiles.

- Duplicate user accounts are prevented.
- Role assignments are enforced correctly.
- User search and filtering functions operate correctly.
- All account modifications are recorded in audit logs.
- User information is stored and retrieved accurately.

5.4 Organization Management Module

5.4.1 Module Description

The Organization Management Module enables the creation, configuration, and administration of organizations within the ProcessMind AI platform. It provides complete isolation of organizational data, allowing multiple organizations to securely share the same application infrastructure while ensuring that users can only access resources belonging to their own organization.

The module manages organization profiles, departments, business settings, workflow configurations, subscription information, and organizational preferences.

5.4.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-041	The system shall allow a System Administrator to register a new organization.	High
FR-042	The system shall generate a unique Organization ID for every organization.	High
FR-043	The system shall allow Organization Administrators to update organization information.	High
FR-044	The system shall allow administrators to configure organization settings.	Medium
FR-045	The system shall support multiple organizations within the same platform.	High
FR-046	The system shall isolate data belonging to different organizations.	High
FR-047	The system shall prevent users from accessing data belonging to other organizations.	High

FR-048	The system shall allow administrators to create departments within an organization.	High
FR-049	The system shall allow administrators to update department information.	Medium
FR-050	The system shall allow administrators to deactivate departments.	Medium
FR-051	The system shall associate users with departments.	High
FR-052	The system shall allow administrators to configure organization-specific approval rules.	High
FR-053	The system shall maintain organization-specific workflow settings.	High
FR-054	The system shall support organization-specific notification settings.	Medium
FR-055	The system shall allow administrators to configure business hours and working days.	Medium
FR-056	The system shall allow administrators to upload organization branding elements such as logos.	Low
FR-057	The system shall maintain subscription information for each organization.	Medium
FR-058	The system shall allow administrators to archive inactive organizations.	Low
FR-059	The system shall maintain audit logs for organization-level changes.	High
FR-060	The system shall prevent duplicate organization names where configured by business rules.	Medium

5.4.3 Organization Information

Each organization shall maintain the following information:

Field	Description
Organization ID	Unique system-generated identifier
Organization Name	Registered organization name
Industry	Business sector or industry
Registration Number	Organization registration number (optional)
Organization Email	Primary contact email
Contact Number	Primary contact number

Address	Organization address
Logo	Organization logo
Time Zone	Organization time zone
Working Days	Configurable business days
Business Hours	Daily operating hours
Subscription Plan	Current subscription tier
Organization Status	Active, Inactive, Archived
Created Date	Organization creation date
Updated Date	Last modification date

5.4.4 Department Information

Each department shall contain:

Field	Description
Department ID	Unique department identifier
Department Name	Name of the department
Department Manager	Assigned manager
Description	Department description
Organization ID	Associated organization
Status	Active or Inactive

5.4.5 Preconditions

- System Administrator is authenticated.
- Organization registration request is valid.
- Database connection is available.
- Organization information passes validation.

5.4.6 Postconditions

- Organization is successfully created or updated.

- Departments are stored correctly.
- Organization settings are applied.
- Audit logs are updated.
- Organization becomes available for user registration.

5.4.7 Business Rules

Rule ID	Business Rule
BR-ORG-001	Every organization shall have a unique Organization ID.
BR-ORG-002	Each user shall belong to only one organization.
BR-ORG-003	Users shall only access data belonging to their organization.
BR-ORG-004	Departments shall belong to exactly one organization.
BR-ORG-005	Organization Administrators shall only manage their own organization.
BR-ORG-006	System Administrators may manage all organizations.
BR-ORG-007	Organization settings shall override system defaults where applicable.

5.4.8 Exception Handling

Exception	Expected System Response
Duplicate organization	Reject registration and notify administrator.
Invalid organization information	Display validation errors.
Unauthorized organization access	Return HTTP 403 Forbidden.
Missing required organization data	Prevent organization creation.
Database unavailable	Return internal server error and record the incident.

5.4.9 Acceptance Criteria

The Organization Management Module shall be considered complete when:

- Multiple organizations can be registered successfully.
- Organization data remains completely isolated between tenants.
- Departments can be created, updated, and managed.
- Organization settings are stored and applied correctly.

- Users cannot access resources belonging to another organization.
- Organization administrators can manage only their own organization.
- All organization-level changes are recorded in audit logs.

5.5 SOP Intelligence Engine

5.5.1 Module Description

The SOP Intelligence Engine is the core Artificial Intelligence module of ProcessMind AI. It is responsible for understanding, interpreting, and transforming natural-language Standard Operating Procedures (SOPs) into structured business process definitions that can be used for workflow generation, validation, simulation, and execution.

Unlike traditional workflow platforms that require users to manually create workflows, the SOP Intelligence Engine enables organizations to upload existing SOP documents and automatically extract business logic using Artificial Intelligence.

The engine utilizes Large Language Models (LLMs) to identify business rules, conditions, actions, approvals, roles, dependencies, decision points, and workflow sequences.

The generated structured process definition serves as the foundation for all downstream modules within the platform.

5.5.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-061	The system shall allow users to upload SOP documents in supported formats.	High
FR-062	The system shall support Plain Text (.txt) SOP documents.	High
FR-063	The system shall support PDF documents.	High
FR-064	The system shall support Microsoft Word (.docx) documents.	High
FR-065	The system shall validate uploaded document formats before processing.	High
FR-066	The system shall reject unsupported file formats.	High
FR-067	The system shall extract readable text from uploaded documents.	High
FR-068	The system shall preprocess extracted text before AI analysis.	High
FR-069	The system shall send SOP content to the AI reasoning engine for analysis.	High
FR-070	The AI shall identify workflow start points.	High
FR-071	The AI shall identify workflow end points.	High

FR-072	The AI shall identify business actions described in the SOP.	High
FR-073	The AI shall identify business conditions within the SOP.	High
FR-074	The AI shall identify approval requirements.	High
FR-075	The AI shall identify organizational roles involved in the process.	High
FR-076	The AI shall identify dependencies between workflow steps.	High
FR-077	The AI shall detect sequential workflow activities.	High
FR-078	The AI shall detect conditional branching logic.	High
FR-079	The AI shall identify parallel workflow activities where applicable.	Medium
FR-080	The AI shall identify workflow inputs and outputs.	Medium
FR-081	The AI shall generate a structured process definition in JSON format.	High
FR-082	The system shall display the extracted process definition for user review.	High
FR-083	The system shall allow users to manually edit AI-generated process definitions before workflow generation.	High
FR-084	The system shall store both the original SOP and the extracted process definition.	High
FR-085	The system shall maintain version history for uploaded SOPs.	Medium
FR-086	The system shall support re-analysis of updated SOP documents.	Medium
FR-087	The system shall log AI processing results for auditing purposes.	Medium
FR-088	The system shall notify users when SOP analysis is completed successfully.	Medium
FR-089	The system shall display meaningful error messages if AI processing fails.	High
FR-090	The system shall allow users to retry SOP analysis after correcting errors.	Medium

5.5.3 Input

The SOP Intelligence Engine shall accept the following inputs:

Input	Description
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Plain Text	User-entered SOP text
PDF	Uploaded PDF document
DOCX	Uploaded Microsoft Word document
SOP Metadata	Title, description, department, category
Organization Information	Organization context for AI reasoning

5.5.4 Output

The engine shall produce the following outputs:

Output	Description
Structured Workflow JSON	Machine-readable workflow definition
Business Rules	Extracted business logic
Conditions	Decision conditions
Workflow Steps	Ordered list of process activities
Approval Chain	Approval hierarchy
Roles	Organizational roles involved
Dependencies	Relationships between workflow steps
AI Analysis Report	Summary of extracted information

5.5.5 AI Processing Flow



5.5.6 Business Rules

Rule ID	Business Rule
BR-SOP-001	Only supported document formats shall be processed.
BR-SOP-002	Uploaded SOP documents shall belong to the user's organization.
BR-SOP-003	Original SOP documents shall remain unchanged after analysis.
BR-SOP-004	Users shall review AI-generated process definitions before deployment.
BR-SOP-005	Every uploaded SOP shall maintain version history.
BR-SOP-006	AI-generated process definitions may be manually modified before workflow generation.
BR-SOP-007	AI analysis results shall be stored for future reference and auditing.

5.5.7 Exception Handling

Exception	Expected System Response
Unsupported file format	Reject upload and display supported formats.
Empty document	Reject processing and request valid content.
Corrupted PDF/DOCX	Display document parsing error.
AI service unavailable	Notify user and allow retry later.
AI processing timeout	Cancel processing and display timeout message.
Invalid AI response	Log the error and request manual review.
Network interruption	Preserve uploaded document and resume processing when possible.

5.5.8 Acceptance Criteria

The SOP Intelligence Engine shall be considered complete when:

- Users can successfully upload supported SOP documents.
- Text is correctly extracted from uploaded documents.
- AI accurately identifies business rules, conditions, actions, approvals, and roles.
- Structured process definitions are successfully generated.
- Users can review and edit AI-generated results.
- Original SOP documents and generated process definitions are securely stored.
- SOP version history is maintained.

- AI processing failures are handled gracefully.
- Audit logs capture all SOP analysis activities.

5.6 Workflow Generation Engine

5.6.1 Module Description

The Workflow Generation Engine is responsible for automatically converting structured business process definitions into executable workflow models.

The engine receives AI-generated process definitions from the SOP Intelligence Engine and constructs workflow structures consisting of activities, decision nodes, approval chains, transitions, dependencies, and execution paths.

Unlike traditional workflow platforms that require users to manually design workflows, the Workflow Generation Engine automatically creates workflow diagrams and machine-executable workflow definitions.

The generated workflow serves as the foundation for workflow validation, simulation, execution, monitoring, and optimization.

5.6.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-091	The system shall generate workflows automatically from AI-generated process definitions.	High
FR-092	The system shall create workflow start nodes automatically.	High
FR-093	The system shall create workflow end nodes automatically.	High
FR-094	The system shall generate sequential workflow activities.	High
FR-095	The system shall generate decision nodes for conditional logic.	High
FR-096	The system shall generate approval nodes based on identified approval requirements.	High
FR-097	The system shall generate parallel execution branches where applicable.	Medium
FR-098	The system shall automatically connect workflow nodes based on extracted dependencies.	High
FR-099	The system shall assign execution order to workflow activities.	High
FR-100	The system shall generate workflow transitions between activities.	High
FR-101	The system shall assign workflow roles to individual activities.	High

FR-102	The system shall assign approval responsibilities to workflow steps.	High
FR-103	The system shall generate workflow metadata including name, description, version, and owner.	Medium
FR-104	The system shall generate a visual workflow diagram.	High
FR-105	The system shall allow users to edit generated workflow diagrams before deployment.	High
FR-106	The system shall support drag-and-drop editing of workflow nodes.	Medium
FR-107	The system shall allow users to add custom workflow activities manually.	Medium
FR-108	The system shall allow users to remove generated workflow activities.	Medium
FR-109	The system shall allow users to modify workflow connections.	Medium
FR-110	The system shall save workflow definitions in executable JSON format.	High
FR-111	The system shall support workflow versioning.	Medium
FR-112	The system shall allow users to duplicate existing workflows.	Low
FR-113	The system shall maintain draft workflows before deployment.	High
FR-114	The system shall store workflow generation history.	Medium
FR-115	The system shall notify users when workflow generation is successfully completed.	Medium

5.6.3 Workflow Components

Each generated workflow shall consist of the following components:

Component	Description
Start Node	Beginning of the workflow
End Node	Completion of the workflow
Activity Node	Business task or operation
Decision Node	Conditional branching
Approval Node	Human approval step
Transition	Connection between workflow nodes
Role Assignment	Responsible organizational role
Conditions	Business rules controlling execution
Workflow Metadata	Name, version, owner, timestamps

5.6.4 Input

The Workflow Generation Engine shall accept:

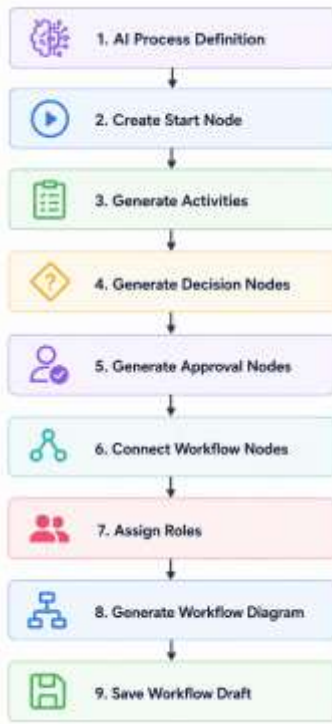
Input	Description
Process Definition JSON	Structured output from SOP Intelligence Engine
Organization Settings	Organization-specific workflow rules
User Configuration	Optional workflow customization
Approval Rules	Organization approval policies

5.6.5 Output

The engine shall generate:

Output	Description
Workflow Diagram	Visual representation of workflow
Executable Workflow JSON	Machine-readable workflow definition
Activity List	Ordered workflow activities
Decision Tree	Conditional execution logic
Approval Chain	Approval hierarchy
Workflow Metadata	Version, owner, timestamps

5.6.6 Workflow Generation Process



5.6.7 Business Rules

Rule ID	Business Rule
BR-WG-001	Every workflow shall contain exactly one Start Node.
BR-WG-002	Every workflow shall contain at least one End Node.
BR-WG-003	Every workflow activity shall be reachable from the Start Node.
BR-WG-004	Workflow activities shall follow the execution sequence extracted from the SOP.
BR-WG-005	Decision nodes shall contain at least two execution paths.
BR-WG-006	Approval nodes shall have assigned approvers before deployment.
BR-WG-007	Generated workflows shall remain editable until deployment.
BR-WG-008	Every workflow version shall be uniquely identifiable.

5.6.8 Exception Handling

Exception	Expected System Response
Missing process definition	Prevent workflow generation and display an error message.
Invalid workflow structure	Reject generation and request manual review.
Missing approval role	Notify user and highlight the affected workflow node.

Circular workflow detected	Stop generation and report validation error.
Database failure	Roll back workflow creation and log the incident.
Workflow generation failure	Preserve input data and allow users to retry.

5.6.9 Acceptance Criteria

The Workflow Generation Engine shall be considered complete when:

- Workflows are automatically generated from AI process definitions.
- Generated workflows accurately represent SOP logic.
- Visual workflow diagrams are displayed successfully.
- Users can edit workflow diagrams before deployment.
- Executable workflow definitions are stored correctly.
- Workflow version history is maintained.
- Workflow generation failures are handled gracefully.
- Generated workflows are ready for validation and simulation.

5.7 Process Validation Engine

5.7.1 Module Description

The Process Validation Engine is responsible for verifying the correctness, completeness, consistency, and quality of AI-generated workflows before deployment.

The engine analyzes workflow structures, business rules, approval chains, execution paths, and organizational constraints to identify logical errors, inconsistencies, missing information, and potential execution failures.

Its primary objective is to ensure that generated workflows are reliable, executable, compliant with organizational policies, and free from structural defects.

The validation engine generates a comprehensive validation report containing detected issues, warnings, recommendations, and an overall workflow quality score.

5.7.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-116	The system shall validate every generated workflow before deployment.	High
FR-117	The system shall verify that every workflow contains a Start Node.	High

FR-118	The system shall verify that every workflow contains at least one End Node.	High
FR-119	The system shall detect unreachable workflow activities.	High
FR-120	The system shall detect dead-end workflow paths.	High
FR-121	The system shall detect circular dependencies between workflow activities.	High
FR-122	The system shall identify duplicate workflow activities.	Medium
FR-123	The system shall verify that every approval activity has an assigned approver.	High
FR-124	The system shall verify that all referenced organizational roles exist.	High
FR-125	The system shall identify missing workflow transitions.	High
FR-126	The system shall validate conditional branches for logical consistency.	High
FR-127	The system shall detect conflicting business rules.	High
FR-128	The system shall detect overlapping workflow conditions.	High
FR-129	The system shall identify incomplete workflow definitions.	High
FR-130	The system shall identify unused workflow activities.	Medium
FR-131	The system shall verify workflow execution order.	High
FR-132	The system shall verify workflow dependency relationships.	High
FR-133	The system shall generate validation warnings for potential risks.	Medium
FR-134	The system shall generate validation errors for critical workflow issues.	High
FR-135	The system shall calculate an overall workflow validation score.	Medium
FR-136	The system shall generate a detailed validation report.	High
FR-137	The system shall allow users to review validation results before deployment.	High
FR-138	The system shall prevent deployment when critical validation errors exist.	High
FR-139	The system shall allow deployment when only warnings are present, subject to administrator approval.	Medium
FR-140	The system shall maintain validation history for every workflow version.	Medium

5.7.3 Validation Types

The Process Validation Engine shall perform the following validations:

Validation Type	Description
Structural Validation	Validates workflow structure and connectivity
Logical Validation	Detects inconsistent business rules
Approval Validation	Verifies approval chains and approvers
Role Validation	Confirms referenced organizational roles exist
Dependency Validation	Checks workflow dependencies
Execution Validation	Verifies executable workflow paths
Completeness Validation	Ensures all required workflow components exist
Compliance Validation	Validates organization-specific business rules

5.7.4 Validation Report

The validation report shall include:

Report Item	Description
Workflow Name	Name of the workflow
Validation Date	Date and time of validation
Workflow Version	Version being validated
Validation Score	Overall workflow quality score
Errors	Critical issues preventing deployment
Warnings	Non-critical recommendations
Passed Checks	Successful validation checks
Recommendations	Suggested improvements

5.7.5 Validation Workflow



5.7.6 Business Rules

Rule ID	Business Rule
BR-VAL-001	Every workflow shall pass structural validation before deployment.
BR-VAL-002	Workflows containing critical validation errors shall not be deployed.
BR-VAL-003	Every approval activity shall have at least one assigned approver.
BR-VAL-004	Every workflow activity shall be reachable from the Start Node.
BR-VAL-005	Every execution path shall eventually reach an End Node.
BR-VAL-006	Circular workflow dependencies shall not be permitted.
BR-VAL-007	Validation reports shall be stored for auditing purposes.
BR-VAL-008	Validation shall be re-executed whenever a workflow is modified.

5.7.7 Exception Handling

Exception	Expected System Response
Missing Start Node	Display validation error and prevent deployment.
Missing End Node	Display validation error and prevent deployment.
Circular dependency detected	Highlight affected workflow nodes and reject deployment.
Undefined organizational role	Display validation warning or error depending on severity.
Missing approval assignment	Prevent deployment until resolved.
Workflow corruption	Reject validation and notify administrator.
Validation engine failure	Log the incident and allow validation retry.

5.7.8 Acceptance Criteria

The Process Validation Engine shall be considered complete when:

- All generated workflows are validated before deployment.
- Structural and logical workflow issues are detected accurately.
- Critical validation errors prevent deployment.
- Validation reports clearly identify issues and recommendations.
- Workflow quality scores are generated successfully.
- Validation history is maintained for all workflow versions.
- Users can review validation results before deployment.
- Validation automatically re-executes after workflow modifications.

5.8 Process Simulation Engine

5.8.1 Module Description

The Process Simulation Engine enables organizations to simulate workflow execution before deployment. It evaluates how workflows behave under different business scenarios by virtually executing workflow logic without affecting production data.

The engine predicts execution paths, approval sequences, completion times, resource utilization, and possible failure points based on user-defined input scenarios and workflow conditions.

Simulation helps organizations validate workflow behavior, reduce operational risks, optimize business processes, and ensure that workflows produce the expected outcomes before being activated.

5.8.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-141	The system shall allow users to simulate workflows before deployment.	High
FR-142	The system shall allow users to provide custom input values for simulation.	High
FR-143	The system shall execute workflow logic using simulation data without modifying production data.	High
FR-144	The system shall evaluate workflow conditions during simulation.	High
FR-145	The system shall simulate approval workflows.	High
FR-146	The system shall simulate conditional branches.	High
FR-147	The system shall simulate parallel workflow execution where applicable.	Medium
FR-148	The system shall display the workflow execution path followed during simulation.	High
FR-149	The system shall estimate workflow completion time.	Medium
FR-150	The system shall estimate the number of approval steps required.	Medium
FR-151	The system shall identify workflow bottlenecks during simulation.	Medium
FR-152	The system shall identify workflow activities that are never executed for the given scenario.	Medium
FR-153	The system shall display simulation results in a user-friendly format.	High
FR-154	The system shall generate a simulation report for each execution.	High
FR-155	The system shall allow users to compare multiple simulation scenarios.	Medium
FR-156	The system shall support repeated simulations using different input values.	High
FR-157	The system shall preserve previous simulation history.	Medium
FR-158	The system shall notify users when simulation is completed successfully.	Medium
FR-159	The system shall prevent deployment if simulation identifies critical execution failures.	High
FR-160	The system shall allow users to export simulation reports.	Low

5.8.3 Simulation Inputs

The Process Simulation Engine shall accept the following inputs:

Input	Description
Workflow Definition	Validated workflow selected for simulation
Input Variables	User-provided business values
Simulation Scenario	Test case selected by the user
Organization Settings	Organization-specific rules
Approval Configuration	Approval hierarchy used during simulation

5.8.4 Simulation Outputs

The engine shall generate the following outputs:

Output	Description
Execution Path	Activities executed during simulation
Decision Results	Outcomes of workflow conditions
Approval Chain	Simulated approval sequence
Completion Time	Estimated execution duration
Bottlenecks	Identified process delays
Workflow Status	Simulation result
Simulation Report	Detailed simulation summary

5.8.5 Simulation Workflow



5.8.6 Business Rules

Rule ID	Business Rule
BR-SIM-001	Simulations shall not modify production workflow data.
BR-SIM-002	Only validated workflows may be simulated.
BR-SIM-003	Every simulation shall generate a simulation report.
BR-SIM-004	Simulation history shall be stored for future reference.
BR-SIM-005	Multiple simulation scenarios may be executed for the same workflow.
BR-SIM-006	Critical simulation failures shall be reported before deployment.
BR-SIM-007	Simulation results shall remain available until manually deleted or archived.

5.8.7 Exception Handling

Exception	Expected System Response
Invalid workflow selected	Prevent simulation and notify the user.
Missing simulation inputs	Display validation errors and request required data.
Simulation execution failure	Stop simulation and display an error message.
Workflow corruption	Reject simulation and log the incident.
System timeout	Cancel simulation and allow the user to retry.
AI reasoning failure	Notify the user and preserve workflow data.

5.8.8 Acceptance Criteria

The Process Simulation Engine shall be considered complete when:

- Users can successfully simulate validated workflows.
- Simulation accurately follows workflow logic.
- Execution paths are displayed correctly.
- Completion time estimates are generated.
- Approval chains are simulated accurately.
- Simulation reports are generated successfully.
- Multiple simulation scenarios can be compared.
- Production workflow data remains unchanged during simulation.
- Critical simulation failures prevent unsafe workflow deployment.

5.9 Workflow Execution Engine

5.9.1 Module Description

The Workflow Execution Engine is responsible for executing deployed workflows in accordance with their defined business logic, workflow structure, approval rules, and organizational policies.

The engine manages the complete lifecycle of workflow instances, including workflow initiation, task routing, state transitions, decision evaluation, approval coordination, exception handling, workflow completion, and execution history.

Each workflow instance is executed independently while maintaining complete traceability, auditability, and data integrity.

The Workflow Execution Engine acts as the operational core of ProcessMind AI and ensures that business processes are executed reliably, consistently, and securely.

5.9.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-161	The system shall execute only deployed workflows.	High
FR-162	The system shall create a new workflow instance for each execution request.	High
FR-163	The system shall assign a unique Workflow Instance ID to every execution.	High
FR-164	The system shall initialize workflow execution from the Start Node.	High
FR-165	The system shall execute workflow activities according to the defined execution sequence.	High
FR-166	The system shall evaluate workflow conditions during execution.	High
FR-167	The system shall automatically route execution based on condition evaluation results.	High
FR-168	The system shall create human approval tasks when approval nodes are reached.	High
FR-169	The system shall pause workflow execution while waiting for required approvals.	High
FR-170	The system shall resume workflow execution after approval decisions are received.	High
FR-171	The system shall reject workflows when approval is denied according to workflow rules.	High
FR-172	The system shall execute parallel workflow branches simultaneously where defined.	Medium
FR-173	The system shall merge parallel workflow branches before continuing execution.	Medium
FR-174	The system shall update workflow state after every completed activity.	High
FR-175	The system shall record execution timestamps for each workflow activity.	High
FR-176	The system shall assign workflow tasks to the appropriate users or roles.	High
FR-177	The system shall support reassignment of workflow tasks by authorized users.	Medium

FR-178	The system shall automatically complete workflow execution when the End Node is reached.	High
FR-179	The system shall terminate workflow execution if an unrecoverable system error occurs.	High
FR-180	The system shall maintain complete execution history for every workflow instance.	High
FR-181	The system shall allow authorized users to view workflow execution progress.	High
FR-182	The system shall support cancellation of active workflow instances by authorized administrators.	Medium
FR-183	The system shall maintain workflow execution logs for auditing purposes.	High
FR-184	The system shall notify users of workflow status changes.	Medium
FR-185	The system shall automatically archive completed workflow instances based on retention policies.	Medium

5.9.3 Workflow States

Each workflow instance shall maintain one of the following states:

State	Description
Draft	Workflow definition created but not deployed
Active	Workflow deployed and ready for execution
Running	Workflow currently executing
Waiting for Approval	Execution paused pending approval
Completed	Workflow finished successfully
Rejected	Workflow terminated due to rejection
Cancelled	Workflow cancelled by an authorized user
Failed	Workflow terminated due to a system error
Archived	Workflow retained for historical purposes

5.9.4 Execution Inputs

The Workflow Execution Engine shall accept the following inputs:

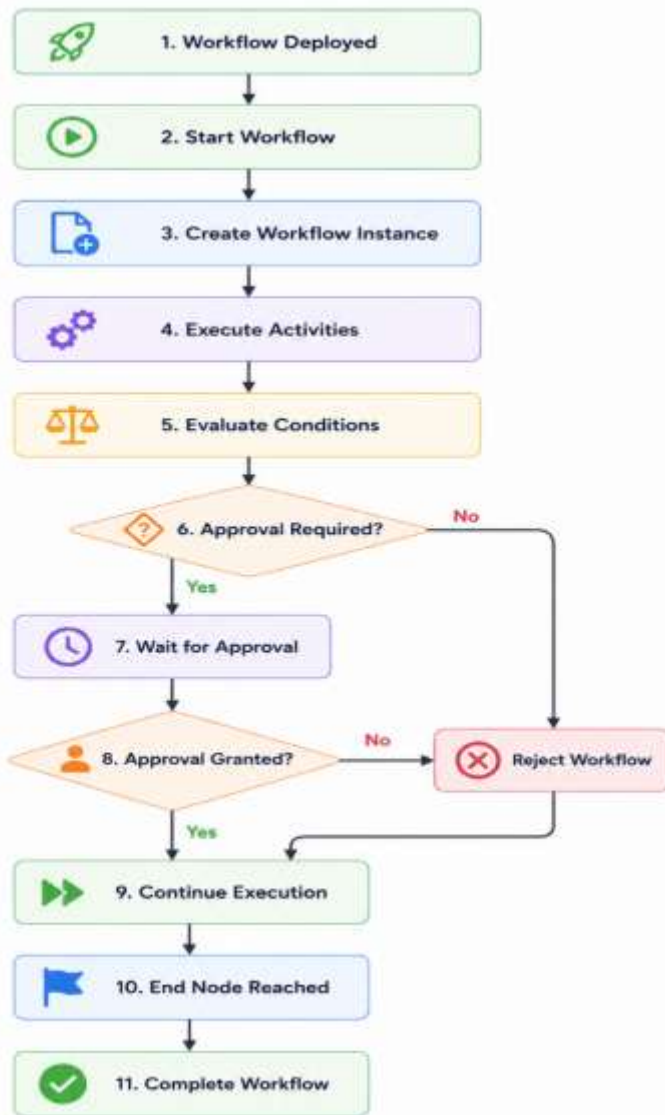
Input	Description
Workflow Definition	Deployed workflow model
Workflow Instance Request	User request to start workflow
Business Data	Input values required by the workflow
Approval Decisions	Responses from approvers
Organization Configuration	Organization-specific workflow settings

5.9.5 Execution Outputs

The Workflow Execution Engine shall produce:

Output	Description
Workflow Instance	Active workflow execution
Task Assignments	Generated workflow tasks
Approval Requests	Human approval tasks
Workflow Status	Current execution state
Execution History	Complete activity log
Notifications	Workflow events and updates

5.9.6 Workflow Lifecycle



5.9.7 Business Rules

Rule ID	Business Rule
BR-EXEC-001	Only deployed workflows may be executed.
BR-EXEC-002	Every workflow execution shall create a unique workflow instance.

BR-EXEC-003	Workflow execution shall always begin at the Start Node.
BR-EXEC-004	Workflow execution shall terminate at an End Node, cancellation, rejection, or failure.
BR-EXEC-005	Approval tasks shall suspend workflow execution until completed.
BR-EXEC-006	Every workflow activity shall be recorded in the execution history.
BR-EXEC-007	Workflow execution logs shall be retained for auditing purposes.
BR-EXEC-008	Only authorized users may cancel active workflow instances.

5.9.8 Exception Handling

Exception	Expected System Response
Workflow not deployed	Prevent execution and notify the user.
Missing workflow input data	Reject execution request and request required information.
Approval timeout	Escalate or terminate according to workflow rules.
System failure during execution	Preserve workflow state and record failure details.
Invalid workflow state	Reject operation and log the incident.
Database failure	Roll back incomplete transactions where applicable.
User assignment unavailable	Notify administrator and pause execution if required.

5.9.9 Acceptance Criteria

The Workflow Execution Engine shall be considered complete when:

- Deployed workflows execute successfully.
- Workflow instances are created correctly.
- Workflow states transition correctly throughout execution.
- Approval stages pause and resume execution appropriately.
- Conditional routing follows workflow logic accurately.
- Execution history is maintained for every workflow instance.
- Workflow cancellation and rejection operate correctly.
- Audit logs accurately record execution events.
- Workflow completion is detected automatically.
- Failed workflow executions are handled gracefully without data corruption.

5.10 Human Approval Engine

5.10.1 Module Description

The Human Approval Engine is responsible for managing approval-based workflow activities within ProcessMind AI. It enables the routing of approval requests to designated users or organizational roles, collects approval decisions, enforces approval rules, manages escalation procedures, and maintains a complete audit trail of all approval activities.

The module supports configurable approval hierarchies, multiple approval levels, sequential and parallel approvals, delegation of approval authority, reminders, deadlines, and escalation mechanisms.

It integrates directly with the Workflow Execution Engine to pause workflow execution until all required approval decisions have been received.

5.10.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-186	The system shall generate approval requests when a workflow reaches an approval activity.	High
FR-187	The system shall assign approval requests to the designated approver(s).	High
FR-188	The system shall support single-level approval workflows.	High
FR-189	The system shall support multi-level approval workflows.	High
FR-190	The system shall support sequential approval processes.	High
FR-191	The system shall support parallel approval processes.	Medium
FR-192	The system shall allow approvers to approve workflow requests.	High
FR-193	The system shall allow approvers to reject workflow requests.	High
FR-194	The system shall allow approvers to request additional information before making a decision.	Medium
FR-195	The system shall allow approvers to provide comments with approval decisions.	Medium
FR-196	The system shall record approval timestamps.	High
FR-197	The system shall record the identity of every approver.	High
FR-198	The system shall automatically continue workflow execution after all required approvals are received.	High
FR-199	The system shall terminate or redirect workflows according to configured rejection rules.	High
FR-200	The system shall support configurable approval deadlines.	Medium

FR-201	The system shall send reminder notifications for pending approvals.	Medium
FR-202	The system shall automatically escalate overdue approval requests.	High
FR-203	The system shall support delegation of approval authority to another authorized user.	Medium
FR-204	The system shall maintain a complete approval history for each workflow instance.	High
FR-205	The system shall display the current approval status to authorized users.	High
FR-206	The system shall prevent unauthorized users from approving workflow requests.	High
FR-207	The system shall allow administrators to configure approval rules and escalation policies.	Medium
FR-208	The system shall notify workflow initiators of approval decisions.	Medium
FR-209	The system shall allow administrators to reassign pending approval requests.	Medium
FR-210	The system shall maintain audit logs for all approval activities.	High

5.10.3 Approval Types

The system shall support the following approval mechanisms:

Approval Type	Description
Single Approval	One designated approver makes the decision.
Sequential Approval	Multiple approvers review the request in a predefined order.
Parallel Approval	Multiple approvers review the request simultaneously.
Multi-Level Approval	Workflow passes through multiple organizational approval levels.
Delegated Approval	Approval authority is temporarily assigned to another authorized user.

5.10.4 Approval Statuses

Each approval request shall maintain one of the following statuses:

Status	Description
Pending	Waiting for approver action
Approved	Request approved

Rejected	Request rejected
Escalated	Forwarded to a higher authority
Delegated	Assigned to another approver
Expired	Approval deadline exceeded
Cancelled	Approval request cancelled

5.10.5 Approval Workflow



5.10.6 Business Rules

Rule ID	Business Rule
BR-APP-001	Every approval request shall have at least one assigned approver.
BR-APP-002	Only authorized approvers may approve or reject requests.
BR-APP-003	Workflow execution shall remain suspended until required approvals are completed.
BR-APP-004	Approval comments shall be recorded when provided.
BR-APP-005	All approval activities shall be stored in the audit log.
BR-APP-006	Escalation shall occur automatically when approval deadlines are exceeded.
BR-APP-007	Delegated approvers shall possess equivalent approval permissions.
BR-APP-008	Approval history shall remain immutable after workflow completion.

5.10.7 Exception Handling

Exception	Expected System Response
Assigned approver does not exist	Prevent workflow execution and notify the administrator.
Unauthorized approval attempt	Reject the request and return HTTP 403 Forbidden.
Approval deadline exceeded	Trigger configured escalation policy.
Delegated approver unavailable	Notify the administrator and request reassignment.
Workflow cancelled during approval	Cancel all pending approval requests.
Notification failure	Log the incident and retry notification delivery.
Database failure	Preserve workflow state and record the error for recovery.

5.10.8 Acceptance Criteria

The Human Approval Engine shall be considered complete when:

- Approval requests are automatically generated during workflow execution.
- Single-level and multi-level approvals function correctly.
- Sequential and parallel approval workflows operate as configured.
- Workflow execution pauses and resumes appropriately.
- Approval deadlines and escalations are enforced.
- Delegation and reassignment functions operate correctly.
- Complete approval histories and audit logs are maintained.
- Unauthorized users cannot approve workflow requests.
- Workflow initiators receive approval outcome notifications.

5.11 Analytics Dashboard

5.11.1 Module Description

The Analytics Dashboard provides real-time visualization and analysis of workflow execution, approval performance, process efficiency, operational trends, and organizational metrics.

The module collects execution data from workflow instances and presents meaningful insights through dashboards, reports, charts, and key performance indicators (KPIs). It enables managers, administrators, and process analysts to monitor business operations, identify bottlenecks, evaluate workflow effectiveness, and support data-driven decision-making.

The Analytics Dashboard also serves as the primary data source for the AI Process Optimization Engine.

5.11.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-211	The system shall provide a real-time analytics dashboard.	High
FR-212	The system shall display the total number of active workflows.	High
FR-213	The system shall display the total number of completed workflows.	High
FR-214	The system shall display the total number of pending workflows.	High
FR-215	The system shall display the total number of rejected workflows.	Medium
FR-216	The system shall calculate workflow completion rates.	High
FR-217	The system shall calculate average workflow completion time.	High
FR-218	The system shall calculate average approval processing time.	High
FR-219	The system shall identify workflow bottlenecks.	High
FR-220	The system shall identify the most delayed workflow activities.	Medium
FR-221	The system shall display department-wise workflow statistics.	Medium
FR-222	The system shall display user workload statistics.	Medium
FR-223	The system shall display approval performance metrics.	Medium
FR-224	The system shall display workflow execution trends over time.	Medium
FR-225	The system shall support filtering analytics by workflow type, department, user, and date range.	High
FR-226	The system shall allow users to search workflow analytics.	Medium
FR-227	The system shall generate workflow performance reports.	High
FR-228	The system shall allow authorized users to export reports in PDF and CSV formats.	Medium
FR-229	The system shall present analytics using charts, graphs, and summary cards.	Medium
FR-230	The system shall refresh dashboard data automatically at configurable intervals.	Medium

5.11.3 Dashboard Components

The Analytics Dashboard shall include the following components:

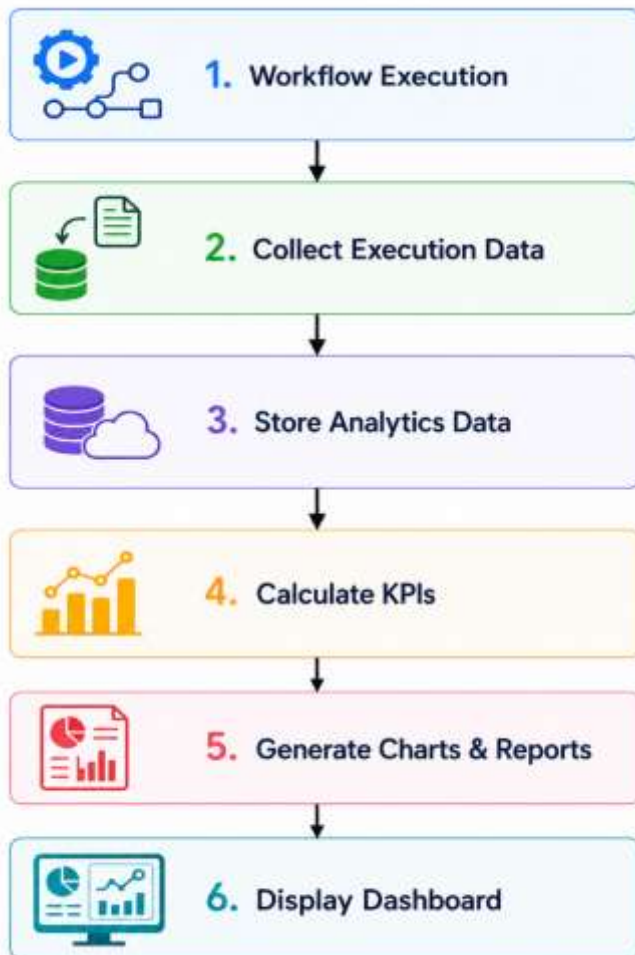
Component	Description
KPI Cards	Display high-level workflow metrics
Workflow Status Summary	Active, Pending, Completed, Rejected workflows
Approval Metrics	Approval counts and processing times
Workflow Trend Charts	Workflow execution over time
Bottleneck Analysis	Delayed workflow activities
Department Statistics	Workflow performance by department
User Performance	Task assignments and workload
Report Center	Exportable analytics reports

5.11.4 Key Performance Indicators (KPIs)

The dashboard shall calculate and display the following KPIs:

KPI	Description
Active Workflows	Number of workflows currently executing
Completed Workflows	Number of successfully completed workflows
Pending Workflows	Number of workflows awaiting action
Rejected Workflows	Number of rejected workflow instances
Average Completion Time	Average duration to complete workflows
Approval Time	Average approval processing time
Bottleneck Count	Number of identified workflow bottlenecks
Workflow Success Rate	Percentage of successfully completed workflows

5.11.5 Analytics Workflow



5.11.6 Business Rules

Rule ID	Business Rule
BR-ANA-001	Dashboard metrics shall reflect the most recent workflow execution data.
BR-ANA-002	Users shall only access analytics for organizations they belong to.
BR-ANA-003	Sensitive workflow information shall only be visible to authorized roles.
BR-ANA-004	KPI calculations shall use verified workflow execution data only.
BR-ANA-005	Generated reports shall include the report generation timestamp.

BR-ANA-006	Historical analytics data shall remain available according to organizational retention policies.
BR-ANA-007	Exported reports shall preserve data integrity and formatting.

5.11.7 Exception Handling

Exception	Expected System Response
No workflow data available	Display an empty-state dashboard with an informative message.
Invalid filter criteria	Reject the request and request valid filter values.
Report generation failure	Notify the user and log the incident.
Export failure	Preserve report data and allow export retry.
Database unavailable	Display a service unavailable message and record the error.
Insufficient permissions	Return HTTP 403 Forbidden.

5.11.8 Acceptance Criteria

The Analytics Dashboard shall be considered complete when:

- Workflow statistics are displayed accurately.
- KPI calculations are correct.
- Dashboard filters function properly.
- Workflow bottlenecks are identified.
- Reports can be generated and exported.
- Analytics update automatically according to configuration.
- Users only access authorized organizational analytics.
- Dashboard performance remains responsive under normal operating conditions.

5.12 Process Optimization Engine

5.12.1 Module Description

The Process Optimization Engine is an AI-powered component responsible for continuously analyzing workflow execution history, identifying operational inefficiencies, and recommending improvements to business processes.

The engine evaluates workflow performance, approval patterns, execution delays, resource utilization, and process bottlenecks to generate intelligent optimization recommendations.

Unlike traditional workflow platforms that only automate predefined workflows, the Process Optimization Engine continuously learns from execution data and assists organizations in improving efficiency, reducing operational costs, minimizing delays, and simplifying business processes.

This module utilizes Artificial Intelligence to support data-driven process improvement while allowing administrators and process analysts to review recommendations before implementation.

5.12.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-231	The system shall analyze completed workflow execution history.	High
FR-232	The system shall identify workflow bottlenecks automatically.	High
FR-233	The system shall identify activities with the longest execution times.	High
FR-234	The system shall identify approval stages causing significant delays.	High
FR-235	The system shall detect redundant workflow activities.	Medium
FR-236	The system shall identify duplicate approval steps.	Medium
FR-237	The system shall identify frequently rejected workflow paths.	Medium
FR-238	The system shall recommend workflow simplifications.	High
FR-239	The system shall recommend approval hierarchy improvements.	Medium
FR-240	The system shall estimate the expected time savings of recommended improvements.	Medium
FR-241	The system shall estimate workflow efficiency improvements.	Medium
FR-242	The system shall prioritize optimization recommendations based on business impact.	Medium
FR-243	The system shall generate AI-powered optimization reports.	High
FR-244	The system shall allow users to review optimization recommendations.	High
FR-245	The system shall allow administrators to accept or reject optimization recommendations.	Medium

FR-246	The system shall maintain optimization history for each workflow.	Medium
FR-247	The system shall compare workflow performance before and after optimization.	Medium
FR-248	The system shall support continuous optimization as workflow execution data increases.	High
FR-249	The system shall notify administrators when new optimization opportunities are identified.	Medium
FR-250	The system shall generate an overall workflow efficiency score.	Medium

5.12.3 Optimization Types

The Process Optimization Engine shall support the following optimization analyses:

Optimization Type	Description
Bottleneck Analysis	Identify slow workflow activities
Approval Optimization	Reduce unnecessary approval steps
Workflow Simplification	Remove redundant activities
Time Optimization	Reduce workflow completion time
Resource Optimization	Improve workload distribution
Process Efficiency Analysis	Evaluate overall workflow performance
AI Recommendations	Suggest improvements using AI reasoning

5.12.4 Optimization Inputs

The engine shall analyze:

Input	Description
Workflow Execution History	Historical workflow instances
Approval Records	Approval decisions and durations
Analytics Data	Workflow KPIs
User Activity Logs	Workflow interaction history

Organization Rules	Business constraints and policies
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5.12.5 Optimization Outputs

The engine shall generate:

Output	Description
Bottleneck Report	Identified process delays
Optimization Recommendations	AI-generated improvement suggestions
Estimated Time Savings	Predicted reduction in execution time
Efficiency Score	Workflow performance rating
Comparison Report	Before vs. after optimization analysis
Optimization History	Record of previous recommendations

5.12.6 Optimization Workflow



5.12.7 Business Rules

Rule ID	Business Rule
BR-OPT-001	Optimization recommendations shall be generated using completed workflow execution data only.
BR-OPT-002	AI recommendations shall not automatically modify deployed workflows.
BR-OPT-003	Administrators shall review recommendations before implementation.
BR-OPT-004	Workflow optimization history shall be retained for auditing purposes.
BR-OPT-005	Optimization reports shall include estimated business impact where applicable.
BR-OPT-006	The system shall preserve historical workflow versions after optimization.
BR-OPT-007	Recommendations shall respect organization-specific business rules and compliance requirements.

5.12.8 Exception Handling

Exception	Expected System Response
Insufficient workflow history	Notify the user that additional execution data is required.
AI optimization failure	Preserve existing workflow and log the error.
Missing analytics data	Generate a partial optimization report with appropriate warnings.
Database unavailable	Record the incident and notify the administrator.
Invalid workflow history	Exclude invalid records from analysis and continue processing.
Recommendation generation timeout	Cancel optimization and allow retry.

5.12.9 Acceptance Criteria

The Process Optimization Engine shall be considered complete when:

- Workflow bottlenecks are identified accurately.
- AI-generated recommendations are produced successfully.
- Optimization reports include estimated business impact.
- Administrators can review recommendations before implementation.

- Workflow efficiency scores are calculated correctly.
- Historical optimization records are maintained.
- The system supports continuous optimization as additional workflow data becomes available.
- Optimization recommendations never modify deployed workflows automatically.

5.13 Notification Module

5.13.1 Module Description

The Notification Module is responsible for delivering timely and relevant notifications to users regarding workflow events, approval requests, system activities, and organizational updates.

The module supports multiple notification channels, including in-application notifications and email notifications. It ensures that users are informed about workflow progress, pending tasks, approval requests, deadlines, reminders, escalations, and system events.

The Notification Module integrates with the Workflow Execution Engine, Human Approval Engine, and Administration Module to provide real-time communication throughout the workflow lifecycle.

5.13.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-251	The system shall generate notifications for workflow status changes.	High
FR-252	The system shall notify users when a new workflow task is assigned.	High
FR-253	The system shall notify approvers when an approval request is assigned.	High
FR-254	The system shall notify workflow initiators when approval decisions are made.	High
FR-255	The system shall notify users when workflows are completed successfully.	Medium
FR-256	The system shall notify users when workflows are rejected.	High
FR-257	The system shall notify administrators when workflow execution fails.	High
FR-258	The system shall send reminder notifications for pending workflow tasks.	Medium
FR-259	The system shall send reminder notifications for pending approvals.	Medium
FR-260	The system shall notify administrators when approval deadlines are exceeded.	Medium

FR-261	The system shall notify users when workflows are cancelled.	Medium
FR-262	The system shall support in-application notifications.	High
FR-263	The system shall support email notifications.	High
FR-264	The system shall allow users to mark notifications as read or unread.	Medium
FR-265	The system shall display notification history to users.	Medium
FR-266	The system shall allow users to delete notifications.	Low
FR-267	The system shall allow users to configure notification preferences.	Medium
FR-268	The system shall categorize notifications by type and priority.	Medium
FR-269	The system shall maintain delivery status for each notification.	Medium
FR-270	The system shall log notification delivery events for auditing purposes.	Medium

5.13.3 Notification Types

The system shall support the following notification types:

Notification Type	Description
Workflow Assigned	User receives a new workflow task
Approval Request	Approval action is required
Workflow Completed	Workflow finished successfully
Workflow Rejected	Workflow has been rejected
Workflow Cancelled	Workflow has been cancelled
Reminder	Pending task or approval reminder
Escalation	Approval deadline exceeded
System Alert	Important administrative or system event

5.13.4 Notification Channels

Channel	Description
In-App Notification	Displayed within the ProcessMind AI dashboard
Email Notification	Sent to the user's registered email address

5.13.5 Notification Workflow



5.13.6 Business Rules

Rule ID	Business Rule
BR-NOT-001	Notifications shall only be sent to authorized recipients.
BR-NOT-002	Notification delivery shall be recorded in the audit log.
BR-NOT-003	Users shall receive only notifications relevant to their assigned workflows and roles.
BR-NOT-004	Failed notification delivery attempts shall be logged for retry or administrator review.
BR-NOT-005	Users may configure their personal notification preferences where permitted.
BR-NOT-006	Notification history shall be retained according to organizational retention policies.
BR-NOT-007	Critical workflow notifications shall be prioritized over informational notifications.

5.13.7 Exception Handling

Exception	Expected System Response
Email delivery failure	Record the failure and retry according to retry policy.
Invalid recipient	Skip delivery and notify the administrator if necessary.
Notification service unavailable	Queue notifications until the service becomes available.
Database unavailable	Preserve pending notifications and retry later.
User has disabled notification type	Respect user preferences unless the notification is marked as mandatory.
Duplicate notification event	Prevent duplicate notification delivery.

5.13.8 Acceptance Criteria

The Notification Module shall be considered complete when:

- Users receive workflow-related notifications promptly.
- Approval requests trigger notifications correctly.
- Reminder and escalation notifications function as configured.
- In-app and email notifications are delivered successfully.
- Notification preferences are respected.
- Notification history is available to users.
- Delivery events are logged for auditing purposes.
- Failed notification deliveries are handled gracefully.

5.14 Administration Module

5.14.1 Module Description

The Administration Module provides centralized management and configuration capabilities for the ProcessMind AI platform. It enables authorized administrators to manage organizations, users, system settings, AI configurations, workflow configurations, audit logs, monitoring, backups, and platform maintenance.

The module ensures that the platform remains secure, configurable, maintainable, and scalable while supporting operational governance and administrative oversight.

Different administrative capabilities are provided based on administrator roles to maintain security and organizational separation.

5.14.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
FR-271	The system shall provide an administrative dashboard for authorized administrators.	High
FR-272	The system shall allow System Administrators to manage organizations.	High
FR-273	The system shall allow Organization Administrators to manage users within their organization.	High
FR-274	The system shall allow administrators to configure workflow settings.	High
FR-275	The system shall allow administrators to configure organization-specific approval policies.	Medium
FR-276	The system shall allow administrators to configure notification settings.	Medium
FR-277	The system shall allow administrators to configure AI processing settings.	Medium
FR-278	The system shall allow administrators to view workflow execution statistics.	Medium
FR-279	The system shall provide access to system audit logs.	High
FR-280	The system shall record all administrative actions in audit logs.	High
FR-281	The system shall allow administrators to search and filter audit logs.	Medium
FR-282	The system shall display overall platform health status.	Medium
FR-283	The system shall display AI service availability status.	Medium
FR-284	The system shall display database connectivity status.	Medium
FR-285	The system shall allow administrators to configure system maintenance settings.	Low
FR-286	The system shall allow administrators to archive completed workflows according to retention policies.	Medium
FR-287	The system shall support manual backup initiation by authorized administrators.	Medium
FR-288	The system shall support restoration from previously created backups.	Medium
FR-289	The system shall maintain system configuration version history.	Low
FR-290	The system shall support future integration of subscription and license management.	Low

5.14.3 Administrative Functions

The Administration Module shall provide the following capabilities:

Function	Description
Organization Management	Create, update, archive, and manage organizations
User Administration	Manage users, roles, and permissions
Workflow Administration	Configure workflow execution settings
AI Configuration	Configure AI-related settings and parameters
Notification Configuration	Configure notification channels and preferences
Audit Management	View and search system audit logs
System Monitoring	Monitor platform health and service availability
Backup & Recovery	Perform backup and restore operations
Configuration Management	Manage system-wide settings
Maintenance	Configure maintenance-related operations

5.14.4 Administrative Roles

Role	Responsibilities
System Administrator	Full access to all organizations, system settings, AI configuration, monitoring, backups, and platform maintenance.
Organization Administrator	Manage users, workflows, departments, and settings within their own organization only.

5.14.5 Administration Workflow



5.14.6 Business Rules

Rule ID	Business Rule
BR-ADM-001	Only authorized administrators shall access administrative functions.
BR-ADM-002	Every administrative action shall be recorded in the audit log.
BR-ADM-003	Organization Administrators shall manage only their own organization.
BR-ADM-004	System Administrators shall have access to all organizations and platform resources.
BR-ADM-005	Critical system configuration changes shall require confirmation before being applied.
BR-ADM-006	Backup and recovery operations shall be available only to authorized administrators.
BR-ADM-007	Administrative actions shall preserve data integrity and system consistency.

5.14.7 Exception Handling

Exception	Expected System Response
Unauthorized administrative access	Reject the request and return HTTP 403 Forbidden.
Invalid system configuration	Reject the configuration and display validation errors.
Backup failure	Notify the administrator and log the failure.
Restore failure	Preserve existing data and notify the administrator.
Audit log retrieval failure	Display an error message and record the incident.
AI service unavailable	Display service status and disable AI-dependent administrative functions where necessary.
Database unavailable	Prevent configuration changes and notify the administrator.

5.14.8 Acceptance Criteria

The Administration Module shall be considered complete when:

- Authorized administrators can successfully access the administration dashboard.
- Organization and user administration functions operate correctly.
- Workflow and AI settings can be configured securely.
- Audit logs accurately record all administrative activities.
- System health information is available.
- Backup and restore operations function correctly.
- Administrative permissions are enforced according to user roles.
- Unauthorized administrative access is prevented.

Chapter 6: AI Functional Requirements

6.1 Overview

Artificial Intelligence (AI) is the core enabling technology of ProcessMind AI. Unlike traditional workflow automation platforms that require users to manually design and configure workflows, ProcessMind AI leverages AI to understand Standard Operating Procedures (SOPs), reason about business logic, generate workflows, validate process structures, simulate execution scenarios, and recommend process optimizations.

This chapter specifies the functional requirements for all AI-related capabilities within the platform. These requirements define how AI services shall interact with users, workflow components, and organizational data while ensuring reliability, explainability, security, and human oversight.

The AI layer is designed as an independent service abstraction, allowing the platform to support multiple AI providers in the future without requiring significant architectural modifications.

The primary objectives of the AI layer are to:

- Understand natural-language SOPs.
- Extract structured business knowledge.
- Generate executable workflow definitions.
- Validate business process logic.
- Simulate workflow execution.
- Analyze workflow performance.
- Recommend workflow optimizations.
- Assist users through intelligent reasoning and decision support.

The AI layer is intended to augment human decision-making rather than replace it. All AI-generated outputs that affect workflow definitions or business processes shall be subject to user review and approval before deployment.

6.2 AI Architecture

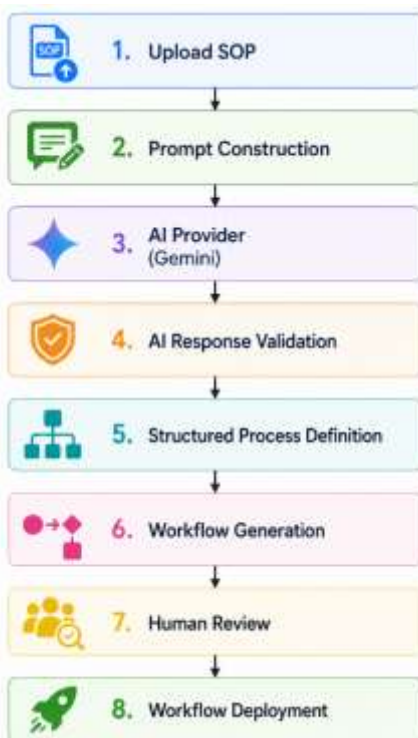
The AI subsystem shall consist of the following logical components:

- AI Service Layer
- Prompt Management
- SOP Understanding Engine
- Workflow Reasoning Engine

- Validation Engine
- Simulation Engine
- Optimization Engine
- AI Response Validator
- AI Provider Interface

These components work together to provide intelligent assistance throughout the workflow lifecycle.

6.3 AI Processing Workflow



6.4 AI Functional Modules

The AI subsystem consists of the following modules:

1. AI Provider Management
2. Prompt Management
3. SOP Understanding
4. AI Workflow Reasoning
5. AI Response Validation
6. AI Confidence Evaluation
7. Human-in-the-Loop Review
8. AI Monitoring and Logging

9. AI Failure Recovery

10. AI Learning and Improvement (Future)

The following sections specify the functional requirements for each module.

6.2 AI Provider Management

6.2.1 Module Description

The AI Provider Management module is responsible for managing communication between ProcessMind AI and external Artificial Intelligence service providers.

The module abstracts AI provider implementations behind a common service interface, enabling the platform to support multiple Large Language Model (LLM) providers without modifying business logic.

The initial implementation shall integrate with the Google Gemini API. However, the architecture shall support future integration with additional AI providers such as OpenAI, Anthropic Claude, Microsoft Azure OpenAI, self-hosted models, and other compatible AI services.

The module is also responsible for managing AI requests, authentication, retries, timeout handling, provider selection, usage monitoring, and error recovery.

6.2.2 Functional Requirements

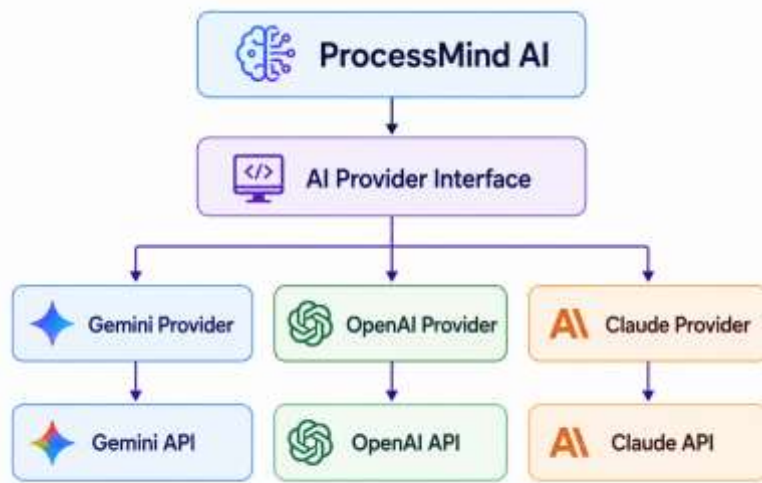
Requirement ID	Functional Requirement	Priority
AI-FR-001	The system shall communicate with AI providers through an abstract AI service interface.	High
AI-FR-002	The system shall support Google Gemini as the default AI provider.	High
AI-FR-003	The system architecture shall allow future integration of additional AI providers without modifying business logic.	High
AI-FR-004	The system shall securely store AI provider API credentials using environment variables or a secure secret management mechanism.	High
AI-FR-005	The system shall authenticate requests before communicating with the AI provider.	High
AI-FR-006	The system shall validate AI requests before submission.	Medium
AI-FR-007	The system shall support configurable request timeout values.	Medium
AI-FR-008	The system shall automatically retry failed AI requests according to configurable retry policies.	Medium
AI-FR-009	The system shall detect AI service unavailability.	High

AI-FR-010	The system shall gracefully handle AI provider failures without causing system crashes.	High
AI-FR-011	The system shall record AI request execution time.	Medium
AI-FR-012	The system shall record AI response status for monitoring purposes.	Medium
AI-FR-013	The system shall monitor AI usage statistics.	Medium
AI-FR-014	The system shall maintain AI request logs for auditing purposes.	Medium
AI-FR-015	The system shall support configurable AI model selection.	Medium
AI-FR-016	The system shall prevent unauthorized access to AI provider credentials.	High
AI-FR-017	The system shall allow administrators to configure AI provider settings.	Medium
AI-FR-018	The system shall notify administrators when AI provider connectivity fails.	Medium
AI-FR-019	The system shall support future AI provider failover mechanisms.	Low
AI-FR-020	The system shall maintain compatibility with future versions of supported AI provider APIs.	Medium

6.2.3 Supported AI Providers

AI Provider	Support Status
Google Gemini	Supported (Default)
OpenAI GPT	Future
Anthropic Claude	Future
Azure OpenAI	Future
Ollama (Local Models)	Future
Custom Enterprise Models	Future

6.2.4 AI Provider Architecture



6.2.5 Business Rules

Rule ID	Business Rule
BR-AI-001	Business logic shall not communicate directly with any AI provider.
BR-AI-002	AI providers shall be accessed only through the AI Provider Interface.
BR-AI-003	AI credentials shall never be stored in source code.
BR-AI-004	AI request failures shall not cause data corruption.
BR-AI-005	All AI requests shall be logged for monitoring and troubleshooting.
BR-AI-006	AI provider configuration changes shall be restricted to authorized administrators.

6.2.6 Exception Handling

Exception	Expected System Response
Invalid API credentials	Reject the request, notify the administrator, and log the incident.
AI provider unavailable	Return an appropriate error message and preserve pending requests where applicable.
Request timeout	Retry according to configured retry policy before reporting failure.
Invalid AI response	Reject the response and request regeneration if applicable.
Network failure	Retry the request and log connectivity issues.

Rate limit exceeded	Delay or queue requests according to configured throttling policies.
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6.2.7 Acceptance Criteria

The AI Provider Management module shall be considered complete when:

- AI requests are routed through the AI Provider Interface.
- Google Gemini integration functions correctly.
- AI credentials are securely managed.
- AI request failures are handled gracefully.
- AI usage statistics are recorded.
- AI request logs are maintained.
- The architecture supports future integration of additional AI providers without requiring changes to business logic.

6.3 Prompt Management

6.3.1 Module Description

The Prompt Management module is responsible for constructing, managing, validating, versioning, and optimizing prompts used for communication with Artificial Intelligence providers.

Instead of embedding prompts directly within application source code, the system shall maintain prompts as configurable and reusable resources. This enables administrators and developers to improve AI behavior without modifying application logic.

The module supports prompt templates, variable substitution, prompt versioning, prompt validation, prompt testing, and prompt execution history.

Prompt Management ensures consistency, maintainability, and reliability across all AI-powered features within ProcessMind AI.

6.3.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-021	The system shall maintain prompt templates for all AI-powered operations.	High
AI-FR-022	The system shall support dynamic variable substitution within prompts.	High

AI-FR-023	The system shall construct prompts using SOP content and workflow context.	High
AI-FR-024	The system shall validate prompt completeness before submission to the AI provider.	High
AI-FR-025	The system shall prevent submission of empty or invalid prompts.	High
AI-FR-026	The system shall maintain version history for prompt templates.	Medium
AI-FR-027	The system shall allow administrators to update prompt templates.	Medium
AI-FR-028	The system shall support multiple prompt templates for different AI tasks.	High
AI-FR-029	The system shall allow prompt selection based on the requested AI operation.	High
AI-FR-030	The system shall sanitize user input before constructing prompts.	High
AI-FR-031	The system shall include organization-specific context when generating prompts where applicable.	Medium
AI-FR-032	The system shall support configurable AI generation parameters such as temperature and token limits.	Medium
AI-FR-033	The system shall record prompt execution history for auditing and troubleshooting.	Medium
AI-FR-034	The system shall allow administrators to test prompt templates before production use.	Medium
AI-FR-035	The system shall support prompt preview before AI execution.	Medium
AI-FR-036	The system shall prevent unauthorized modification of production prompt templates.	High
AI-FR-037	The system shall maintain prompt metadata including version, creator, creation date, and last modification date.	Medium
AI-FR-038	The system shall support prompt rollback to a previous version.	Low
AI-FR-039	The system shall log prompt execution failures.	Medium
AI-FR-040	The system shall reuse standardized prompt templates across multiple AI modules.	High

6.3.3 Prompt Categories

The platform shall maintain separate prompt templates for different AI operations.

Prompt Category	Purpose
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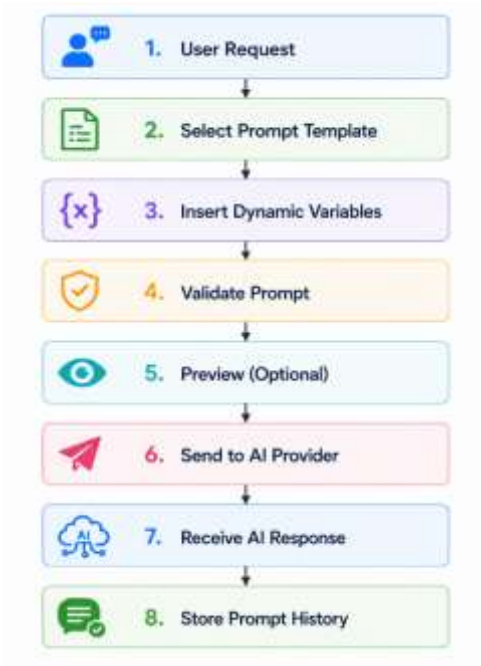
SOP Understanding	Extract workflow logic from SOP documents
Workflow Generation	Generate executable workflow definitions
Workflow Validation	Detect workflow issues and inconsistencies
Process Simulation	Simulate workflow execution scenarios
Process Optimization	Generate optimization recommendations
AI Assistant	Respond to user questions and guidance
Report Generation	Generate AI-powered summaries and reports

6.3.4 Prompt Structure

Each prompt template shall contain:

Component	Description
System Instructions	Define AI behavior and response format
Context	Organization and workflow context
User Input	SOP or business process content
Output Format	Expected structured response format
Validation Rules	Constraints applied to AI output

6.3.5 Prompt Processing Workflow



6.3.6 Business Rules

Rule ID	Business Rule
BR-PROMPT-001	All AI interactions shall use predefined prompt templates.
BR-PROMPT-002	Prompt templates shall be version-controlled.
BR-PROMPT-003	Prompt templates shall only be modified by authorized administrators.
BR-PROMPT-004	User input shall be sanitized before inclusion in prompts.
BR-PROMPT-005	Prompt execution history shall be retained for auditing and troubleshooting.
BR-PROMPT-006	Production prompt templates shall be tested before activation.
BR-PROMPT-007	Prompt rollback shall preserve historical versions.

6.3.7 Exception Handling

Exception	Expected System Response
Missing prompt template	Prevent AI execution and notify the administrator.
Invalid prompt variables	Reject prompt construction and display validation errors.
Unauthorized prompt modification	Return HTTP 403 Forbidden.
Prompt validation failure	Prevent submission to the AI provider.
Prompt execution failure	Log the incident and allow retry.
Prompt version conflict	Preserve the active version and require administrator resolution.

6.3.8 Acceptance Criteria

The Prompt Management module shall be considered complete when:

- Prompt templates are managed independently of application code.
- Dynamic variables are correctly substituted into prompts.
- Prompt templates are version-controlled.
- Prompt validation prevents invalid AI requests.
- Prompt execution history is recorded.
- Administrators can safely manage and test prompt templates.
- Prompt rollback functions correctly.
- Prompt templates are reusable across multiple AI-powered modules.

6.4 SOP Understanding Engine

6.4.1 Module Description

The SOP Understanding Engine is the primary Artificial Intelligence component responsible for interpreting Standard Operating Procedures (SOPs) written in natural language and transforming them into structured business knowledge.

The engine analyzes uploaded SOP documents to identify business activities, decision points, conditions, organizational roles, approval requirements, dependencies, inputs, outputs, exceptions, and business rules.

Rather than simply extracting text, the engine performs semantic reasoning to understand the intent and operational logic of the process. The resulting structured representation serves as the foundation for workflow generation, validation, simulation, execution, and optimization.

The module shall support future enhancements such as multilingual SOP analysis, industry-specific reasoning models, and organization-specific knowledge bases.

6.4.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-041	The system shall analyze SOP documents written in natural language.	High
AI-FR-042	The system shall extract business activities from the SOP.	High
AI-FR-043	The system shall identify workflow start events.	High
AI-FR-044	The system shall identify workflow end events.	High
AI-FR-045	The system shall identify decision points and conditional branches.	High
AI-FR-046	The system shall extract business conditions from the SOP.	High
AI-FR-047	The system shall identify approval requirements.	High
AI-FR-048	The system shall identify organizational roles involved in the process.	High
AI-FR-049	The system shall identify workflow dependencies between activities.	High
AI-FR-050	The system shall identify workflow inputs and outputs.	Medium
AI-FR-051	The system shall extract business rules described in the SOP.	High
AI-FR-052	The system shall identify exception handling procedures described in the SOP.	Medium
AI-FR-053	The system shall identify required documents and supporting artifacts referenced in the SOP.	Medium
AI-FR-054	The system shall classify workflow activities according to predefined activity types.	Medium

AI-FR-055	The system shall detect ambiguous or incomplete SOP instructions.	High
AI-FR-056	The system shall identify missing process information where possible.	Medium
AI-FR-057	The system shall generate a structured process representation in JSON format.	High
AI-FR-058	The system shall assign confidence scores to extracted entities and relationships.	High
AI-FR-059	The system shall preserve traceability between extracted information and the original SOP text.	High
AI-FR-060	The system shall allow users to review and edit AI-generated process representations before workflow generation.	High

6.4.3 AI Knowledge Extraction

The SOP Understanding Engine shall extract the following knowledge elements:

Knowledge Element	Description
Workflow Activities	Individual business tasks
Decision Points	Conditional workflow branches
Business Rules	Organizational policies and constraints
Organizational Roles	Users or departments responsible for activities
Approval Requirements	Required approval stages
Dependencies	Relationships between workflow activities
Inputs	Required information for execution
Outputs	Results produced by workflow activities
Exceptions	Error-handling or alternative execution paths
Documents	Forms, invoices, contracts, or other referenced documents

6.4.4 AI Processing Workflow



6.4.5 Structured Output Model

The generated process representation shall include:

Output	Description
Workflow Name	Name of the identified process
Activities	Ordered list of workflow tasks
Conditions	Decision logic
Roles	Responsible users or departments
Approvals	Required approval stages
Business Rules	Extracted organizational rules
Dependencies	Relationships among activities
Exceptions	Alternative execution paths
Confidence Scores	AI confidence for extracted information

6.4.6 Business Rules

Rule ID	Business Rule
BR-SOPAI-001	AI-generated process models shall preserve the meaning of the original SOP.
BR-SOPAI-002	Confidence scores shall be generated for all extracted entities.
BR-SOPAI-003	Ambiguous SOP sections shall be highlighted for user review.
BR-SOPAI-004	Users shall review AI-generated process models before workflow generation.
BR-SOPAI-005	Original SOP documents shall remain unchanged throughout AI processing.
BR-SOPAI-006	Traceability shall be maintained between extracted elements and source text.
BR-SOPAI-007	AI shall not fabricate business rules that are unsupported by the SOP content.

6.4.7 Exception Handling

Exception	Expected System Response
Unsupported language	Notify the user that the language is not currently supported.
Poor document quality	Highlight unreadable sections and request a higher-quality document.
Ambiguous instructions	Flag the affected sections for manual review.
Missing workflow information	Generate a partial process model and report missing elements.
AI interpretation failure	Preserve uploaded data, log the incident, and allow reprocessing.
Invalid AI response	Reject the generated model and request regeneration.

6.4.8 Acceptance Criteria

The SOP Understanding Engine shall be considered complete when:

- SOPs are successfully interpreted into structured process models.
- Workflow activities, roles, approvals, conditions, and business rules are accurately extracted.
- Confidence scores are assigned to extracted information.
- Ambiguous or incomplete instructions are identified.
- Users can review and modify AI-generated outputs.
- Traceability between the original SOP and extracted knowledge is preserved.
- Structured outputs are suitable for downstream workflow generation and validation.

6.5 AI Workflow Reasoning Engine

6.5.1 Module Description

The AI Workflow Reasoning Engine is responsible for transforming structured business knowledge into logically consistent and executable workflow models through Artificial Intelligence reasoning.

Unlike rule-based workflow generators, this module performs semantic reasoning to determine the optimal workflow structure by analyzing business objectives, organizational policies, dependencies, decision logic, approval chains, exceptions, and execution constraints.

The engine acts as an intelligent business analyst by interpreting extracted knowledge, identifying implicit workflow relationships, resolving ambiguities where possible, and constructing workflow models that accurately represent organizational processes.

The generated workflow serves as the primary input for workflow validation, simulation, and execution.

6.5.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-061	The system shall reason over structured process knowledge to generate executable workflow logic.	High
AI-FR-062	The system shall determine the optimal execution sequence for workflow activities.	High
AI-FR-063	The system shall infer missing workflow transitions where sufficient evidence exists.	High
AI-FR-064	The system shall construct workflow decision trees based on identified business conditions.	High
AI-FR-065	The system shall identify opportunities for parallel workflow execution.	Medium
AI-FR-066	The system shall determine approval routing based on extracted organizational roles and business rules.	High
AI-FR-067	The system shall infer workflow dependencies between activities.	High
AI-FR-068	The system shall identify alternative execution paths for exception scenarios.	Medium
AI-FR-069	The system shall reason about workflow termination conditions.	High
AI-FR-070	The system shall ensure that generated workflows satisfy identified business rules.	High
AI-FR-071	The system shall identify logical inconsistencies before workflow generation.	High

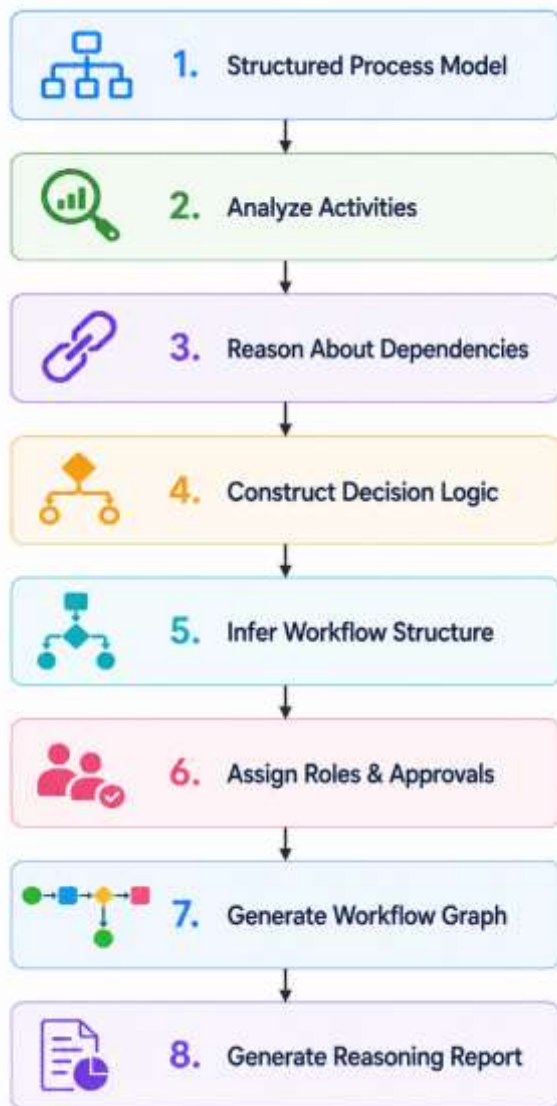
AI-FR-072	The system shall recommend clarification when business rules are ambiguous.	High
AI-FR-073	The system shall generate workflow metadata describing reasoning decisions.	Medium
AI-FR-074	The system shall provide explanations for AI-generated workflow decisions.	High
AI-FR-075	The system shall produce machine-readable workflow representations suitable for execution.	High
AI-FR-076	The system shall preserve reasoning traceability between workflow elements and source SOP content.	High
AI-FR-077	The system shall assign confidence scores to generated workflow structures.	High
AI-FR-078	The system shall allow users to review AI reasoning before workflow deployment.	High
AI-FR-079	The system shall support organization-specific reasoning rules in future releases.	Medium
AI-FR-080	The system shall generate reasoning logs for auditing and debugging purposes.	Medium

6.5.3 AI Reasoning Capabilities

The AI Workflow Reasoning Engine shall perform the following reasoning tasks:

Reasoning Capability	Description
Process Sequencing	Determine the logical order of activities
Dependency Reasoning	Identify activity dependencies
Decision Reasoning	Build conditional execution paths
Approval Reasoning	Determine approval hierarchy
Exception Reasoning	Handle alternative execution paths
Compliance Reasoning	Ensure workflows satisfy business policies
Optimization Awareness	Generate workflows suitable for later optimization

6.5.4 AI Reasoning Workflow



6.5.5 AI Outputs

The engine shall generate the following outputs:

Output	Description
Workflow Graph	AI-generated executable workflow structure
Decision Tree	Conditional workflow logic
Approval Chain	Inferred approval routing
Execution Sequence	Ordered workflow activities
Reasoning Report	Explanation of AI decisions
Confidence Scores	Confidence assigned to workflow components

Traceability Map	Mapping between workflow elements and SOP content
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6.5.6 Business Rules

Rule ID	Business Rule
BR-REASON-001	AI-generated workflows shall remain consistent with the original SOP.
BR-REASON-002	AI shall provide explanations for significant reasoning decisions.
BR-REASON-003	Workflow reasoning shall preserve identified business rules and organizational constraints.
BR-REASON-004	Users shall review AI-generated workflows before deployment.
BR-REASON-005	Confidence scores shall accompany generated workflow structures.
BR-REASON-006	AI shall highlight assumptions made during reasoning.
BR-REASON-007	Traceability between workflow elements and SOP content shall be maintained.

6.5.7 Exception Handling

Exception	Expected System Response
Conflicting business rules	Highlight conflicts and request user review.
Ambiguous workflow sequence	Generate multiple candidate sequences or request clarification.
Missing dependency information	Produce a partial workflow and identify missing information.
AI reasoning failure	Preserve extracted process model and allow regeneration.
Low confidence result	Flag the workflow for mandatory manual review.
Invalid workflow graph	Reject the generated workflow and log the incident.

6.5.8 Acceptance Criteria

The AI Workflow Reasoning Engine shall be considered complete when:

- AI successfully constructs executable workflow logic.
- Workflow sequencing reflects business intent.
- Decision trees and approval paths are generated accurately.
- Reasoning explanations accompany AI-generated workflows.
- Confidence scores are assigned to workflow structures.
- Users can review AI-generated reasoning before deployment.

- Traceability between workflow elements and source SOP content is preserved.
- Generated workflows are compatible with validation, simulation, and execution modules.

6.6 AI Response Validation Engine

6.6.1 Module Description

The AI Response Validation Engine is responsible for validating all responses generated by Artificial Intelligence before they are used by ProcessMind AI.

The engine verifies that AI-generated outputs are syntactically correct, structurally complete, logically consistent, semantically valid, and compliant with system rules.

The validation process prevents malformed, incomplete, hallucinated, or unsafe AI responses from affecting workflow generation, workflow execution, or business operations.

The engine acts as the quality assurance layer between the AI Provider and the remaining system modules.

6.6.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-081	The system shall validate every AI response before processing it.	High
AI-FR-082	The system shall verify that AI responses conform to the expected JSON schema.	High
AI-FR-083	The system shall detect incomplete AI responses.	High
AI-FR-084	The system shall detect malformed AI responses.	High
AI-FR-085	The system shall validate required workflow fields.	High
AI-FR-086	The system shall validate workflow relationships and dependencies.	High
AI-FR-087	The system shall validate approval structures generated by AI.	High
AI-FR-088	The system shall validate extracted business rules for completeness.	Medium
AI-FR-089	The system shall detect contradictory workflow logic.	High
AI-FR-090	The system shall detect duplicate workflow activities generated by AI.	Medium
AI-FR-091	The system shall detect unsupported workflow constructs.	Medium
AI-FR-092	The system shall validate workflow graph connectivity.	High
AI-FR-093	The system shall reject AI responses that fail validation.	High
AI-FR-094	The system shall request AI regeneration when validation fails.	Medium

AI-FR-095	The system shall allow manual review of rejected AI responses.	Medium
AI-FR-096	The system shall log all validation failures.	Medium
AI-FR-097	The system shall assign a validation status to every AI response.	Medium
AI-FR-098	The system shall support configurable validation rules.	Low
AI-FR-099	The system shall generate detailed validation reports for AI outputs.	Medium
AI-FR-100	The system shall maintain validation history for audit and troubleshooting purposes.	Medium

6.6.3 Validation Types

The AI Response Validation Engine shall perform the following validations:

Validation Type	Description
Schema Validation	Verify JSON structure and required fields
Syntax Validation	Detect malformed responses
Logical Validation	Detect inconsistent workflow logic
Semantic Validation	Ensure AI output matches business intent
Dependency Validation	Verify workflow relationships
Approval Validation	Validate approval chains
Safety Validation	Detect unsafe or prohibited content
Completeness Validation	Ensure all required information is present

6.6.4 AI Validation Workflow



6.6.5 Validation Status

Each AI response shall receive one of the following validation statuses:

Status	Description
Valid	Response passed all validation checks
Valid with Warnings	Minor issues detected but response is acceptable
Review Required	Manual review is required before use
Invalid	Response failed validation and cannot be used
Regeneration Required	AI response must be regenerated

6.6.6 Business Rules

Rule ID	Business Rule
BR-AIVAL-001	Every AI response shall undergo validation before use.
BR-AIVAL-002	Invalid AI responses shall never be used for workflow generation or execution.
BR-AIVAL-003	AI responses failing validation shall be logged for analysis.
BR-AIVAL-004	Critical validation failures shall trigger AI response regeneration.

BR-AIVAL-005	Manual review shall be required for low-confidence or partially valid responses.
BR-AIVAL-006	Validation reports shall be retained for auditing purposes.
BR-AIVAL-007	Validation rules shall be configurable by authorized administrators where supported.

6.6.7 Exception Handling

Exception	Expected System Response
Invalid JSON response	Reject the response and request regeneration.
Missing required fields	Mark the response as invalid and notify the user.
Contradictory workflow logic	Prevent processing and require manual review.
Unsupported output format	Reject the response and log the incident.
Validation engine failure	Stop downstream processing and notify the administrator.
AI regeneration failure	Preserve previous valid data and request user intervention.

6.6.8 Acceptance Criteria

The AI Response Validation Engine shall be considered complete when:

- Every AI response is validated before use.
- Invalid responses are rejected automatically.
- Validation reports accurately identify detected issues.
- AI regeneration is triggered when required.
- Validation history is maintained for auditing.
- Manual review is supported for uncertain AI outputs.
- Only validated AI outputs proceed to downstream workflow generation and execution.

6.7 AI Confidence Evaluation

6.7.1 Module Description

The AI Confidence Evaluation module is responsible for measuring the reliability of AI-generated outputs and determining the level of human intervention required before the outputs are accepted by the system.

For every AI-generated response, the module calculates confidence scores based on multiple evaluation criteria, including response completeness, consistency, validation results, reasoning quality, semantic alignment with the original SOP, and historical AI performance.

The calculated confidence score enables ProcessMind AI to distinguish between highly reliable AI outputs that may proceed with minimal intervention and lower-confidence outputs that require additional review or clarification.

This module improves transparency, trustworthiness, and governance while supporting responsible AI practices.

6.7.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-101	The system shall calculate a confidence score for every AI-generated response.	High
AI-FR-102	The system shall calculate confidence scores using multiple evaluation criteria.	High
AI-FR-103	The system shall evaluate semantic similarity between the AI response and the original SOP.	High
AI-FR-104	The system shall incorporate AI Response Validation results into the confidence score.	High
AI-FR-105	The system shall reduce confidence scores when inconsistencies are detected.	High
AI-FR-106	The system shall increase confidence scores when all validation checks pass successfully.	Medium
AI-FR-107	The system shall assign confidence scores to individual workflow components.	Medium
AI-FR-108	The system shall assign an overall confidence score to the generated workflow.	High
AI-FR-109	The system shall categorize confidence levels (High, Medium, Low).	High
AI-FR-110	The system shall require manual review for AI outputs below a configurable confidence threshold.	High
AI-FR-111	The system shall display confidence scores to authorized users.	High
AI-FR-112	The system shall record confidence evaluation history.	Medium
AI-FR-113	The system shall support configurable confidence thresholds.	Medium

AI-FR-114	The system shall explain the primary factors affecting confidence scores.	High
AI-FR-115	The system shall notify users when low-confidence results require review.	Medium
AI-FR-116	The system shall support confidence recalculation after manual edits.	Medium
AI-FR-117	The system shall preserve confidence evaluation reports for auditing.	Medium
AI-FR-118	The system shall allow administrators to configure confidence evaluation policies.	Low
AI-FR-119	The system shall maintain traceability between confidence scores and evaluated AI outputs.	Medium
AI-FR-120	The system shall prevent automatic deployment of workflows that fail confidence policy requirements.	High

6.7.3 Confidence Levels

The system shall classify AI confidence into the following categories:

Confidence Level	Score Range	Description
High	90–100%	AI output is highly reliable and requires minimal review.
Medium	70–89%	AI output is acceptable but should be reviewed before deployment.
Low	Below 70%	AI output requires mandatory manual review and correction.

6.7.4 Confidence Evaluation Criteria

The confidence score shall consider the following factors:

Evaluation Criterion	Description
Response Completeness	Presence of all required workflow information
Structural Validity	Compliance with expected workflow schema
Logical Consistency	Absence of conflicting workflow logic
Semantic Accuracy	Alignment with the original SOP
Business Rule Compliance	Compliance with extracted business rules
Validation Results	Outcome of AI Response Validation
Historical Performance	Previous AI accuracy for similar workflows

6.7.5 Confidence Evaluation Workflow



6.7.6 Business Rules

Rule ID	Business Rule
BR-CONF-001	Every AI-generated response shall receive a confidence score.
BR-CONF-002	Low-confidence responses shall require manual review before workflow generation.
BR-CONF-003	Confidence scores shall be recalculated whenever AI outputs are modified.
BR-CONF-004	Confidence evaluation reports shall be retained for auditing purposes.
BR-CONF-005	Administrators may configure confidence thresholds within defined system limits.
BR-CONF-006	Confidence evaluation shall consider both AI-generated content and validation results.

BR-CONF-007	Confidence scores shall not replace human review where organizational policies require manual approval.
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6.7.7 Exception Handling

Exception	Expected System Response
Confidence calculation failure	Prevent automatic workflow progression and notify the administrator.
Missing validation results	Delay confidence evaluation until validation is completed.
Invalid confidence score	Reject the evaluation and log the incident.
Threshold configuration error	Apply system default thresholds and notify the administrator.
Evaluation engine failure	Require manual review of AI outputs before further processing.

6.7.8 Acceptance Criteria

The AI Confidence Evaluation module shall be considered complete when:

- Every AI-generated response receives a confidence score.
- Confidence scores accurately reflect validation outcomes.
- Low-confidence outputs are automatically flagged for manual review.
- Confidence reports explain key factors influencing the score.
- Confidence thresholds are configurable by administrators.
- Confidence evaluation history is maintained for auditing.
- Workflow deployment complies with confidence evaluation policies.

6.8 Human-in-the-Loop (HITL) Review

6.8.1 Module Description

The Human-in-the-Loop (HITL) Review module ensures that human users remain responsible for reviewing, validating, correcting, and approving AI-generated outputs before they are used in business operations.

Rather than replacing human expertise, ProcessMind AI augments human decision-making by generating intelligent recommendations while allowing authorized users to review and modify AI outputs.

The module supports collaborative review, manual corrections, approval workflows, feedback collection, and traceability of all user interventions. It provides transparency into AI reasoning and ensures accountability for AI-assisted business decisions.

6.8.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-121	The system shall require human review for AI outputs that fall below the configured confidence threshold.	High
AI-FR-122	The system shall allow authorized users to review AI-generated workflow definitions before deployment.	High
AI-FR-123	The system shall display AI reasoning explanations during review.	High
AI-FR-124	The system shall display confidence scores for AI-generated workflow elements.	High
AI-FR-125	The system shall allow reviewers to edit AI-generated workflow components.	High
AI-FR-126	The system shall allow reviewers to modify business rules generated by AI.	Medium
AI-FR-127	The system shall allow reviewers to modify approval chains before deployment.	Medium
AI-FR-128	The system shall allow reviewers to accept AI recommendations without modification.	Medium
AI-FR-129	The system shall allow reviewers to reject AI-generated outputs.	High
AI-FR-130	The system shall require reviewers to provide a reason when rejecting AI-generated outputs.	Medium
AI-FR-131	The system shall record all manual modifications made to AI-generated content.	High
AI-FR-132	The system shall maintain version history for reviewed AI outputs.	High
AI-FR-133	The system shall recalculate confidence scores after manual modifications where applicable.	Medium
AI-FR-134	The system shall collect reviewer feedback regarding AI output quality.	Medium
AI-FR-135	The system shall maintain complete review history for auditing purposes.	High
AI-FR-136	The system shall identify the reviewer responsible for approving AI-generated outputs.	High
AI-FR-137	The system shall support multiple review cycles before workflow deployment.	Medium
AI-FR-138	The system shall prevent deployment of AI-generated workflows until mandatory reviews are completed.	High

AI-FR-139	The system shall notify reviewers when AI-generated outputs require review.	Medium
AI-FR-140	The system shall generate review reports summarizing reviewer decisions and modifications.	Medium

6.8.3 Review Actions

Authorized reviewers shall be able to perform the following actions:

Action	Description
Accept	Approve AI-generated output without changes
Edit	Modify AI-generated workflow elements
Reject	Reject AI-generated output
Request Regeneration	Request the AI to generate a new result
Add Comments	Provide reviewer comments and feedback
Save Draft	Save review progress before completion

6.8.4 Review Workflow



6.8.5 Review Information

The review interface shall display:

Information	Description
Original SOP	Uploaded source document
AI Output	Generated workflow or analysis
Confidence Scores	AI confidence for each component
AI Reasoning	Explanation of AI decisions
Validation Results	AI Response Validation findings
Reviewer Comments	Comments from reviewers
Version History	Previous review versions

6.8.6 Business Rules

Rule ID	Business Rule
BR-HITL-001	AI-generated workflows requiring review shall not be deployed until approved by an authorized reviewer.
BR-HITL-002	Manual modifications shall be recorded with user identity and timestamp.
BR-HITL-003	Every rejection shall include a reviewer-provided reason.
BR-HITL-004	Review history shall be retained for auditing purposes.
BR-HITL-005	Reviewer permissions shall be enforced using Role-Based Access Control (RBAC).
BR-HITL-006	AI reasoning explanations shall be available during the review process.
BR-HITL-007	Manual edits shall not overwrite previous versions of AI-generated outputs.

6.8.7 Exception Handling

Exception	Expected System Response
Unauthorized review attempt	Reject the request and return HTTP 403 Forbidden.
Missing reviewer comments on rejection	Prevent submission until a reason is provided.
Concurrent review conflict	Notify reviewers and require conflict resolution.
Version conflict	Preserve both versions and request manual reconciliation.
Review session timeout	Save draft changes where possible and notify the reviewer.
Review history unavailable	Log the incident and notify the administrator.

6.8.8 Acceptance Criteria

The Human-in-the-Loop Review module shall be considered complete when:

- Authorized users can review AI-generated outputs.
- AI reasoning and confidence scores are displayed during review.
- Users can edit, accept, reject, or regenerate AI-generated results.
- Manual modifications are recorded with full traceability.
- Review history and version history are maintained.
- Mandatory reviews prevent deployment until completed.
- Reviewer feedback is captured for future analysis.
- All review activities are recorded for auditing.

6.9 AI Monitoring and Logging

6.9.1 Module Description

The AI Monitoring and Logging module is responsible for continuously monitoring, recording, and analyzing all AI-related activities within ProcessMind AI.

The module provides operational visibility into AI service performance, request processing, response quality, latency, resource utilization, token consumption, API usage, costs, and system health.

Monitoring data enables administrators to identify performance issues, troubleshoot failures, optimize AI usage, evaluate service reliability, and ensure compliance with organizational governance policies.

Comprehensive logging also supports auditing, debugging, incident investigation, and continuous improvement of AI-powered features.

6.9.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-141	The system shall log every AI request submitted to an AI provider.	High
AI-FR-142	The system shall log every AI response received from an AI provider.	High
AI-FR-143	The system shall record AI request timestamps.	High
AI-FR-144	The system shall record AI response times.	High
AI-FR-145	The system shall monitor AI request latency.	High
AI-FR-146	The system shall record AI processing duration for each request.	High
AI-FR-147	The system shall record AI provider information for every request.	Medium
AI-FR-148	The system shall monitor AI API success and failure rates.	High
AI-FR-149	The system shall record AI request and response sizes.	Medium
AI-FR-150	The system shall monitor token consumption for each AI request.	Medium
AI-FR-151	The system shall estimate AI usage costs where supported by the provider.	Medium
AI-FR-152	The system shall monitor AI provider availability.	High
AI-FR-153	The system shall generate alerts for repeated AI failures.	High
AI-FR-154	The system shall log AI validation failures.	Medium
AI-FR-155	The system shall log confidence evaluation results.	Medium
AI-FR-156	The system shall maintain AI operational metrics.	Medium

AI-FR-157	The system shall allow administrators to search AI logs.	Medium
AI-FR-158	The system shall support configurable AI log retention periods.	Low
AI-FR-159	The system shall generate AI performance reports.	Medium
AI-FR-160	The system shall display AI monitoring dashboards for administrators.	Medium

6.9.3 AI Metrics

The monitoring module shall collect the following operational metrics:

Metric	Description
Total AI Requests	Total number of AI requests processed
Successful Requests	Number of successful AI responses
Failed Requests	Number of failed AI requests
Average Response Time	Average AI response latency
Average Processing Time	Total AI processing duration
Token Usage	Tokens consumed per request and overall
Estimated AI Cost	Estimated API usage cost
Validation Success Rate	Percentage of responses passing validation
Average Confidence Score	Average confidence across AI outputs
Provider Availability	AI service uptime and availability

6.9.4 Monitoring Workflow



6.9.5 Log Information

Each AI log entry shall include:

Field	Description
Request ID	Unique identifier
Timestamp	Request and response time
User ID	Requesting user
Organization ID	Organization initiating the request
AI Provider	AI service used
AI Model	Model used for processing
Prompt Template Version	Version of prompt template used
Response Status	Success or failure
Response Time	Processing latency
Confidence Score	AI confidence evaluation
Validation Status	Validation result
Error Details	Failure information (if applicable)

6.9.6 Business Rules

Rule ID	Business Rule
BR-MON-001	Every AI request shall be logged.
BR-MON-002	AI logs shall not expose sensitive information such as API keys or user passwords.
BR-MON-003	Monitoring data shall only be accessible to authorized administrators.
BR-MON-004	Critical AI failures shall trigger administrator alerts.
BR-MON-005	AI operational metrics shall be retained according to organizational retention policies.
BR-MON-006	AI monitoring shall not significantly impact application performance.
BR-MON-007	Monitoring dashboards shall display near real-time operational metrics.

6.9.7 Exception Handling

Exception	Expected System Response
AI logging failure	Continue AI processing and log the failure through the system logging service.
Monitoring service unavailable	Buffer monitoring events and synchronize them when the service becomes available.
Dashboard data unavailable	Display the most recent available metrics and notify administrators.
Metric calculation failure	Mark affected metrics as unavailable and record the incident.
Log storage failure	Notify administrators and preserve logs in temporary storage where possible.

6.9.8 Acceptance Criteria

The AI Monitoring and Logging module shall be considered complete when:

- All AI requests and responses are logged.
- AI performance metrics are collected accurately.
- Token usage and estimated costs are monitored.
- AI monitoring dashboards display operational metrics.

- Critical AI failures trigger alerts.
- Administrators can search and review AI logs.
- Monitoring data is retained according to configured policies.
- Sensitive information is protected within AI logs.

6.10 AI Failure Recovery

6.10.1 Module Description

The AI Failure Recovery module is responsible for detecting, managing, and recovering from failures that occur during AI processing.

The module ensures that temporary AI service interruptions, invalid responses, provider outages, timeout errors, and communication failures do not compromise the integrity, availability, or reliability of ProcessMind AI.

Recovery strategies include automatic retries, request queuing, fallback mechanisms, user notifications, manual intervention, and comprehensive failure logging.

The module enhances system resilience by allowing business operations to continue safely while minimizing the impact of AI service disruptions.

6.10.2 Functional Requirements

Requirement ID	Functional Requirement	Priority
AI-FR-161	The system shall detect AI service failures automatically.	High
AI-FR-162	The system shall detect AI request timeouts.	High
AI-FR-163	The system shall detect malformed AI responses.	High
AI-FR-164	The system shall automatically retry failed AI requests according to configurable retry policies.	High
AI-FR-165	The system shall stop retrying after the maximum retry limit is reached.	Medium
AI-FR-166	The system shall preserve the original user request when AI processing fails.	High
AI-FR-167	The system shall queue failed AI requests for later retry where appropriate.	Medium
AI-FR-168	The system shall notify users when AI processing cannot be completed.	High
AI-FR-169	The system shall notify administrators of repeated AI service failures.	High
AI-FR-170	The system shall log all AI failures for auditing and troubleshooting.	High
AI-FR-171	The system shall prevent partially generated workflows from being deployed.	High

AI-FR-172	The system shall allow users to manually restart failed AI operations.	Medium
AI-FR-173	The system shall support configurable recovery policies.	Medium
AI-FR-174	The system shall maintain recovery history for failed AI operations.	Medium
AI-FR-175	The system shall support future failover to alternative AI providers.	Low
AI-FR-176	The system shall preserve workflow state during AI recovery operations.	High
AI-FR-177	The system shall generate recovery reports for administrators.	Medium
AI-FR-178	The system shall support graceful degradation when AI services are unavailable.	High
AI-FR-179	The system shall allow manual completion of AI-assisted processes when recovery is not possible.	Medium
AI-FR-180	The system shall ensure that AI failures do not corrupt workflow or organizational data.	High

6.10.3 Failure Types

The AI Failure Recovery module shall handle the following failure categories:

Failure Type	Description
Provider Unavailable	AI service cannot be reached
Network Failure	Communication failure between the application and AI provider
Timeout	AI response exceeds configured timeout
Invalid Response	Malformed or incomplete AI output
Rate Limit Exceeded	AI provider rejects requests due to usage limits
Authentication Failure	Invalid or expired API credentials
Internal AI Error	AI provider returns an internal server error

6.10.4 Recovery Workflow



6.10.5 Recovery Strategies

Strategy	Description
Automatic Retry	Retry transient failures using configurable policies
Request Queue	Store eligible failed requests for later processing
Manual Retry	Allow users or administrators to restart failed operations
Graceful Degradation	Disable AI-dependent features while allowing other system functions to continue
Provider Failover	Route requests to an alternative AI provider (future enhancement)
Manual Completion	Allow users to complete processes without AI assistance

6.10.6 Business Rules

Rule ID	Business Rule
BR-REC-001	AI failures shall not corrupt workflow or organizational data.
BR-REC-002	Failed AI requests shall be logged with complete diagnostic information.
BR-REC-003	Maximum retry attempts shall be configurable.
BR-REC-004	Users shall be informed when AI processing cannot be completed.
BR-REC-005	Administrators shall be notified of repeated or critical AI failures.
BR-REC-006	Workflow execution shall not continue using invalid AI-generated outputs.
BR-REC-007	Recovery operations shall preserve auditability and traceability.

6.10.7 Exception Handling

Exception	Expected System Response
Retry limit exceeded	Stop retry attempts, notify the user, and log the incident.
Recovery queue unavailable	Record the failure and require manual retry.
Provider failover unavailable	Continue with manual recovery procedures.
Persistent AI outage	Enable graceful degradation and notify administrators.
Corrupted recovery data	Reject recovery and preserve original workflow data.
Recovery service failure	Log the incident and require administrator intervention.

6.10.8 Acceptance Criteria

The AI Failure Recovery module shall be considered complete when:

- AI failures are detected automatically.
- Retry policies are executed correctly.
- Failed requests can be queued and retried.
- Users and administrators receive appropriate notifications.
- Workflow and organizational data remain protected during failures.
- Recovery history is maintained.

Chapter 7: Database Requirements

7.1 Database Overview

7.1.1 Purpose

The database is responsible for securely storing, organizing, and managing all persistent data required by ProcessMind AI.

It serves as the central repository for organizational information, user accounts, Standard Operating Procedures (SOPs), AI-generated process definitions, workflow configurations, workflow execution history, approval records, analytics data, notifications, audit logs, and system configuration.

The database shall provide high data integrity, consistency, security, scalability, and availability while supporting concurrent access by multiple organizations in a multi-tenant Software-as-a-Service (SaaS) environment.

7.1.2 Database Management System

The primary database management system shall be:

Property	Specification
Database Management System	PostgreSQL
Database Type	Relational Database Management System (RDBMS)
Deployment	Cloud-hosted PostgreSQL
ORM Framework	Spring Data JPA / Hibernate
SQL Standard	PostgreSQL SQL
Transaction Support	ACID Compliant
Multi-Tenant Support	Shared Database with Tenant Isolation

7.1.3 Database Objectives

The database shall support the following objectives:

- Store organizational business information securely.
- Maintain workflow execution history.
- Store AI-generated workflow definitions.
- Preserve workflow version history.
- Maintain complete audit trails.
- Support workflow analytics.

- Enable AI optimization using historical workflow data.
- Ensure data consistency across all modules.
- Support future scalability and enterprise integrations.

7.1.4 Data Categories

The database shall store the following categories of information:

Data Category	Description
Organization Data	Organizations, departments, settings
User Data	Users, roles, permissions, profiles
Authentication Data	Login credentials, sessions, tokens
SOP Data	Uploaded SOP documents and versions
AI Data	AI requests, responses, prompts, confidence scores
Workflow Data	Workflow definitions, versions, diagrams
Execution Data	Workflow instances, task execution history
Approval Data	Approval requests and decisions
Notification Data	User notifications
Analytics Data	Workflow metrics and reports
Audit Data	System audit logs
Configuration Data	System settings and AI configurations

7.1.5 Database Architecture

The database architecture follows a normalized relational design to minimize redundancy while maintaining high performance and scalability.

The architecture consists of multiple logical domains:

1. Authentication Domain
2. Organizations Domain
3. Workflow Domain
4. AI Domain
5. Execution Domain

6. Analytics Domain
7. Administration Domain

Each domain contains related entities that collectively support the platform's business processes.

7.1.6 Multi-Tenant Database Strategy

ProcessMind AI shall implement a shared database with logical tenant isolation.

Each organization shall have its own isolated data while sharing the same database infrastructure.

Every organization-owned record shall contain an Organization ID to enforce tenant isolation.

The application layer shall ensure that users can access only data belonging to their own organization.

7.1.7 Data Integrity

The database shall enforce the following integrity requirements:

- Primary Key constraints
- Foreign Key constraints
- Unique constraints
- Check constraints
- NOT NULL constraints
- Referential integrity
- Transaction consistency

7.1.8 Transaction Management

The database shall support ACID transactions.

Every critical operation, including workflow deployment, approval processing, user creation, workflow execution, and AI result storage, shall execute within transactional boundaries to ensure data consistency.

Transactions shall be rolled back automatically if any operation fails.

7.1.9 Database Scalability

The database architecture shall support future growth through:

- Database indexing
- Connection pooling
- Read replicas
- Database partitioning
- Horizontal scaling (future)
- Query optimization
- Efficient transaction management

7.1.10 Database Security

The database shall implement the following security measures:

- Encrypted database connections (TLS)
- Password hashing using BCrypt
- Role-based database access
- Secure credential management
- Audit logging
- Backup encryption
- Protection against SQL Injection through parameterized queries and ORM frameworks

7.1.11 Database Backup

The platform shall support:

- Automatic scheduled backups
- Manual backups
- Point-in-time recovery (where supported)

- Backup verification
- Disaster recovery procedures

7.1.12 Database Retention Policy

Historical records shall be retained according to configurable organizational policies.

Retention policies shall apply to:

- Workflow history
- AI logs
- Audit logs
- Notifications
- Analytics data
- Approval history

Archived data shall remain accessible to authorized administrators where required.

7.1.13 Acceptance Criteria

The database shall be considered compliant when:

- Data integrity is maintained.
- Multi-tenant isolation is enforced.
- Transactions execute reliably.
- Database performance meets system requirements.
- Backup and recovery procedures function successfully.
- Security controls protect sensitive information.
- Historical data is retained according to configured policies.

7.2 Entity Definitions

7.2.1 Overview

The ProcessMind AI database is organized into logical business domains. Each entity represents a business object required to support workflow management, AI processing, user management, analytics, notifications, approvals, and system administration.

The database follows normalization principles (up to Third Normal Form where applicable) while maintaining performance through appropriate indexing and optimized relationships.

The primary domains are:

- Identity & Access Management
- Organization Management
- SOP Management
- Workflow Management
- Workflow Execution
- AI Management
- Analytics
- Notification
- Administration
- Audit & Logging

Identity & Access Management

Entity: Organization

Purpose

Stores information about organizations (tenants) using ProcessMind AI.

Primary Key

- organization_id (UUID)

Attributes

Attribute	Type	Description
organization_id	UUID	Unique identifier
organization_name	VARCHAR(255)	Organization name
organization_code	VARCHAR(50)	Unique organization code
industry	VARCHAR(100)	Industry sector
email	VARCHAR(255)	Contact email

phone	VARCHAR(30)	Contact phone
address	TEXT	Address
subscription_plan	VARCHAR(50)	Current subscription
status	ENUM	Active, Suspended, Archived
created_at	TIMESTAMP	Creation date
updated_at	TIMESTAMP	Last update

Relationships

- One Organization → Many Departments
- One Organization → Many Users
- One Organization → Many SOPs
- One Organization → Many Workflows

Entity: Department

Purpose

Stores organizational departments.

Primary Key

department_id

Foreign Keys

organization_id

Attributes

- department_name
- description
- manager_id
- status
- created_at

Relationships

- One Organization → Many Departments
- One Department → Many Users

Entity: Role**Purpose**

Stores application roles.

Examples

- Super Admin
- Organization Admin
- Process Manager
- Employee
- Reviewer
- Approver

Primary Key

role_id

Entity: Permission**Purpose**

Stores system permissions.

Examples

- CREATE_WORKFLOW
- DELETE_SOP
- DEPLOY_WORKFLOW
- MANAGE_USERS

Primary Key

permission_id

Entity: RolePermission**Purpose**

Many-to-Many relationship between Roles and Permissions.

Entity: User**Purpose**

Stores user accounts.

Primary Key

user_id

Foreign Keys

- organization_id
- department_id
- role_id

Attributes

- first_name
- last_name
- email
- password_hash
- profile_picture
- designation
- status
- last_login
- created_at

Relationships

- User creates SOPs
- User owns workflows
- User approves requests
- User reviews AI output

SOP Management

Entity: SOP

Purpose

Stores uploaded Standard Operating Procedures.

Attributes

- sop_id
- organization_id
- title
- description
- current_version
- status
- uploaded_by
- uploaded_at

Entity: SOPVersion

Purpose

Stores version history.

Attributes

- version_number
- file_location
- extracted_text
- ai_processing_status
- created_at

Workflow Management

Entity: Workflow

Purpose

Stores workflow definitions.

Attributes

- workflow_id
- organization_id
- sop_id
- workflow_name
- status
- created_by
- created_at

Entity: WorkflowVersion

Purpose

Stores workflow versions.

Attributes

- workflow_json
- version_number
- validation_score
- confidence_score
- deployment_status

Entity: WorkflowNode

Purpose

Stores workflow nodes.

Examples

- Task
- Approval
- Decision
- Start
- End

Attributes

- node_type
- position
- label
- configuration

Entity: WorkflowEdge

Purpose

Stores connections between nodes.

Attributes

- source_node
- target_node
- condition
- priority

Workflow Execution

Entity: WorkflowInstance

Purpose

Stores every workflow execution.

Attributes

- instance_id
- workflow_version
- initiated_by
- status

- started_at
- completed_at

Entity: WorkflowTask

Purpose

Stores runtime tasks.

Attributes

- task_name
- assigned_user
- status
- due_date
- completed_at

Entity: ApprovalRequest

Purpose

Stores approval requests.

Attributes

- approver_id
- approval_level
- decision
- comments
- decided_at

Entity: ApprovalHistory

Purpose

Stores approval history.

AI Domain

Entity: AIRequest

Purpose

Stores every request sent to Gemini.

Attributes

- request_id
- prompt_template
- request_payload
- model
- token_count
- status

Entity: AIResponse

Purpose

Stores AI responses.

Attributes

- response_text
- confidence
- latency
- validation_status

Entity: PromptTemplate

Purpose

Stores prompt templates.

Attributes

- template_name
- version
- template_body
- task_type
- status

Entity: PromptVersion

Purpose

Stores prompt version history.

Entity: AIValidationResult

Purpose

Stores validation reports.

Attributes

- schema_status
- logical_status
- semantic_status
- validation_score

Entity: AIConfidence

Purpose

Stores confidence evaluations.

Attributes

- confidence_score
- confidence_level
- explanation

Entity: AIReview

Purpose

Stores Human-in-the-Loop review.

Attributes

- reviewer

- decision
- comments
- reviewed_at

Analytics

Entity: AnalyticsRecord

Purpose

Stores workflow analytics.

Metrics

- completion_time
- approval_time
- bottlenecks
- execution_count

Entity: OptimizationRecommendation

Purpose

Stores AI optimization suggestions.

Attributes

- recommendation
- estimated_time_saved
- estimated_cost_saved
- accepted

Notifications

Entity: Notification

Purpose

Stores notifications.

Attributes

- title
- message
- recipient
- channel
- status

Administration

Entity: SystemConfiguration

Purpose

Stores system-wide settings.

Examples

- AI timeout
- Retry policy
- Confidence threshold
- Notification configuration

Entity: OrganizationSettings

Purpose

Stores organization-specific settings.

Audit

Entity: AuditLog

Purpose

Stores every important system event.

Examples

- Login
- Workflow deployed

- Approval
- AI review
- User creation

Attributes

- action
- actor
- timestamp
- IP address
- entity_type
- entity_id

Entity: ActivityLog

Purpose

Stores general user activities.

Entity: ErrorLog

Purpose

Stores application errors.

Entity: SystemHealth

Purpose

Stores monitoring metrics.

Examples

- CPU
- Memory
- AI latency
- API uptime

Entity Relationship Summary

Domain	Entities
Identity & Access	Organization, Department, User, Role, Permission, RolePermission
SOP	SOP, SOPVersion
Workflow	Workflow, WorkflowVersion, WorkflowNode, WorkflowEdge
Execution	WorkflowInstance, WorkflowTask, ApprovalRequest, ApprovalHistory
AI	AIRequest, AIResponse, PromptTemplate, PromptVersion, AIValidationResult, AIConfidence, AIReview
Analytics	AnalyticsRecord, OptimizationRecommendation
Notification	Notification
Administration	SystemConfiguration, OrganizationSettings
Audit	AuditLog, ActivityLog, ErrorLog, SystemHealth

7.3 Relationship Specifications

7.3.1 Overview

This section defines the relationships among the database entities used within ProcessMind AI.

Relationships establish referential integrity between entities and ensure that business data remains consistent throughout the lifecycle of organizations, users, workflows, AI processing, approvals, analytics, and system administration.

All relationships shall be enforced through foreign key constraints unless otherwise specified.

7.3.2 Organization Relationships

Organization → Department

Property	Value
Relationship Type	One-to-Many (1:N)
Parent Entity	Organization
Child Entity	Department
Foreign Key	Department.organization_id

Description

An organization may contain multiple departments.

Each department shall belong to exactly one organization.

Organization → User

Property	Value
Relationship Type	One-to-Many
Foreign Key	User.organization_id

Each organization may contain multiple users.

Each user belongs to exactly one organization.

Organization → SOP

Property	Value
Relationship Type	One-to-Many
Foreign Key	SOP.organization_id

Each organization may own multiple SOPs.

Organization → Workflow

Property	Value
Relationship Type	One-to-Many
Foreign Key	Workflow.organization_id

Each organization may own multiple workflows.

7.3.3 Department Relationships

Department → User

Property	Value
----------	-------

Relationship Type	One-to-Many
Foreign Key	User.department_id

A department may contain multiple users.

Each user belongs to one department.

7.3.4 Role Relationships

Role → User

Property	Value
Relationship Type	One-to-Many
Foreign Key	User.role_id

Each role may be assigned to multiple users.

Role → Permission

Property	Value
Relationship Type	Many-to-Many
Junction Entity	RolePermission

A role may contain multiple permissions.

A permission may belong to multiple roles.

7.3.5 SOP Relationships

SOP → SOPVersion

Property	Value
Relationship Type	One-to-Many
Foreign Key	SOPVersion.sop_id

Each SOP may have multiple versions.

SOP → Workflow

Property	Value
Relationship Type	One-to-Many
Foreign Key	Workflow.sop_id

One SOP may generate multiple workflow versions over time.

7.3.6 Workflow Relationships

Workflow → WorkflowVersion

Property	Value
Relationship Type	One-to-Many
Foreign Key	WorkflowVersion.workflow_id

A workflow may contain multiple versions.

WorkflowVersion → WorkflowNode

Property	Value
Relationship Type	One-to-Many
Foreign Key	WorkflowNode.workflow_version_id

Each workflow version contains multiple workflow nodes.

WorkflowNode → WorkflowEdge

Property	Value
Relationship Type	One-to-Many (Source Node)

Each workflow node may connect to multiple outgoing edges.

WorkflowEdge → WorkflowNode

Property	Value
----------	-------

Relationship Type	Many-to-One (Target Node)
-------------------	---------------------------

Each workflow edge connects to one destination node.

7.3.7 Workflow Execution Relationships

WorkflowVersion → WorkflowInstance

Property	Value
Relationship Type	One-to-Many

Each deployed workflow version may be executed many times.

WorkflowInstance → WorkflowTask

Property	Value
Relationship Type	One-to-Many

Each workflow execution generates multiple runtime tasks.

WorkflowTask → ApprovalRequest

Property	Value
Relationship Type	One-to-One or One-to-Many

Approval tasks generate one or more approval requests.

ApprovalRequest → ApprovalHistory

Property	Value
Relationship Type	One-to-Many

Each approval request maintains its approval history.

7.3.8 AI Relationships

PromptTemplate → AIRequest

Property	Value
Relationship Type	One-to-Many

One prompt template may generate many AI requests.

AIRequest → AIResponse

Property	Value
Relationship Type	One-to-One

Each AI request produces one AI response.

AIResponse → AIValidationResult

Property	Value
Relationship Type	One-to-One

Every AI response has one validation result.

AIResponse → AIConfidence

Property	Value
Relationship Type	One-to-One

Each AI response receives one confidence evaluation.

AIResponse → AIReview

Property	Value
Relationship Type	One-to-Many

AI responses may undergo multiple review cycles.

7.3.9 Analytics Relationships

WorkflowInstance → AnalyticsRecord

Property	Value
Relationship Type	One-to-One

Each completed workflow execution contributes one analytics record.

Workflow → OptimizationRecommendation

Property	Value
Relationship Type	One-to-Many

A workflow may receive multiple optimization recommendations over time.

7.3.10 Notification Relationships

User → Notification

Property	Value
Relationship Type	One-to-Many

Each user may receive multiple notifications.

7.3.11 Audit Relationships

User → AuditLog

Property	Value
Relationship Type	One-to-Many

Every user action may create an audit log entry.

User → ActivityLog

Property	Value
Relationship Type	One-to-Many

User activities shall be recorded for operational monitoring.

SystemConfiguration → OrganizationSettings

Property	Value
Relationship Type	One-to-Many

Global system settings provide defaults for organization-specific settings.

7.3.12 Relationship Summary

Parent Entity	Child Entity	Cardinality
Organization	Department	1:N
Organization	User	1:N
Organization	SOP	1:N
Organization	Workflow	1:N
Department	User	1:N
Role	User	1:N
Role	Permission	M:N
SOP	SOPVersion	1:N
SOP	Workflow	1:N
Workflow	WorkflowVersion	1:N
WorkflowVersion	WorkflowNode	1:N
WorkflowNode	WorkflowEdge	1:N
WorkflowVersion	WorkflowInstance	1:N
WorkflowInstance	WorkflowTask	1:N
WorkflowTask	ApprovalRequest	1:N
ApprovalRequest	ApprovalHistory	1:N
PromptTemplate	AIRequest	1:N
AIRequest	AIResponse	1:1
AIResponse	AIValidationResult	1:1
AIResponse	AIConfidence	1:1
AIResponse	AIReview	1:N

WorkflowInstance	AnalyticsRecord	1:1
Workflow	OptimizationRecommendation	1:N
User	Notification	1:N
User	AuditLog	1:N
User	ActivityLog	1:N

7.3.13 Acceptance Criteria

The relationship model shall be considered complete when:

- All entities are connected through defined relationships.
- Referential integrity is enforced using foreign keys.
- Cardinality accurately reflects business rules.
- No orphan records exist for mandatory relationships.
- Relationship definitions align with the Entity Relationship Diagram (ERD).
- Multi-tenant isolation is maintained across all organization-owned entities.

7.4 Data Dictionary

7.4.1 Overview

The Data Dictionary defines the metadata of all database entities used within ProcessMind AI. It specifies the structure, data types, constraints, default values, and descriptions of database attributes to ensure consistency during implementation and maintenance.

All database objects shall follow standardized naming conventions and data integrity rules.

7.4.2 Naming Conventions

Item	Convention	Example
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Tables	Singular PascalCase	User, WorkflowInstance
Columns	snake_case	organization_id
Primary Keys	<entity>_id	user_id
Foreign Keys	<referenced_entity>_id	workflow_id
Boolean Fields	Prefix with is_ or has_	is_active
Date Fields	Suffix _at	created_at
Version Fields	Suffix _version	prompt_version

7.4.3 Standard Data Types

PostgreSQL Type	Usage
UUID	Primary and foreign keys
VARCHAR(n)	Short text
TEXT	Long text
BOOLEAN	True/False values
INTEGER	Numeric counters
BIGINT	Large numeric values
DECIMAL	Percentages and monetary values
TIMESTAMP WITH TIME ZONE	Date and time
DATE	Calendar date
JSONB	AI responses, workflow definitions, metadata
BYTEA	Binary files (if stored in DB)
ENUM	Controlled value lists

7.4.4 Standard Audit Columns

Unless otherwise specified, every business entity shall include the following audit attributes.

Column	Type	Nullable	Description
created_at	TIMESTAMP	No	Record creation timestamp
updated_at	TIMESTAMP	No	Last modification timestamp
created_by	UUID	Yes	User who created the record
updated_by	UUID	Yes	User who last modified the record

version	INTEGER	No	Optimistic locking version
is_deleted	BOOLEAN	No	Soft delete indicator

7.4.5 Entity Data Dictionary

Organization

Column	Type	Nullable	Constraints	Description
organization_id	UUID	No	PK	Organization identifier
organization_name	VARCHAR(255)	No	Unique	Organization name
organization_code	VARCHAR(50)	No	Unique	Internal organization code
industry	VARCHAR(100)	Yes	—	Industry sector
email	VARCHAR(255)	No	Unique	Contact email
phone	VARCHAR(30)	Yes	—	Contact number
address	TEXT	Yes	—	Organization address
subscription_plan	VARCHAR(50)	No	—	Subscription plan
status	ENUM	No	—	Organization status

Department

Column	Type	Nullable	Constraints	Description
department_id	UUID	No	PK	Department identifier
organization_id	UUID	No	FK	Parent organization
department_name	VARCHAR(255)	No	—	Department name
manager_id	UUID	Yes	FK	Department manager

description	TEXT	Yes	—	Department description
status	ENUM	No	—	Department status

User

Column	Type	Nullable	Constraints	Description
user_id	UUID	No	PK	User identifier
organization_id	UUID	No	FK	Organization
department_id	UUID	Yes	FK	Department
role_id	UUID	No	FK	Assigned role
first_name	VARCHAR(100)	No	—	First name
last_name	VARCHAR(100)	No	—	Last name
email	VARCHAR(255)	No	Unique	Login email
password_hash	VARCHAR(255)	No	—	BCrypt password
designation	VARCHAR(100)	Yes	—	Job title
status	ENUM	No	—	User status
last_login	TIMESTAMP	Yes	—	Last login

SOP

Column	Type	Nullable	Constraints	Description
sop_id	UUID	No	PK	SOP identifier
organization_id	UUID	No	FK	Organization
title	VARCHAR(255)	No	—	SOP title

description	TEXT	Yes	—	Description
current_version	INTEGER	No	—	Current version
uploaded_by	UUID	No	FK	User who uploaded
status	ENUM	No	—	SOP status

Workflow

Column	Type	Nullable	Constraints	Description
workflow_id	UUID	No	PK	Workflow identifier
organization_id	UUID	No	FK	Organization
sop_id	UUID	No	FK	Source SOP
workflow_name	VARCHAR(255)	No	—	Workflow name
status	ENUM	No	—	Workflow status
current_version	INTEGER	No	—	Active version

WorkflowVersion

Column	Type	Nullable	Constraints	Description
workflow_version_id	UUID	No	PK	Version identifier
workflow_id	UUID	No	FK	Parent workflow
version_number	INTEGER	No	—	Version number
workflow_definition	JSONB	No	—	Complete workflow graph
confidence_score	DECIMAL(5,2)	Yes	—	AI confidence
validation_score	DECIMAL(5,2)	Yes	—	Validation score
deployment_status	ENUM	No	—	Deployment state

WorkflowInstance

Column	Type	Nullable	Constraints	Description
workflow_instance_id	UUID	No	PK	Workflow execution
workflow_version_id	UUID	No	FK	Workflow version

initiated_by	UUID	No	FK	Initiating user
status	ENUM	No	—	Execution status
started_at	TIMESTAMP	No	—	Start time
completed_at	TIMESTAMP	Yes	—	Completion time

AIRequest

Column	Type	Nullable	Constraints	Description
ai_request_id	UUID	No	PK	AI request
prompt_template_id	UUID	No	FK	Prompt used
provider	VARCHAR(100)	No	—	Gemini/OpenAI
model	VARCHAR(100)	No	—	Model name
prompt	TEXT	No	—	Submitted prompt
status	ENUM	No	—	Request status

AIResponse

Column	Type	Nullable	Constraints	Description
ai_response_id	UUID	No	PK	AI response
ai_request_id	UUID	No	FK	Related request
response	JSONB	No	—	Structured response
confidence_score	DECIMAL(5,2)	Yes	—	AI confidence
validation_status	ENUM	No	—	Validation result
latency_ms	INTEGER	Yes	—	Processing latency

Notification

Column	Type	Nullable	Constraints	Description
notification_id	UUID	No	PK	Notification

user_id	UUID	No	FK	Recipient
title	VARCHAR(255)	No	—	Notification title
message	TEXT	No	—	Content
channel	ENUM	No	—	Email/In-App
status	ENUM	No	—	Delivery status

AuditLog

Column	Type	Nullable	Constraints	Description
audit_log_id	UUID	No	PK	Audit entry
user_id	UUID	Yes	FK	User
action	VARCHAR(255)	No	—	Performed action
entity_type	VARCHAR(100)	No	—	Affected entity
entity_id	UUID	Yes	—	Affected record
ip_address	VARCHAR(50)	Yes	—	Source IP
timestamp	TIMESTAMP	No	—	Event time

7.4.6 Common Constraints

Constraint	Description
Primary Key	Every entity shall have a UUID primary key.
Foreign Key	Referential integrity shall be enforced.
Unique	Emails, organization codes, and role names shall be unique.
NOT NULL	Mandatory business fields shall not be null.
CHECK	Enum values and numeric ranges shall be validated.
Soft Delete	Business entities shall use logical deletion where appropriate.

7.4.7 Acceptance Criteria

The data dictionary shall be considered complete when:

- Every major entity has documented attributes.
- Data types align with PostgreSQL standards.
- Constraints enforce business rules.
- Naming conventions are applied consistently.
- Audit columns are standardized across business entities.
- The documented schema aligns with the ER Diagram and implementation.

7.5 Constraints & Integrity Rules

7.5.1 Overview

The database shall enforce integrity constraints to ensure that all stored information remains accurate, consistent, reliable, and compliant with business rules.

Constraints shall be implemented using PostgreSQL constraint mechanisms, transaction management, application-level validation, and database triggers where appropriate.

These rules apply across all ProcessMind AI modules, including authentication, workflow management, AI processing, approvals, analytics, notifications, and auditing.

7.5.2 Entity Integrity

Every entity shall have a unique primary key.

Rule ID	Constraint
DB-CON-001	Every table shall contain exactly one Primary Key.
DB-CON-002	Primary Keys shall use UUID data type.
DB-CON-003	Primary Keys shall never be updated after record creation.
DB-CON-004	Primary Keys shall uniquely identify every record.

7.5.3 Referential Integrity

Foreign Key relationships shall ensure valid relationships between entities.

Rule ID	Constraint
DB-CON-005	Every Foreign Key shall reference an existing Primary Key.
DB-CON-006	Orphan records shall not be permitted.
DB-CON-007	Referential integrity shall be enforced using PostgreSQL Foreign Keys.
DB-CON-008	Invalid Foreign Key values shall be rejected.

7.5.4 Mandatory Fields

Required business information shall not be omitted.

Rule ID	Constraint
DB-CON-009	Mandatory fields shall be defined using NOT NULL constraints.
DB-CON-010	Business-critical entities shall not be created with incomplete information.
DB-CON-011	Required relationships shall exist before dependent records are created.

Examples include:

- Organization Name
- User Email
- Workflow Name
- Workflow Status
- AI Request Status

7.5.5 Uniqueness Constraints

Certain business attributes shall remain unique.

Rule ID	Constraint
DB-CON-012	Organization Code shall be unique.
DB-CON-013	User Email shall be unique within the system.
DB-CON-014	Role Name shall be unique.
DB-CON-015	Permission Name shall be unique.
DB-CON-016	Prompt Template Name and Version combination shall be unique.

7.5.6 Domain Constraints

Column values shall comply with predefined domains.

Rule ID	Constraint
DB-CON-017	ENUM values shall only contain predefined values.
DB-CON-018	Confidence Score shall be between 0 and 100.
DB-CON-019	Validation Score shall be between 0 and 100.
DB-CON-020	Retry Count shall be zero or greater.
DB-CON-021	Workflow Version Number shall be greater than zero.

7.5.7 Business Integrity Rules

Business processes shall maintain logical consistency.

Rule ID	Constraint
DB-CON-022	Every Workflow shall belong to one Organization.
DB-CON-023	Every Workflow Version shall belong to one Workflow.
DB-CON-024	Every Workflow Instance shall reference a deployed Workflow Version.
DB-CON-025	Every Approval Request shall reference an existing Workflow Task.
DB-CON-026	Every AI Response shall reference an existing AI Request.
DB-CON-027	Every Notification shall reference an existing User.
DB-CON-028	Every Audit Log entry shall reference a valid system event.

7.5.8 Cascade Rules

The database shall apply controlled cascading operations.

Parent Entity	Child Entity	Delete Behavior
Organization	Department	RESTRICT
Organization	User	RESTRICT
Organization	Workflow	RESTRICT
SOP	SOP Version	CASCADE
Workflow	Workflow Version	CASCADE

Workflow Version	Workflow Node	CASCADE
Workflow Version	Workflow Instance	RESTRICT
Workflow Instance	Workflow Task	CASCADE
Workflow Task	Approval Request	CASCADE
AI Request	AI Response	CASCADE

Explanation

Historical business records shall not be deleted automatically unless explicitly permitted by business policy.

7.5.9 Soft Delete Policy

To preserve historical information:

Rule ID	Constraint
DB-CON-029	Business entities shall use logical deletion where appropriate.
DB-CON-030	Deleted records shall remain available for auditing.
DB-CON-031	Soft-deleted records shall be excluded from normal queries.

Entities using soft delete include:

- User
- Workflow
- SOP
- Organization
- Notification

7.5.10 Transaction Integrity

Critical business operations shall execute atomically.

Rule ID	Constraint
DB-CON-032	Workflow deployment shall execute within a single transaction.
DB-CON-033	Approval processing shall execute within a single transaction.
DB-CON-034	User creation shall execute within a single transaction.
DB-CON-035	AI processing results shall be stored atomically.
DB-CON-036	Failed transactions shall be rolled back automatically.

7.5.11 Optimistic Locking

To prevent concurrent update conflicts:

Rule ID	Constraint
DB-CON-037	Business entities shall support optimistic locking.
DB-CON-038	Record version numbers shall be incremented after successful updates.
DB-CON-039	Conflicting updates shall be rejected and require retry.

7.5.12 Audit Integrity

Audit information shall remain immutable.

Rule ID	Constraint
DB-CON-040	Audit Log records shall not be modified after creation.
DB-CON-041	Audit Log records shall not be physically deleted.
DB-CON-042	Every administrative operation shall generate an audit record.

7.5.13 Multi-Tenant Integrity

Tenant isolation shall be enforced.

Rule ID	Constraint
DB-CON-043	Every organization-owned record shall contain Organization ID.
DB-CON-044	Users shall access only data belonging to their organization.
DB-CON-045	Cross-organization access shall be prohibited except for System Administrators.

7.5.14 Data Validation Rules

Application-level validation shall complement database constraints.

Examples include:

Field	Validation Rule
Email	Valid email format

Password	Minimum security requirements
Confidence Score	0–100
Workflow Name	Maximum 255 characters
Prompt Template	Cannot be empty
API Key	Must conform to configured provider requirements

7.5.15 Acceptance Criteria

The database integrity model shall be considered complete when:

- Primary and foreign key constraints are enforced.
- Referential integrity prevents orphan records.
- Business rules are reflected through database constraints.
- Transactions preserve consistency.
- Soft deletion preserves historical records.
- Audit data remains immutable.
- Tenant isolation is maintained.
- Concurrent updates are handled safely through optimistic locking.

7.6 Indexing Strategy

7.6.1 Overview

The Indexing Strategy defines how database indexes shall be designed and maintained to optimize query performance, reduce response times, and support efficient data retrieval.

Indexes shall be created for frequently queried columns, foreign keys, search fields, and analytical operations while balancing read performance with write overhead.

The strategy shall be periodically reviewed and optimized based on application usage patterns.

7.6.2 Objectives

The indexing strategy shall achieve the following objectives:

- Improve query execution performance.
- Reduce response time for user operations.

- Optimize workflow search and filtering.
- Accelerate AI log retrieval.
- Improve analytics performance.
- Support efficient multi-tenant data access.
- Minimize unnecessary full table scans.

7.6.3 Primary Key Indexes

Every Primary Key shall be indexed automatically.

Index ID	Entity	Indexed Column
IDX-PK-001	Organization	organization_id
IDX-PK-002	Department	department_id
IDX-PK-003	User	user_id
IDX-PK-004	SOP	sop_id
IDX-PK-005	Workflow	workflow_id
IDX-PK-006	WorkflowVersion	workflow_version_id
IDX-PK-007	WorkflowInstance	workflow_instance_id
IDX-PK-008	AIRequest	ai_request_id
IDX-PK-009	AIResponse	ai_response_id
IDX-PK-010	Notification	notification_id

7.6.4 Foreign Key Indexes

Foreign Keys shall be indexed to improve join performance.

Index ID	Indexed Column
IDX-FK-001	organization_id
IDX-FK-002	department_id
IDX-FK-003	role_id
IDX-FK-004	workflow_id
IDX-FK-005	workflow_version_id
IDX-FK-006	sop_id
IDX-FK-007	user_id

IDX-FK-008	ai_request_id
IDX-FK-009	prompt_template_id

7.6.5 Unique Indexes

The following business attributes shall be uniquely indexed.

Column	Reason
organization_code	Organization identification
email	User authentication
role_name	RBAC
permission_name	Permission management
prompt_name + version	Prompt versioning

7.6.6 Search Indexes

The following fields shall be indexed to support search functionality.

Entity	Indexed Field	Purpose
SOP	title	SOP search
Workflow	workflow_name	Workflow search
User	first_name	User lookup
User	last_name	User lookup
Department	department_name	Department search
Notification	title	Notification search

7.6.7 Multi-Tenant Indexes

Since ProcessMind AI is a SaaS application, Organization ID shall be indexed.

Entity	Indexed Field
User	organization_id

Department	organization_id
SOP	organization_id
Workflow	organization_id
WorkflowInstance	organization_id
Notification	organization_id
AuditLog	organization_id

This allows rapid filtering of organization-specific data.

7.6.8 Composite Indexes

Composite indexes shall optimize common filtering scenarios.

Composite Index	Purpose
(organization_id, email)	User authentication
(organization_id, workflow_name)	Workflow search
(organization_id, status)	Active workflow retrieval
(workflow_version_id, status)	Workflow execution lookup
(assigned_user_id, status)	Pending task retrieval
(approver_id, status)	Pending approvals
(user_id, created_at)	Notification history

7.6.9 Full-Text Search Indexes

The platform shall support PostgreSQL Full-Text Search for textual content.

Entity	Indexed Fields
SOP	title + extracted_text
Workflow	workflow_name + description
PromptTemplate	template_name + template_body
AuditLog	action
Notification	title + message

This enables:

- Keyword search
- Phrase search

- AI document search
- SOP lookup

7.6.10 JSONB Indexes

JSONB columns shall use GIN indexes.

Entity	JSON Column	Index Type
WorkflowVersion	workflow_definition	GIN
AIResponse	response	GIN
SystemConfiguration	configuration	GIN

These indexes support efficient querying within structured JSON data.

7.6.11 Analytics Indexes

The following indexes shall improve reporting performance.

Indexed Column	Purpose
created_at	Time-series reports
completed_at	Workflow analytics
status	Dashboard statistics
confidence_score	AI analytics
validation_status	AI monitoring
execution_time	Performance analysis

7.6.12 Audit Indexes

Audit queries shall be optimized using indexes.

Indexed Column	Purpose
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timestamp	Audit history
user_id	User activity
entity_type	Entity filtering
action	Audit reporting

7.6.13 AI Monitoring Indexes

Indexes shall support AI monitoring dashboards.

Indexed Column	Purpose
provider	Provider performance
model	Model comparison
latency_ms	Performance metrics
request_timestamp	AI timeline
validation_status	Validation analytics

7.6.14 Index Maintenance

The database shall support periodic index maintenance.

Maintenance activities include:

- Rebuilding fragmented indexes.
- Updating PostgreSQL statistics using ANALYZE.
- Reclaiming storage using VACUUM.
- Monitoring unused indexes.
- Removing redundant indexes.

These tasks shall be scheduled during low-traffic periods to minimize operational impact.

7.6.15 Business Rules

Rule ID	Rule
DB-IDX-001	Primary Keys shall always be indexed.
DB-IDX-002	Foreign Keys shall be indexed unless justified otherwise.
DB-IDX-003	Frequently searched columns shall be indexed.
DB-IDX-004	Full-text indexes shall support document search.
DB-IDX-005	JSONB fields shall use GIN indexes where querying is required.
DB-IDX-006	Indexes shall be periodically reviewed for effectiveness.
DB-IDX-007	Indexes shall not be created unnecessarily to avoid write-performance degradation.

7.6.16 Acceptance Criteria

The indexing strategy shall be considered complete when:

- Primary and foreign key indexes exist.
- Frequently executed queries use appropriate indexes.
- Search operations complete efficiently.
- Analytics queries perform within acceptable response times.
- Multi-tenant filtering is optimized.
- Full-text search supports SOP discovery.
- JSONB queries perform efficiently.
- Index maintenance procedures are documented and implemented.

7.7 Backup & Recovery Requirements

7.7.1 Overview

The Backup and Recovery Requirements define the strategies, policies, and procedures used to protect organizational data against accidental deletion, system failures, hardware failures, software defects, cyberattacks, and natural disasters.

The backup system shall ensure that critical business data can be restored with minimal data loss and acceptable service downtime.

These requirements apply to all persistent data managed by ProcessMind AI, including organizational data, workflows, SOPs, AI processing records, approvals, audit logs, notifications, and analytics.

7.7.2 Objectives

The backup and recovery strategy shall achieve the following objectives:

- Protect organizational data.
- Minimize service interruption.
- Reduce data loss.
- Support disaster recovery.
- Ensure business continuity.
- Maintain regulatory compliance.
- Verify backup integrity.

7.7.3 Backup Scope

The following data shall be included in scheduled backups.

Data Category	Included
Organizations	Yes
Users	Yes
Departments	Yes
Roles & Permissions	Yes
SOPs	Yes
Workflow Definitions	Yes
Workflow Versions	Yes
Workflow Execution History	Yes
Approval Records	Yes
AI Requests	Yes
AI Responses	Yes
Prompt Templates	Yes
Notifications	Yes

Analytics Data	Yes
Audit Logs	Yes
System Configuration	Yes
Organization Settings	Yes

7.7.4 Backup Types

The system shall support the following backup strategies.

Backup Type	Description
Full Backup	Complete database backup
Incremental Backup	Backup of changes since the previous backup
Differential Backup	Backup of changes since the last full backup
Configuration Backup	System and application configuration
File Backup	Uploaded SOP documents and related files

7.7.5 Backup Schedule

The following backup schedule shall be implemented.

Backup Type	Frequency
Full Backup	Weekly
Incremental Backup	Daily
Configuration Backup	After configuration changes
File Backup	Daily
Audit Log Archive	Monthly

The schedule may be modified by authorized administrators.

7.7.6 Recovery Objectives

The platform shall define recovery objectives.

Metric	Target
Recovery Point Objective (RPO)	≤ 24 hours
Recovery Time Objective (RTO)	≤ 4 hours

Backup Verification	100% successful verification
Backup Availability	99.9%

Note: These values represent initial deployment targets and may be adjusted for production environments based on organizational service-level agreements (SLAs).

7.7.7 Recovery Process

The recovery process shall follow the sequence below.



7.7.8 Backup Security

The following security controls shall apply.

Requirement	Description
Encryption	Backups shall be encrypted at rest.
Secure Transfer	Backup transmission shall use encrypted communication channels (TLS).
Access Control	Only authorized administrators may access backups.
Integrity Verification	Backup integrity shall be verified before restoration.
Backup Logging	All backup operations shall be logged.

7.7.9 Backup Retention Policy

The system shall retain backups according to the following policy.

Backup Type	Retention Period
Daily Backups	30 days

Weekly Backups	12 weeks
Monthly Backups	12 months
Annual Backups	7 years (configurable)

Retention periods may be adjusted to comply with organizational or regulatory requirements.

7.7.10 Disaster Recovery

The disaster recovery strategy shall support:

- Database recovery.
- File storage recovery.
- Configuration restoration.
- AI configuration recovery.
- Workflow restoration.
- Audit log preservation.
- Notification recovery.

Recovery procedures shall be documented and periodically tested.

7.7.11 Recovery Validation

After restoration, the following validation checks shall be performed.

Validation	Description
	Verify referential integrity and constraints
Workflow Integrity	Verify workflow definitions and versions
User Accounts	Verify user authentication data
AI Data	Verify AI requests and responses
Audit Logs	Verify audit history
File Storage	Verify uploaded SOP documents

The system shall not return to production until validation is successfully completed.

7.7.12 Business Rules

Rule ID	Rule
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DB-BKP-001	Scheduled backups shall execute automatically according to the configured schedule.
DB-BKP-002	Backup files shall be encrypted.
DB-BKP-003	Backup integrity shall be verified before restoration.
DB-BKP-004	Only authorized administrators shall perform restore operations.
DB-BKP-005	All backup and recovery operations shall be recorded in the audit log.
DB-BKP-006	Backup failures shall generate administrator alerts.
DB-BKP-007	Disaster recovery procedures shall be tested periodically.

7.7.13 Exception Handling

Exception	Expected System Response
Backup failure	Log the failure, notify administrators, and retry according to the backup policy.
Backup corruption	Reject the backup and restore from the most recent verified backup.
Insufficient storage	Notify administrators and suspend new backups until storage is available.
Recovery validation failure	Prevent production use until validation succeeds.
Unauthorized restore attempt	Reject the request and record the incident in the audit log.
File restoration failure	Continue database recovery where safe and report missing files.

7.7.14 Acceptance Criteria

The backup and recovery strategy shall be considered complete when:

- Scheduled backups execute successfully.
- Backup files are encrypted and protected.
- Backup integrity is verified before restoration.
- Recovery procedures restore all required data.
- Recovery objectives (RPO/RTO) are met under normal operating conditions.
- Backup and recovery events are recorded in audit logs.
- Disaster recovery procedures are documented and tested periodically.

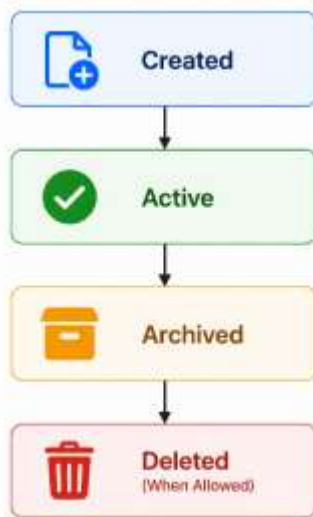
7.8 Data Retention & Archival Policy

7.8.1 Overview

The Data Retention and Archival Policy defines how ProcessMind AI manages the lifecycle of stored data throughout its existence.

The policy ensures that business information is retained for operational, legal, auditing, and analytical purposes while preventing unnecessary database growth.

Data shall progress through the following lifecycle:



The policy applies to all organizations using the platform while allowing organization-specific retention settings where supported.

7.8.2 Objectives

The retention policy shall:

- Preserve important business records.
- Reduce database storage growth.
- Improve database performance.
- Maintain auditability.
- Support regulatory compliance.
- Reduce operational costs.
- Enable historical analytics.

7.8.3 Data Classification

All stored data shall be classified according to its business importance.

Classification	Description
Operational Data	Data actively used by the application
Historical Data	Completed workflows and historical records
Audit Data	Security and audit information
AI Processing Data	AI requests, responses, prompts, confidence scores
Configuration Data	Organization and system settings
Temporary Data	Cached or intermediate processing data

7.8.4 Retention Periods

The default retention periods shall be as follows.

Data Category	Retention Period	Archive	Permanent Deletion
Organizations	Lifetime	No	Administrator only
Users	Lifetime (while active)	Yes	Configurable
SOPs	Lifetime	Yes	Administrator only
Workflow Definitions	Lifetime	Yes	Administrator only
Workflow Execution History	5 Years	Yes	Configurable
Approval History	5 Years	Yes	Configurable
AI Requests	2 Years	Yes	Configurable
AI Responses	2 Years	Yes	Configurable
Prompt Templates	Lifetime	Yes	Never (default)
Notifications	1 Year	Yes	Automatic
Analytics Records	5 Years	Yes	Configurable
Audit Logs	7 Years	Yes	Restricted
Error Logs	1 Year	Yes	Automatic
System Health Metrics	1 Year	Yes	Automatic

These values represent recommended defaults and may be overridden by organizational or regulatory policies where permitted.

7.8.5 Archival Strategy

Archived data shall:

- Remain read-only.
- Be removed from active operational queries.
- Remain searchable by authorized users.
- Be restorable when required.
- Maintain referential integrity.

Archived data may be stored in:

- Archive database
- Cold storage
- Encrypted object storage
- Long-term backup storage

7.8.6 Data Lifecycle



7.8.7 Archival Rules

Rule ID	Rule
DB-ARC-001	Only completed workflow instances may be archived.
DB-ARC-002	Archived records shall remain read-only.
DB-ARC-003	Archived records shall preserve relationships with associated entities.
DB-ARC-004	Archived data shall remain available for reporting and auditing where authorized.

DB-ARC-005	Archiving shall not interrupt active system operations.
DB-ARC-006	Archival operations shall be logged.

7.8.8 Secure Deletion

When permanent deletion is permitted:

- The deletion shall comply with organizational retention policies.
- All related records shall be handled according to referential integrity rules.
- Deletion operations shall be audited.
- Sensitive data shall be securely erased where applicable.

Deletion shall not occur for records subject to:

- Active legal holds.
- Compliance requirements.
- Pending investigations.
- Ongoing workflow execution.

7.8.9 Business Rules

Rule ID	Rule
DB-RET-001	Active workflow data shall never be archived.
DB-RET-002	Archived data shall remain immutable unless restored.
DB-RET-003	Retention policies shall be configurable by authorized administrators within permitted limits.
DB-RET-004	Audit logs shall not be deleted before the configured retention period expires.
DB-RET-005	Archival and deletion operations shall generate audit log entries.
DB-RET-006	Archived data shall remain encrypted while stored.
DB-RET-007	Restored archived records shall preserve their original identifiers and relationships.

7.8.10 Exception Handling

Exception	Expected System Response
Archive operation failure	Log the incident, notify the administrator, and retry according to policy.

Archive storage unavailable	Delay archival and retain data in the active database until storage becomes available.
Restore failure	Preserve archived data and notify the administrator.
Attempt to delete protected records	Reject the request and record the event.
Retention policy conflict	Apply the stricter policy and notify administrators.
Archive integrity verification failure	Prevent archival completion until verification succeeds.

7.8.11 Acceptance Criteria

The Data Retention and Archival Policy shall be considered complete when:

- Data retention periods are enforced according to policy.
- Eligible data is archived automatically or through authorized administrative actions.
- Archived data remains secure, immutable, and recoverable.
- Permanent deletion occurs only when permitted.
- Retention, archival, restoration, and deletion events are recorded in audit logs.
- The policy supports organizational and regulatory compliance requirements.

7.9 Database Security Requirements

7.9.1 Overview

The Database Security Requirements define the controls, policies, and mechanisms used to protect the confidentiality, integrity, and availability of data stored within ProcessMind AI.

The database security architecture shall implement defense-in-depth principles through authentication, authorization, encryption, auditing, network security, tenant isolation, and secure database configuration.

These requirements apply to all database environments, including development, testing, staging, and production.

7.9.2 Security Objectives

The database security architecture shall achieve the following objectives:

- Protect sensitive organizational data.
- Prevent unauthorized database access.
- Ensure tenant isolation.
- Maintain data integrity.
- Protect credentials and secrets.
- Support auditing and forensic investigations.
- Comply with security best practices.

7.9.3 Authentication

Database authentication shall comply with the following requirements.

Requirement ID	Requirement
DB-SEC-001	Every database connection shall be authenticated.
DB-SEC-002	Database credentials shall not be stored in source code.
DB-SEC-003	Database credentials shall be stored using secure environment variables or a secrets management solution.
DB-SEC-004	Administrative database accounts shall be restricted to authorized administrators.
DB-SEC-005	Shared administrator accounts shall not be used.

7.9.4 Authorization

Access shall follow the principle of least privilege.

Requirement ID	Requirement
DB-SEC-006	Database roles shall grant only the minimum required privileges.
DB-SEC-007	Application service accounts shall not possess superuser privileges.
DB-SEC-008	Database administrators shall use separate administrative accounts.
DB-SEC-009	Privileged operations shall be restricted to authorized personnel.
DB-SEC-010	Database permissions shall be reviewed periodically.

7.9.5 Encryption

Sensitive information shall be protected through encryption.

Requirement	Description
Encryption in Transit	All database communication shall use TLS.
Encryption at Rest	Database storage and backups shall be encrypted.
Password Storage	User passwords shall be stored using BCrypt hashes.
API Credentials	API keys and secrets shall be encrypted or securely managed.

7.9.6 Multi-Tenant Security

Tenant isolation shall be enforced.

Requirement ID	Requirement
DB-SEC-011	Every organization-owned record shall include an Organization ID.
DB-SEC-012	Cross-tenant data access shall be prohibited.
DB-SEC-013	Tenant isolation shall be enforced at the application layer.
DB-SEC-014	PostgreSQL Row-Level Security (RLS) should be used in production deployments where appropriate.

7.9.7 Data Integrity

The following controls shall protect database integrity.

Requirement ID	Requirement
DB-SEC-015	Foreign key constraints shall be enforced.
DB-SEC-016	Database transactions shall comply with ACID properties.
DB-SEC-017	Optimistic locking shall prevent concurrent update conflicts.
DB-SEC-018	Audit records shall be immutable.

7.9.8 Secure Database Configuration

The database shall be configured securely.

Security measures include:

- Disable unnecessary database extensions.
- Restrict remote administrative access.
- Use secure configuration defaults.
- Apply security patches regularly.
- Enable database activity logging.
- Disable unused accounts.
- Rotate credentials periodically.

7.9.9 Audit Logging

The database shall maintain security-relevant audit records.

Logged Event

Authentication attempts

Failed login attempts

Administrative actions

Privilege changes

Backup operations

Restore operations

Schema modifications

Logged Event

Sensitive data access

Audit logs shall be protected against unauthorized modification.

7.9.10 Network Security

Database servers shall be protected using network security controls.

These include:

- Firewall restrictions.
- Private network deployment.
- VPN or secure administrative access where required.
- IP allowlists for administrative connections.
- Network segmentation between application and database layers.

7.9.11 Security Monitoring

The system shall monitor:

- Unauthorized login attempts.
- Failed authentication events.
- Suspicious database activity.
- Privilege escalation attempts.
- Excessive query rates.
- Database availability.
- Storage utilization.

Critical events shall generate alerts for administrators.

7.9.12 Business Rules

Rule ID	Rule
DB-SEC-BR-001	Database credentials shall never be exposed in application source code.
DB-SEC-BR-002	Sensitive data shall be encrypted during transmission and storage where appropriate.
DB-SEC-BR-003	Users shall access only data belonging to their organization unless explicitly authorized.
DB-SEC-BR-004	Administrative database operations shall be audited.
DB-SEC-BR-005	Security logs shall be retained according to organizational retention policies.
DB-SEC-BR-006	Database security controls shall be reviewed periodically.
DB-SEC-BR-007	Security incidents shall be investigated and documented.

7.9.13 Exception Handling

Exception	Expected System Response
Invalid database credentials	Reject the connection attempt and log the event.
Unauthorized data access attempt	Deny access, generate an audit record, and notify administrators where appropriate.
TLS negotiation failure	Reject the connection and require a secure connection.
Database account lockout	Prevent authentication until administrative review or configured unlock conditions are met.
Security configuration error	Prevent deployment where possible and notify administrators.
Suspected compromise	Trigger incident response procedures and increase monitoring.

7.9.14 Acceptance Criteria

The database security requirements shall be considered complete when:

- All database connections are authenticated and encrypted.
- Database credentials are securely managed.
- Tenant isolation is enforced.
- Role-based access controls are implemented.
- Security-relevant events are audited.
- Database configurations follow secure defaults.

- Sensitive information is protected in transit and at rest.
- Security monitoring detects and reports critical events.

Chapter 8: REST API Specifications

8.1 API Overview

8.1.1 Purpose

The REST API provides a standardized interface for communication between the ProcessMind AI frontend, backend services, AI modules, and future third-party integrations.

The API enables secure access to system resources while enforcing authentication, authorization, validation, auditing, and tenant isolation.

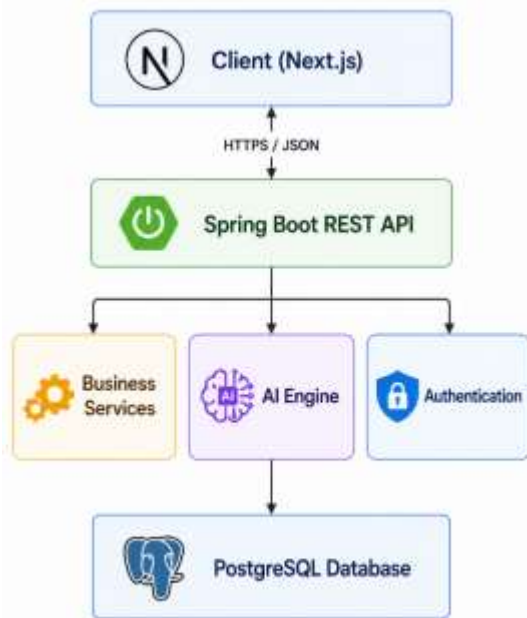
All APIs shall follow REST architectural principles and return structured JSON responses.

8.1.2 Objectives

The API shall:

- Enable secure client-server communication.
- Support CRUD operations.
- Support AI-powered business operations.
- Maintain stateless communication.
- Support integration with future systems.
- Enforce authorization policies.
- Maintain backward compatibility.

8.1.3 Architecture



8.1.4 API Principles

The API shall adhere to the following principles:

- RESTful resource design.
- Stateless requests.
- JSON request and response bodies.
- Standard HTTP methods.
- Consistent error responses.
- Secure authentication.
- API versioning.
- Idempotent operations where applicable.

8.1.5 Base URL

Environment	Base URL
Development	http://localhost:8080/api/v1
Staging	https://staging.processmind.ai/api/v1
Production	https://api.processmind.ai/api/v1

8.1.6 Supported HTTP Methods

Method	Purpose
GET	Retrieve resources
POST	Create resources
PUT	Replace resources
PATCH	Partial updates
DELETE	Remove resources (logical deletion where applicable)

8.1.7 Content Types

Type	Usage
application/json	Standard request and response format
multipart/form-data	File uploads (e.g., SOP documents)
application/pdf	Download reports
application/octet-stream	File downloads

8.1.8 API Versioning

The API shall use URI-based versioning.

Example:

/ api / v1 / workflows

/ api / v2 / workflows

New versions shall not break existing client applications unless explicitly documented.

8.1.9 Authentication

The API shall require authentication using **JWT (JSON Web Token)**.

Protected endpoints shall require the following header:

Authorization: Bearer <JWT_TOKEN>

Public endpoints (example: login and registration) shall not require authentication.

8.1.10 Authorization

Authorization shall be enforced using **Role-Based Access Control (RBAC)**.

Example roles:

- System Administrator
- Organization Administrator
- Process Manager
- Reviewer
- Approver
- Employee

Access to endpoints shall be restricted based on assigned roles and organization membership.

8.1.11 Standard Success Response

```
{
  "success": true,
  "message": "Workflow created successfully.",
  "data": {
    "...": "..."
  },
  "timestamp": "2026-06-13T10:30:00Z"
}
```

8.1.12 Standard Error Response

```
{
  "success": false,
  "errorCode": "WORKFLOW_NOT_FOUND",
  "message": "The requested workflow does not exist.",
  "timestamp": "2026-06-13T10:30:00Z"
}
```

8.1.13 Standard HTTP Status Codes

Status Code	Meaning
200 OK	Request successful
201 Created	Resource created
204 No Content	Successful request with no response body
400 Bad Request	Invalid request
401 Unauthorized	Authentication required
403 Forbidden	Insufficient permissions
404 Not Found	Resource not found
409 Conflict	Business rule conflict
422 Unprocessable Entity	Validation failed
429 Too Many Requests	Rate limit exceeded
500 Internal Server Error	Unexpected server error
503 Service Unavailable	Temporary service outage

8.1.14 Request Validation

All incoming requests shall be validated before processing.

Validation shall include:

- Required fields.
- Data types.
- Maximum lengths.
- Business rules.
- File size.
- File type.
- Tenant ownership verification.

Invalid requests shall return appropriate error responses.

8.1.15 API Security

The API shall implement:

- HTTPS-only communication.
- JWT authentication.
- RBAC authorization.
- Request validation.
- SQL injection prevention.
- Cross-Site Scripting (XSS) protection.
- Cross-Origin Resource Sharing (CORS) configuration.
- Rate limiting.
- Audit logging.
- Input sanitization.

8.1.16 Rate Limiting

To prevent abuse, rate limits shall be applied.

Endpoint Category	Default Limit
Authentication	10 requests/minute/IP
AI Processing	30 requests/minute/user
Standard API	120 requests/minute/user
File Upload	20 requests/hour/user

These limits shall be configurable by administrators.

8.1.17 Business Rules

Rule ID	Rule
API-BR-001	All protected endpoints shall require JWT authentication.
API-BR-002	Users shall only access resources belonging to their organization.
API-BR-003	All requests and responses shall use JSON unless file transfer is required.
API-BR-004	Validation errors shall return structured error responses.
API-BR-005	All API requests shall be logged for auditing where appropriate.
API-BR-006	API versioning shall preserve backward compatibility.
API-BR-007	Sensitive information shall never be exposed in API responses.

8.1.18 Acceptance Criteria

The REST API specification shall be considered complete when:

- All endpoints follow RESTful conventions.
- Authentication and authorization are enforced.
- Requests and responses follow standardized formats.
- Versioning supports backward compatibility.
- Validation protects the system from invalid input.
- API security measures are implemented.
- Rate limiting protects against abuse.
- API behavior is fully documented.

8.2 Authentication & Authorization APIs

8.2.1 Module Overview

The Authentication & Authorization API module provides secure user authentication, session management, password management, token management, and role-based authorization.

Authentication shall be implemented using **JWT (JSON Web Token)** with refresh tokens.

Passwords shall be securely hashed using **BCrypt** before storage.

All protected endpoints shall require a valid JWT access token.

8.2.2 API Endpoints

Method	Endpoint	Description	Authentication
POST	/auth/register	Register organization and administrator	No
POST	/auth/login	User login	No
POST	/auth/logout	Logout current user	Yes
POST	/auth/refresh-token	Refresh expired access token	Yes (Refresh Token)
POST	/auth/forgot-password	Request password reset	No
POST	/auth/reset-password	Reset password	No (Valid Reset Token)
POST	/auth/change-password	Change current password	Yes
GET	/auth/profile	Retrieve authenticated user profile	Yes
PATCH	/auth/profile	Update authenticated user profile	Yes
POST	/auth/verify-email	Verify email address	No
POST	/auth/resend-verification	Resend verification email	No

8.2.3 Register Organization

Endpoint

POST /api/v1/auth/register

Description

Creates a new organization along with its first **Organization Administrator** account.

Request Body

```
{
  "organizationName": "ABC Manufacturing",
  "organizationCode": "ABC001",
  "adminFirstName": "John",
  "adminLastName": "Doe",
  "email": "admin@abc.com",
  "password": "StrongPassword123!"
}
```

Successful Response

201 Created

```
{
  "success": true,
  "message": "Organization registered successfully."
}
```

Validation Rules

- Organization name is required.
- Organization code must be unique.
- Email must be unique.
- Password shall satisfy password policy.

Error Responses

HTTP Code	Reason
400	Invalid request
409	Organization already exists
422	Validation failed

8.2.4 Login

Endpoint

POST /api/v1/auth/login

Request

```
{
  "email": "admin@abc.com",
  "password": "StrongPassword123!"
}
```

Response

```
{
  "accessToken": "JWT_ACCESS_TOKEN",
  "refreshToken": "JWT_REFRESH_TOKEN",
  "expiresIn": 3600,
  "user": {
    "userId": "...",
    "firstName": "John",
    "lastName": "Doe",
    "role": "Organization Admin"
  }
}
```

Business Rules

- Account must be active.
- Email must be verified (if enabled).
- Failed login attempts shall be monitored.
- JWT access token shall be issued upon successful authentication.

8.2.5 Logout

Endpoint

POST /api/v1/auth/logout

Description

Terminates the current authenticated session.

Authorization

Bearer Token Required

Response

```
{
  "success": true,
  "message": "Logged out successfully."
}
```

8.2.6 Refresh Token

Endpoint

POST /api/v1/auth/refresh-token

Purpose

Generates a new access token using a valid refresh token.

Response

```
{
  "accessToken": "...",
  "expiresIn": 3600
}
```

8.2.7 Forgot Password

Endpoint

POST /api/v1/auth/forgot-password

Request

```
{
  "email": "admin@abc.com"
}
```

Response

```
{
  "success": true,
  "message": "Password reset instructions have been sent if the account exists."
}
```

Security Note: The response shall not reveal whether the email address exists in the system.

8.2.8 Reset Password

Endpoint

POST /api/v1/auth/reset-password

Request

```
{
  "token": "RESET_TOKEN",
  "newPassword": "NewPassword123!"
}
```

Business Rules

- Reset token must be valid.
- Reset token must not be expired.
- New password must satisfy password policy.

8.2.9 Change Password

Endpoint

POST /api/v1/auth/change-password

Authorization

Bearer Token Required

Request

```
{
  "currentPassword": "OldPassword123!",
  "newPassword": "NewPassword123!"
}
```

Response

```
{
  "success": true,
  "message": "Password changed successfully."
}
```

8.2.10 Get Profile

Endpoint

GET /api/v1/auth/profile

Authorization

Bearer Token Required

Response

```
{
  "userId": "...",
  "firstName": "John",
  "lastName": "Doe",
  "email": "admin@abc.com",
  "organization": "ABC Manufacturing",
  "role": "Organization Admin"
}
```

8.2.11 Update Profile

Endpoint

PATCH /api/v1/auth/profile

Request

```
{
  "firstName": "John",
  "lastName": "Smith",
  "phone": "+94XXXXXXXXX"
}
```

Response

```
{
  "success": true,
  "message": "Profile updated successfully."
}
```

8.2.12 Verify Email

Endpoint

POST /api/v1/auth/verify-email

Purpose

Verifies a user's email address using a verification token.

8.2.13 Resend Verification Email

Endpoint

POST /api/v1/auth/resend-verification

Purpose

Sends a new verification email to an unverified user.

8.2.14 Authentication Flow



8.2.15 Security Requirements

- Passwords shall be hashed using BCrypt.
- JWT tokens shall be digitally signed.

- Refresh tokens shall be securely stored.
- Login attempts shall be rate-limited.
- Failed login attempts shall be logged.
- Password reset tokens shall expire after a configurable period (example: 15–30 minutes).
- Authentication events shall generate audit log entries.

8.2.16 Business Rules

Rule ID	Rule
AUTH-BR-001	Email addresses shall be unique across the system.
AUTH-BR-002	Passwords shall never be stored in plain text.
AUTH-BR-003	Protected APIs shall require a valid JWT access token.
AUTH-BR-004	Users shall only access resources belonging to their organization.
AUTH-BR-005	Authentication events shall be recorded in the audit log.
AUTH-BR-006	User accounts may be temporarily locked after repeated failed login attempts.
AUTH-BR-007	Refresh tokens shall be invalidated on logout or explicit revocation.

8.2.17 Error Responses

HTTP Status	Error Code	Description
400	INVALID_REQUEST	Request payload is invalid.
401	INVALID_CREDENTIALS	Incorrect email or password.
401	TOKEN_EXPIRED	Access token has expired.
401	INVALID_TOKEN	JWT or refresh token is invalid.
403	ACCOUNT_LOCKED	User account is temporarily locked.
403	EMAIL_NOT_VERIFIED	Email verification is required.
404	USER_NOT_FOUND	Requested user does not exist.
409	DUPLICATE_EMAIL	Email address is already registered.
422	VALIDATION_FAILED	Input validation failed.

8.2.18 Acceptance Criteria

The Authentication & Authorization API module shall be considered complete when:

- Users can securely register and authenticate.
- JWT access and refresh token workflows function correctly.
- Password management features operate securely.
- Role-based authorization is enforced.
- Authentication events are audited.
- Account security measures (rate limiting, lockout, password hashing) are implemented.
- All endpoints return standardized success and error responses.

8.3 Organization Management APIs

8.3.1 Module Overview

The Organization Management API enables administrators to manage organizations, departments, organization settings, subscription information, and organizational preferences.

These APIs support the multi-tenant architecture of ProcessMind AI by ensuring that each organization operates within an isolated environment while allowing authorized administrators to manage organization-specific resources.

Only authorized users with the appropriate administrative privileges shall access these endpoints.

8.3.2 API Endpoints

Organization APIs

Method	Endpoint	Description	Authorization
GET	/organizations	Retrieve organizations (System Admin only)	System Admin
GET	/organizations/{organizationId}	Retrieve organization details	Organization Admin / System Admin
POST	/organizations	Create a new organization	System Admin

PUT	/organizations/{organizationId}	Update organization information	Organization Admin
DELETE	/organizations/{organizationId}	Soft delete organization	System Admin
PATCH	/organizations/{organizationId}/status	Activate or suspend organization	System Admin

Department APIs

Method	Endpoint	Description	Authorization
GET	/departments	List departments	Organization Admin
GET	/departments/{departmentId}	Retrieve department details	Organization Admin
POST	/departments	Create department	Organization Admin
PUT	/departments/{departmentId}	Update department	Organization Admin
DELETE	/departments/{departmentId}	Delete department	Organization Admin

Organization Settings APIs

Method	Endpoint	Description	Authorization
GET	/organization-settings	Retrieve organization settings	Organization Admin
PUT	/organization-settings	Update organization settings	Organization Admin
GET	/organization-settings/ai	Retrieve AI configuration	Organization Admin
PUT	/organization-settings/ai	Update AI configuration	Organization Admin
GET	/organization-settings/security	Retrieve security settings	Organization Admin
PUT	/organization-settings/security	Update security settings	Organization Admin

Subscription APIs

Method	Endpoint	Description	Authorization
GET	/subscriptions	Retrieve subscription information	Organization Admin
PUT	/subscriptions	Update subscription information	System Admin
GET	/subscriptions/usage	Retrieve current usage statistics	Organization Admin

8.3.3 Request Validation

The Organization Management APIs shall validate:

- Organization name.
- Organization code uniqueness.
- Department names.
- Organization ownership.
- Subscription limits.
- Organization status.
- User permissions.

8.3.4 Business Rules

Rule ID	Rule
ORG-BR-001	Each organization shall have a unique organization code.
ORG-BR-002	Users shall access only their own organization's resources unless they are System Administrators.
ORG-BR-003	Organizations with active workflows shall not be permanently deleted.
ORG-BR-004	Department names shall be unique within an organization.
ORG-BR-005	Organization status changes shall be recorded in the audit log.
ORG-BR-006	Organization settings shall only be modified by Organization Administrators or System Administrators.
ORG-BR-007	Subscription limits shall be enforced before creating new users, workflows, or AI processing tasks where applicable.

8.3.5 Error Responses

HTTP Status	Error Code	Description
400	INVALID_REQUEST	Invalid request payload.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	Insufficient permissions.
404	ORGANIZATION_NOT_FOUND	Organization does not exist.
409	DUPLICATE_ORGANIZATION_CODE	Organization code already exists.
422	VALIDATION_FAILED	Validation rules failed.

8.3.6 Acceptance Criteria

The Organization Management API module shall be considered complete when:

- Organizations can be created, updated, retrieved, and managed securely.
- Departments can be managed within their respective organizations.
- Organization settings are configurable by authorized administrators.
- Subscription information and usage statistics are accessible.
- Multi-tenant isolation is enforced.
- All operations are validated and audited.

8.4 User & Role Management APIs

8.4.1 Module Overview

The User & Role Management API provides secure management of users, roles, permissions, departments, and access rights within an organization.

The module enables administrators to create and manage user accounts, assign organizational roles, configure permissions, and enforce Role-Based Access Control (RBAC).

All operations shall be restricted according to the authenticated user's privileges and organizational scope.

8.4.2 User Management APIs

Method	Endpoint	Description	Authorization
GET	/users	Retrieve all users	Organization Admin
GET	/users/{userId}	Retrieve user details	Organization Admin
POST	/users	Create a new user	Organization Admin
PUT	/users/{userId}	Update user information	Organization Admin
PATCH	/users/{userId}/status	Activate, suspend, or deactivate a user	Organization Admin

DELETE	/users/{userId}	Soft delete a user	Organization Admin
GET	/users/search	Search users	Organization Admin
GET	/users/me	Retrieve current authenticated user	Authenticated User
PATCH	/users/me	Update own profile	Authenticated User
POST	/users/bulk-import	Import users from CSV/Excel	Organization Admin
GET	/users/export	Export users	Organization Admin

8.4.3 Role Management APIs

Method	Endpoint	Description	Authorization
GET	/roles	Retrieve available roles	Organization Admin
GET	/roles/{roleId}	Retrieve role details	Organization Admin
POST	/roles	Create custom role	Organization Admin
PUT	/roles/{roleId}	Update role	Organization Admin
DELETE	/roles/{roleId}	Delete custom role	Organization Admin
POST	/roles/{roleId}/clone	Clone an existing role	Organization Admin

8.4.4 Permission Management APIs

Method	Endpoint	Description	Authorization
GET	/permissions	Retrieve all permissions	Organization Admin
GET	/roles/{roleId}/permissions	Retrieve permissions assigned to a role	Organization Admin
PUT	/roles/{roleId}/permissions	Update role permissions	Organization Admin

8.4.5 User Role Assignment APIs

Method	Endpoint	Description	Authorization
POST	/users/{userId}/roles	Assign role to user	Organization Admin
DELETE	/users/{userId}/roles/{roleId}	Remove role from user	Organization Admin
GET	/users/{userId}/roles	Retrieve assigned roles	Organization Admin

8.4.6 Department Assignment APIs

Method	Endpoint	Description	Authorization
POST	/users/{userId}/department	Assign user to department	Organization Admin
PATCH	/users/{userId}/department	Transfer user to another department	Organization Admin
DELETE	/users/{userId}/department	Remove department assignment	Organization Admin

8.4.7 User Search & Filtering APIs

The API shall support searching and filtering users by:

- Name
- Email
- Department
- Role
- Status
- Organization
- Created Date

Example:

GET /users?role = Reviewer&status = ACTIVE&page = 1&size = 20

Pagination, sorting, and filtering shall be supported for all list endpoints.

8.4.8 Request Validation

The User & Role Management APIs shall validate:

- Unique email addresses.
- Required user information.
- Valid department assignment.
- Valid role assignment.
- Existing permissions.
- Organization ownership.

- Password policy (where applicable).

8.4.9 Business Rules

Rule ID	Rule
USER-BR-001	User email addresses shall be unique within the system.
USER-BR-002	Users shall belong to exactly one organization.
USER-BR-003	Users may belong to one department at a time.
USER-BR-004	Only authorized administrators may create or modify user accounts.
USER-BR-005	Role assignments shall comply with RBAC policies.
USER-BR-006	System-defined roles shall not be deleted.
USER-BR-007	Deleted users shall be soft-deleted to preserve audit history.
USER-BR-008	Bulk user imports shall validate all records before processing.
USER-BR-009	Changes to roles and permissions shall be recorded in the audit log.
USER-BR-010	Users shall not access resources outside their organization.

8.4.10 Error Responses

HTTP Status	Error Code	Description
400	INVALID_REQUEST	Invalid request payload.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	Insufficient permissions.
404	USER_NOT_FOUND	Requested user does not exist.
404	ROLE_NOT_FOUND	Requested role does not exist.
404	DEPARTMENT_NOT_FOUND	Requested department does not exist.
409	DUPLICATE_EMAIL	Email address already exists.
409	ROLE_ALREADY_ASSIGNED	Role is already assigned to the user.
422	VALIDATION_FAILED	Request validation failed.

8.4.11 Acceptance Criteria

The User & Role Management API module shall be considered complete when:

- Users can be created, updated, searched, and managed.
- Departments and roles can be assigned appropriately.
- Permissions are managed through RBAC.
- Bulk import and export operations function correctly.
- Multi-tenant isolation is enforced.
- All administrative operations are audited.
- Search, filtering, pagination, and sorting work consistently across endpoints.

8.5 SOP Management APIs

8.5.1 Module Overview

The SOP Management API provides endpoints for uploading, managing, versioning, analyzing, approving, searching, and archiving Standard Operating Procedures (SOPs).

The module serves as the entry point to the AI-powered process intelligence pipeline. Uploaded SOPs are securely stored, version-controlled, and processed by the AI engine to extract business knowledge and generate workflow definitions.

The module supports document lifecycle management while maintaining complete version history and auditability.

8.5.2 SOP Management APIs

Method	Endpoint	Description	Authorization
GET	/sops	Retrieve all SOPs	Organization User
GET	/sops/{sopId}	Retrieve SOP details	Organization User
POST	/sops	Upload a new SOP	Process Manager
PUT	/sops/{sopId}	Update SOP metadata	Process Manager
DELETE	/sops/{sopId}	Archive or soft delete SOP	Organization Admin
GET	/sops/search	Search SOPs	Organization User
GET	/sops/filter	Filter SOPs	Organization User

8.5.3 SOP Version APIs

Method	Endpoint	Description	Authorization
GET	/sops/{sopId}/versions	Retrieve all SOP versions	Organization User
GET	/sops/{sopId}/versions/{versionId}	Retrieve a specific version	Organization User
POST	/sops/{sopId}/versions	Upload a new SOP version	Process Manager
POST	/sops/{sopId}/versions/{versionId}/restore	Restore a previous version	Process Manager

8.5.4 AI Processing APIs

Method	Endpoint	Description	Authorization
POST	/sops/{sopId}/analyze	Analyze SOP using AI	Process Manager
GET	/sops/{sopId}/analysis	Retrieve AI analysis	Organization User
POST	/sops/{sopId}/reanalyze	Re-run AI analysis	Process Manager
GET	/sops/{sopId}/confidence	Retrieve AI confidence scores	Organization User
GET	/sops/{sopId}/validation	Retrieve AI validation results	Organization User

8.5.5 SOP Comparison APIs

Method	Endpoint	Description	Authorization
GET	/sops/{sopId}/compare/{versionId}	Compare SOP versions	Organization User
GET	/sops/{sopId}/changes	View version history and changes	Organization User

8.5.6 SOP Approval APIs

Method	Endpoint	Description	Authorization
POST	/sops/{sopId}/submit	Submit SOP for approval	Process Manager
POST	/sops/{sopId}/approve	Approve SOP	Approver
POST	/sops/{sopId}/reject	Reject SOP	Approver

GET	/sops/pending-approvals	Retrieve pending SOP approvals	Approver
-----	-------------------------	--------------------------------	----------

8.5.7 SOP Archive APIs

Method	Endpoint	Description	Authorization
POST	/sops/{sopId}/archive	Archive SOP	Organization Admin
POST	/sops/{sopId}/restore	Restore archived SOP	Organization Admin
GET	/sops/archived	Retrieve archived SOPs	Organization Admin

8.5.8 Document Upload Requirements

The API shall support:

Feature	Requirement
Supported Formats	PDF, DOCX, TXT
Maximum File Size	Configurable (Default: 25 MB)
Virus Scanning	Required before storage
Duplicate Detection	Optional
Version Control	Automatic
Metadata Extraction	Automatic

8.5.9 Search & Filtering

The API shall support searching by:

- SOP Title
- Department
- Owner
- Status
- Version

- Upload Date
- Tags
- Keywords

Pagination and sorting shall be available for all list operations.

8.5.10 Business Rules

Rule ID	Rule
SOP-BR-001	Every SOP shall belong to one organization.
SOP-BR-002	Every uploaded SOP shall create an initial version.
SOP-BR-003	Previous SOP versions shall remain immutable.
SOP-BR-004	Only authorized users may upload or modify SOPs.
SOP-BR-005	AI analysis shall only begin after successful document validation.
SOP-BR-006	Unsupported document formats shall be rejected.
SOP-BR-007	Every SOP modification shall generate an audit log entry.
SOP-BR-008	Archived SOPs shall remain available for authorized retrieval.
SOP-BR-009	AI-generated analysis shall be associated with the specific SOP version analyzed.
SOP-BR-010	Organization users shall only access SOPs belonging to their organization.

8.5.11 Error Responses

HTTP Status	Error Code	Description
400	INVALID_FILE	Unsupported or malformed document.
400	FILE_TOO_LARGE	File exceeds the configured size limit.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	Insufficient permissions.
404	SOP_NOT_FOUND	SOP does not exist.
409	DUPLICATE_VERSION	Duplicate SOP version detected.
415	UNSUPPORTED_MEDIA_TYPE	Unsupported file format.
422	AI_ANALYSIS_FAILED	AI processing could not complete successfully.

8.5.12 Acceptance Criteria

The SOP Management API module shall be considered complete when:

- SOPs can be uploaded, versioned, searched, and archived.
- AI analysis can be initiated and retrieved.
- SOP versions are immutable and restorable.
- Approval workflows function correctly.
- Search and filtering operate efficiently.
- Audit logs record all significant operations.
- Multi-tenant access restrictions are enforced.

8.6 Workflow Management APIs

8.6.1 Module Overview

The Workflow Management API provides endpoints for creating, generating, editing, validating, simulating, deploying, publishing, importing, exporting, and managing workflows.

The module enables both AI-generated and manually designed workflows while maintaining version control, validation, execution readiness, and auditability.

Every workflow shall progress through a controlled lifecycle from creation to retirement.

8.6.2 Workflow Lifecycle



8.6.3 Workflow Management APIs

Method	Endpoint	Description	Authorization
GET	/workflows	Retrieve workflows	Organization User
GET	/workflows/{workflowId}	Retrieve workflow details	Organization User
POST	/workflows	Create workflow manually	Process Manager
PUT	/workflows/{workflowId}	Update workflow	Process Manager
DELETE	/workflows/{workflowId}	Archive workflow	Organization Admin
GET	/workflows/search	Search workflows	Organization User
GET	/workflows/filter	Filter workflows	Organization User

8.6.4 AI Workflow Generation APIs

Method	Endpoint	Description	Authorization
POST	/workflows/generate	Generate workflow from SOP	Process Manager
POST	/workflows/{workflowId}/regenerate	Regenerate workflow using AI	Process Manager
GET	/workflows/{workflowId}/reasoning	Retrieve AI reasoning report	Organization User
GET	/workflows/{workflowId}/confidence	Retrieve AI confidence scores	Organization User
GET	/workflows/{workflowId}/validation	Retrieve AI validation report	Organization User

8.6.5 Workflow Editing APIs

Method	Endpoint	Description	Authorization
PATCH	/workflows/{workflowId}/nodes	Add or modify workflow nodes	Process Manager
PATCH	/workflows/{workflowId}/edges	Add or modify workflow connections	Process Manager
PATCH	/workflows/{workflowId}/business-rules	Update business rules	Process Manager
PATCH	/workflows/{workflowId}/approvals	Configure approval steps	Process Manager
PATCH	/workflows/{workflowId}/settings	Update workflow settings	Process Manager

8.6.6 Workflow Version APIs

Method	Endpoint	Description	Authorization
GET	/workflows/{workflowId}/versions	Retrieve workflow versions	Organization User
POST	/workflows/{workflowId}/versions	Create workflow version	Process Manager
GET	/workflows/{workflowId}/versions/{versionId}	Retrieve version details	Organization User
POST	/workflows/{workflowId}/versions/{versionId}/restore	Restore workflow version	Process Manager
GET	/workflows/{workflowId}/history	Retrieve workflow history	Organization User

8.6.7 Workflow Validation APIs

Method	Endpoint	Description	Authorization
POST	/workflows/{workflowId}/validate	Validate workflow	Process Manager
GET	/workflows/{workflowId}/validation-report	Retrieve validation report	Organization User
GET	/workflows/{workflowId}/errors	Retrieve validation errors	Organization User

GET	/workflows/{workflowId}/warnings	Retrieve validation warnings	Organization User
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8.6.8 Workflow Simulation APIs

Method	Endpoint	Description	Authorization
POST	/workflows/{workflowId}/simulate	Run workflow simulation	Process Manager
GET	/workflows/{workflowId}/simulation	Retrieve simulation results	Organization User
GET	/workflows/{workflowId}/bottlenecks	Retrieve detected bottlenecks	Organization User
GET	/workflows/{workflowId}/performance	Retrieve predicted performance metrics	Organization User

8.6.9 Workflow Deployment APIs

Method	Endpoint	Description	Authorization
POST	/workflows/{workflowId}/deploy	Deploy workflow	Organization Admin
POST	/workflows/{workflowId}/undeploy	Undeploy workflow	Organization Admin
GET	/workflows/deployed	Retrieve deployed workflows	Organization User
GET	/workflows/{workflowId}/deployment-status	Retrieve deployment status	Organization User

8.6.10 Workflow Import & Export APIs

Method	Endpoint	Description	Authorization
POST	/workflows/import	Import workflow	Process Manager

GET	/workflows/{workflowId}/export/json	Export as JSON	Organization User
GET	/workflows/{workflowId}/export/pdf	Export documentation	Organization User
GET	/workflows/{workflowId}/export/bpmn	Export BPMN 2.0	Organization User

8.6.11 Workflow Search & Filtering

The API shall support searching by:

- Workflow Name
- SOP
- Department
- Status
- Owner
- Version
- Tags
- Creation Date
- Deployment Status

Pagination, filtering, and sorting shall be supported for all list endpoints.

8.6.12 Business Rules

Rule ID	Rule
WF-BR-001	Every workflow shall belong to one organization.
WF-BR-002	Every workflow shall be associated with an SOP.
WF-BR-003	Workflow versions shall be immutable after deployment.
WF-BR-004	Workflows shall pass validation before deployment.
WF-BR-005	Simulation shall not modify production workflow data.
WF-BR-006	AI-generated workflows shall support manual editing before deployment.
WF-BR-007	Workflow deployment shall generate audit log entries.
WF-BR-008	Only one active version of a workflow may be deployed at a time.
WF-BR-009	Archived workflows shall not accept new executions.

WF-BR-010	Users shall only access workflows belonging to their organization.
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8.6.13 Error Responses

HTTP Status	Error Code	Description
400	INVALID_WORKFLOW	Invalid workflow definition.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	Insufficient permissions.
404	WORKFLOW_NOT_FOUND	Workflow does not exist.
409	WORKFLOW_ALREADY_DEPLOYED	Active deployment already exists.
409	VALIDATION_FAILED	Workflow contains validation errors.
422	SIMULATION_FAILED	Workflow simulation failed.
500	DEPLOYMENT_ERROR	Unexpected deployment failure.

8.6.14 Acceptance Criteria

The Workflow Management API module shall be considered complete when:

- Workflows can be created manually and generated by AI.
- Version control preserves workflow history.
- Validation identifies structural and business-rule issues.
- Simulations predict workflow behavior without affecting production data.
- Deployment enforces validation and approval requirements.
- Import and export operations support standard formats.
- Workflow lifecycle transitions are controlled and auditable.
- Multi-tenant access restrictions are enforced.

8.7 Workflow Execution APIs

8.7.1 Module Overview

The Workflow Execution API provides endpoints for executing deployed workflows, managing workflow instances, assigning and completing tasks, processing approvals, handling workflow exceptions, monitoring execution status, and maintaining execution history.

The module enables organizations to execute business processes consistently while ensuring traceability, accountability, and real-time monitoring.

Only deployed workflows may be executed.

8.7.2 Workflow Execution Lifecycle



8.7.3 Workflow Instance APIs

Method	Endpoint	Description	Authorization
POST	/workflow-instances	Start a new workflow instance	Organization User
GET	/workflow-instances	Retrieve workflow instances	Organization User
GET	/workflow-instances/{instanceId}	Retrieve workflow instance details	Organization User
PATCH	/workflow-instances/{instanceId}	Update workflow instance metadata	Process Manager

DELETE	/workflow-instances/{instanceId}	Cancel workflow instance	Process Manager
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8.7.4 Task Management APIs

Method	Endpoint	Description	Authorization
GET	/tasks	Retrieve assigned tasks	Authenticated User
GET	/tasks/{taskId}	Retrieve task details	Task Assignee
PATCH	/tasks/{taskId}/claim	Claim an unassigned task	Authorized User
PATCH	/tasks/{taskId}/start	Start task execution	Task Assignee
PATCH	/tasks/{taskId}/complete	Complete task	Task Assignee
PATCH	/tasks/{taskId}/reassign	Reassign task	Process Manager
PATCH	/tasks/{taskId}/delegate	Delegate task	Process Manager
PATCH	/tasks/{taskId}/cancel	Cancel task	Process Manager

8.7.5 Approval APIs

Method	Endpoint	Description	Authorization
GET	/approvals	Retrieve pending approvals	Approver
GET	/approvals/{approvalId}	Retrieve approval details	Approver
POST	/approvals/{approvalId}/approve	Approve request	Approver
POST	/approvals/{approvalId}/reject	Reject request	Approver
POST	/approvals/{approvalId}/request-changes	Request modifications	Approver
GET	/approvals/history	Retrieve approval history	Process Manager

8.7.6 Workflow Control APIs

Method	Endpoint	Description	Authorization
PATCH	/workflow-instances/{instanceId}/pause	Pause workflow execution	Process Manager
PATCH	/workflow-instances/{instanceId}/resume	Resume workflow execution	Process Manager
PATCH	/workflow-instances/{instanceId}/terminate	Terminate workflow	Organization Admin

PATCH	/workflow-instances/{instanceId}/restart	Restart workflow	Process Manager
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8.7.7 Workflow Monitoring APIs

Method	Endpoint	Description	Authorization
GET	/workflow-instances/{instanceId}/status	Retrieve execution status	Organization User
GET	/workflow-instances/{instanceId}/timeline	Retrieve execution timeline	Organization User
GET	/workflow-instances/{instanceId}/activities	Retrieve executed activities	Organization User
GET	/workflow-instances/{instanceId}/logs	Retrieve execution logs	Process Manager

8.7.8 Exception Management APIs

Method	Endpoint	Description	Authorization
GET	/workflow-instances/{instanceId}/exceptions	Retrieve execution exceptions	Process Manager
POST	/workflow-instances/{instanceId}/retry	Retry failed execution step	Process Manager
POST	/workflow-instances/{instanceId}/skip-task	Skip task (authorized only)	Organization Admin
POST	/workflow-instances/{instanceId}/manual-intervention	Trigger manual intervention	Process Manager

8.7.9 Execution History APIs

Method	Endpoint	Description	Authorization
GET	/workflow-history	Retrieve execution history	Organization User
GET	/workflow-history/{instanceId}	Retrieve execution history details	Organization User
GET	/workflow-history/export	Export execution history	Organization Admin

8.7.10 Search & Filtering

Workflow execution APIs shall support searching and filtering by:

- Workflow Name
- Workflow Instance ID

- Status
- Assigned User
- Department
- Priority
- Due Date
- Start Date
- Completion Date
- Approval Status

Pagination, sorting, and filtering shall be supported.

8.7.11 Business Rules

Rule ID	Rule
EXEC-BR-001	Only deployed workflows may be executed.
EXEC-BR-002	Each workflow instance shall reference a deployed workflow version.
EXEC-BR-003	Tasks shall only be completed by authorized assignees or delegated users.
EXEC-BR-004	Workflow execution history shall be immutable.
EXEC-BR-005	Approval decisions shall be permanently recorded.
EXEC-BR-006	Paused workflows shall not continue until resumed.
EXEC-BR-007	Cancelled workflows shall not resume without an explicit restart action.
EXEC-BR-008	Every execution event shall generate an audit log entry.
EXEC-BR-009	Manual interventions shall require appropriate authorization and justification.
EXEC-BR-010	Users shall only access workflow instances belonging to their organization.

8.7.12 Error Responses

HTTP Status	Error Code	Description
400	INVALID_STATE	Workflow is not in a valid state for the requested operation.
401	UNAUTHORIZED	Authentication required.

403	FORBIDDEN	User lacks the required permissions.
404	WORKFLOW_INSTANCE_NOT_FOUND	Workflow instance does not exist.
404	TASK_NOT_FOUND	Task does not exist.
404	APPROVAL_NOT_FOUND	Approval request does not exist.
409	TASK_ALREADY_COMPLETED	Task has already been completed.
409	WORKFLOW_PAUSED	Workflow is currently paused.
422	APPROVAL_REQUIRED	Required approval has not been completed.

8.7.13 Acceptance Criteria

The Workflow Execution API module shall be considered complete when:

- Deployed workflows can be executed successfully.
- Workflow instances and tasks are created and managed correctly.
- Approval processes function according to configured workflows.
- Execution state transitions (pause, resume, terminate, restart) operate correctly.
- Execution history is preserved and immutable.
- Exceptions can be managed through controlled recovery mechanisms.
- Audit logs capture all execution-related events.
- Multi-tenant access restrictions are enforced.

8.8 AI Processing APIs

8.8.1 Module Overview

The AI Processing API provides intelligent services for document understanding, workflow generation, reasoning, validation, confidence evaluation, optimization, Human-in-the-Loop (HITL) review, AI monitoring, and AI provider management.

These APIs orchestrate communication between the application and supported AI providers while ensuring secure, auditable, and reliable AI-assisted business process automation.

All AI operations shall be logged, versioned, and traceable.

8.8.2 AI Service Categories

The AI Processing APIs are organized into the following service groups:

- SOP Understanding
- Workflow Generation
- AI Reasoning
- AI Validation
- Confidence Evaluation
- Human-in-the-Loop Review
- AI Optimization
- AI Monitoring
- AI Provider Management
- Prompt Management

8.8.3 SOP Understanding APIs

Method	Endpoint	Description	Authorization
POST	/ai/sop/analyze	Analyze uploaded SOP	Process Manager
GET	/ai/sop/{analysisId}	Retrieve analysis results	Organization User
POST	/ai/sop/reanalyze	Re-run SOP analysis	Process Manager
GET	/ai/sop/{analysisId}/entities	Retrieve extracted entities	Organization User
GET	/ai/sop/{analysisId}/business-rules	Retrieve extracted business rules	Organization User

8.8.4 Workflow Generation APIs

Method	Endpoint	Description	Authorization
POST	/ai/workflows/generate	Generate workflow from SOP	Process Manager
POST	/ai/workflows/regenerate	Regenerate workflow	Process Manager
GET	/ai/workflows/{generationId}	Retrieve generated workflow	Organization User
GET	/ai/workflows/{generationId}/reasoning	Retrieve AI reasoning	Organization User

8.8.5 AI Validation APIs

Method	Endpoint	Description	Authorization
POST	/ai/validate	Validate AI-generated workflow	Process Manager

GET	/ai/validation/{validationId}	Retrieve validation report	Organization User
GET	/ai/validation/{validationId}/errors	Retrieve validation errors	Organization User
GET	/ai/validation/{validationId}/warnings	Retrieve validation warnings	Organization User

8.8.6 Confidence Evaluation APIs

Method	Endpoint	Description	Authorization
GET	/ai/confidence/{analysisId}	Retrieve confidence scores	Organization User
GET	/ai/confidence/{analysisId}/details	Retrieve confidence breakdown	Organization User

8.8.7 Human-in-the-Loop (HITL) APIs

Method	Endpoint	Description	Authorization
GET	/ai/reviews/pending	Retrieve AI outputs awaiting review	Reviewer
GET	/ai/reviews/{reviewId}	Retrieve review details	Reviewer
POST	/ai/reviews/{reviewId}/approve	Approve AI output	Reviewer
POST	/ai/reviews/{reviewId}/reject	Reject AI output	Reviewer
POST	/ai/reviews/{reviewId}/request-changes	Request AI regeneration or manual edits	Reviewer

8.8.8 AI Optimization APIs

Method	Endpoint	Description	Authorization
POST	/ai/optimize	Generate workflow optimization suggestions	Process Manager
GET	/ai/optimizations/{optimizationId}	Retrieve optimization results	Organization User

POST	/ai/optimizations/{optimizationId}/apply	Apply accepted optimization	Process Manager
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8.8.9 AI Monitoring APIs

Method	Endpoint	Description	Authorization
GET	/ai/metrics	Retrieve AI performance metrics	Organization Admin
GET	/ai/usage	Retrieve AI usage statistics	Organization Admin
GET	/ai/logs	Retrieve AI processing logs	Organization Admin
GET	/ai/failures	Retrieve AI failure history	Organization Admin
GET	/ai/providers/status	Retrieve AI provider health	Organization Admin

8.8.10 Prompt Management APIs

Method	Endpoint	Description	Authorization
GET	/ai/prompts	Retrieve prompt templates	Organization Admin
POST	/ai/prompts	Create prompt template	Organization Admin
PUT	/ai/prompts/{promptId}	Update prompt template	Organization Admin
GET	/ai/prompts/{promptId}/versions	Retrieve prompt version history	Organization Admin

8.8.11 AI Provider Management APIs

Method	Endpoint	Description	Authorization
GET	/ai/providers	List configured AI providers	System Admin
POST	/ai/providers	Register AI provider	System Admin
PATCH	/ai/providers/{providerId}	Update provider configuration	System Admin
DELETE	/ai/providers/{providerId}	Remove AI provider	System Admin
POST	/ai/providers/{providerId}/test	Test provider connectivity	System Admin

8.8.12 Business Rules

Rule ID	Rule
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AI-BR-001	All AI requests shall be authenticated and authorized.
AI-BR-002	AI operations shall be associated with a specific organization and user.
AI-BR-003	Every AI request and response shall be logged for auditing.
AI-BR-004	AI-generated workflows shall undergo validation before deployment.
AI-BR-005	Confidence scores shall be available for AI-generated outputs.
AI-BR-006	Human review shall be required when confidence falls below configurable thresholds.
AI-BR-007	AI optimization suggestions shall require user approval before application.
AI-BR-008	AI provider failures shall trigger recovery and retry mechanisms where appropriate.
AI-BR-009	Prompt templates shall be version-controlled.
AI-BR-010	Sensitive organizational data shall not be exposed outside configured AI processing policies.

8.8.13 Error Responses

HTTP Status	Error Code	Description
400	INVALID_AI_REQUEST	AI request payload is invalid.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	User lacks permission.
404	ANALYSIS_NOT_FOUND	Requested analysis does not exist.
409	AI_PROVIDER_UNAVAILABLE	Configured AI provider is unavailable.
422	AI_VALIDATION_FAILED	AI output failed validation.
429	AI_RATE_LIMIT_EXCEEDED	AI request rate limit exceeded.
500	AI_PROCESSING_ERROR	Unexpected AI processing failure.
503	AI_SERVICE_UNAVAILABLE	AI service temporarily unavailable.

8.8.14 Acceptance Criteria

The AI Processing API module shall be considered complete when:

- SOPs can be analyzed successfully.

- AI-generated workflows can be created and retrieved.
- Validation and confidence evaluation operate correctly.
- Human-in-the-Loop review workflows are supported.
- AI optimization recommendations can be generated and applied.
- AI provider configurations can be managed securely.
- AI usage, monitoring, and failure data are available.
- All AI activities are logged and auditable.

8.9 Analytics & Reporting APIs

8.9.1 Module Overview

The Analytics & Reporting API provides endpoints for retrieving operational metrics, workflow performance, AI analytics, organizational KPIs, dashboards, reports, and predictive insights.

The module enables organizations to monitor business processes, evaluate AI performance, identify bottlenecks, measure productivity, and make data-driven decisions.

All analytics shall be generated using organization-specific data while maintaining tenant isolation and auditability.

8.9.2 Analytics Categories

The platform shall provide analytics in the following categories:

- Executive Dashboard
- Workflow Performance
- AI Performance
- Task Performance
- User Productivity
- Approval Analytics
- SLA Compliance
- Resource Utilization
- Organization KPIs
- Predictive Analytics

8.9.3 Dashboard APIs

Method	Endpoint	Description	Authorization
GET	/analytics/dashboard	Retrieve executive dashboard	Organization Admin
GET	/analytics/dashboard/workflows	Workflow dashboard	Process Manager
GET	/analytics/dashboard/ai	AI dashboard	Organization Admin
GET	/analytics/dashboard/tasks	Task dashboard	Process Manager
GET	/analytics/dashboard/users	User productivity dashboard	Organization Admin

8.9.4 Workflow Analytics APIs

Method	Endpoint	Description	Authorization
GET	/analytics/workflows	Workflow performance summary	Process Manager
GET	/analytics/workflows/{workflowId}	Workflow performance details	Process Manager
GET	/analytics/workflows/bottlenecks	Detect workflow bottlenecks	Process Manager
GET	/analytics/workflows/trends	Workflow execution trends	Organization Admin
GET	/analytics/workflows/comparison	Compare workflows	Organization Admin

8.9.5 AI Analytics APIs

Method	Endpoint	Description	Authorization
GET	/analytics/ai	AI performance summary	Organization Admin
GET	/analytics/ai/providers	Compare AI providers	System Admin
GET	/analytics/ai/models	Compare AI models	System Admin
GET	/analytics/ai/confidence	Confidence score analytics	Organization Admin
GET	/analytics/ai/failures	AI failure analytics	Organization Admin

8.9.6 Task & User Analytics APIs

Method	Endpoint	Description	Authorization
GET	/analytics/tasks	Task completion statistics	Process Manager
GET	/analytics/tasks/overdue	Overdue task analytics	Process Manager
GET	/analytics/users/productivity	User productivity metrics	Organization Admin

GET	/analytics/users/workload	User workload distribution	Organization Admin
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8.9.7 KPI APIs

Method	Endpoint	Description	Authorization
GET	/analytics/kpis	Retrieve organization KPIs	Organization Admin
GET	/analytics/kpis/{kpild}	Retrieve KPI details	Organization Admin
POST	/analytics/kpis	Create custom KPI	Organization Admin
PUT	/analytics/kpis/{kpild}	Update KPI	Organization Admin
DELETE	/analytics/kpis/{kpild}	Delete custom KPI	Organization Admin

8.9.8 Report Generation APIs

Method	Endpoint	Description	Authorization
POST	/reports/generate	Generate report	Organization User
GET	/reports/{reportId}	Retrieve generated report	Organization User
GET	/reports/history	Retrieve report history	Organization User
DELETE	/reports/{reportId}	Delete generated report	Organization User

8.9.9 Report Export APIs

Method	Endpoint	Description	Authorization
GET	/reports/{reportId}/pdf	Export report as PDF	Organization User
GET	/reports/{reportId}/excel	Export report as Excel	Organization User
GET	/reports/{reportId}/csv	Export report as CSV	Organization User
GET	/reports/{reportId}/json	Export report as JSON	Organization User

8.9.10 Scheduled Report APIs

Method	Endpoint	Description	Authorization
POST	/reports/schedules	Schedule report generation	Organization Admin
GET	/reports/schedules	Retrieve scheduled reports	Organization Admin
PUT	/reports/schedules/{scheduleId}	Update report schedule	Organization Admin

DELETE	/reports/schedules/{scheduleId}	Delete report schedule	Organization Admin
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8.9.11 Predictive Analytics APIs

Method	Endpoint	Description	Authorization
GET	/analytics/predictions/workload	Predict future workload	Organization Admin
GET	/analytics/predictions/bottlenecks	Predict workflow bottlenecks	Process Manager
GET	/analytics/predictions/sla-risk	Predict SLA violations	Organization Admin
GET	/analytics/predictions/optimization	Predict optimization opportunities	Process Manager

8.9.12 Business Rules

Rule ID	Rule
ANA-BR-001	Analytics shall only include data belonging to the authenticated organization.
ANA-BR-002	Historical analytics shall remain available according to the configured retention policy.
ANA-BR-003	Report generation shall be logged for auditing purposes.
ANA-BR-004	KPI calculations shall use validated operational data.
ANA-BR-005	Predictive analytics shall clearly distinguish forecasts from historical metrics.
ANA-BR-006	Scheduled reports shall execute according to configured schedules.
ANA-BR-007	Exported reports shall respect user authorization and data access policies.
ANA-BR-008	AI analytics shall include provider, model, latency, confidence, and failure metrics.
ANA-BR-009	Dashboard data shall be refreshed according to configurable intervals.
ANA-BR-010	Generated reports shall remain accessible according to retention policies.

8.9.13 Error Responses

HTTP Status	Error Code	Description
400	INVALID_REPORT_REQUEST	Invalid report parameters.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	User lacks required permissions.
404	REPORT_NOT_FOUND	Requested report does not exist.
404	KPI_NOT_FOUND	Requested KPI does not exist.

409	REPORT_ALREADY_RUNNING	Report generation is already in progress.
422	ANALYTICS_DATA_UNAVAILABLE	Required analytics data is unavailable.
500	REPORT_GENERATION_FAILED	Unexpected report generation failure.

8.9.14 Acceptance Criteria

The Analytics & Reporting API module shall be considered complete when:

- Executive dashboards provide real-time operational insights.
- Workflow and AI performance metrics are available.
- KPI management supports organizational customization.
- Reports can be generated, scheduled, and exported in multiple formats.
- Predictive analytics provide actionable forecasts.
- Analytics respect tenant isolation and authorization rules.
- Reporting activities are fully audited.

8.10 Notification & Communication APIs

8.10.1 Module Overview

The Notification & Communication API provides endpoints for managing in-app notifications, email notifications, workflow alerts, approval requests, AI processing updates, system announcements, communication templates, and user notification preferences.

The module ensures that users receive timely and relevant information while allowing organizations to configure delivery channels and notification policies.

Notifications shall support both real-time and asynchronous delivery mechanisms.

8.10.2 Notification Categories

The platform shall support the following notification categories:

- Workflow Notifications
- Task Notifications
- Approval Notifications
- AI Processing Notifications
- System Notifications
- Security Notifications
- Administrative Announcements
- SLA & Deadline Alerts

8.10.3 Notification APIs

Method	Endpoint	Description	Authorization
GET	/notifications	Retrieve notifications	Authenticated User
GET	/notifications/{notificationId}	Retrieve notification details	Authenticated User
PATCH	/notifications/{notificationId}/read	Mark notification as read	Authenticated User
PATCH	/notifications/read-all	Mark all notifications as read	Authenticated User
DELETE	/notifications/{notificationId}	Delete notification	Authenticated User
GET	/notifications/unread-count	Retrieve unread notification count	Authenticated User

8.10.4 Workflow Notification APIs

Method	Endpoint	Description	Authorization
GET	/notifications/workflows	Retrieve workflow notifications	Organization User
POST	/notifications/workflows/test	Send test workflow notification	Organization Admin

8.10.5 Approval Notification APIs

Method	Endpoint	Description	Authorization
GET	/notifications/approvals	Retrieve approval notifications	Approver

POST	/notifications/approvals/remind	Send approval reminder	Process Manager
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8.10.6 AI Notification APIs

Method	Endpoint	Description	Authorization
GET	/notifications/ai	Retrieve AI processing notifications	Organization User
POST	/notifications/ai/test	Send AI notification test	Organization Admin

8.10.7 Email Communication APIs

Method	Endpoint	Description	Authorization
POST	/communications/email/send	Send email	Organization Admin
GET	/communications/email/history	Retrieve email history	Organization Admin
POST	/communications/email/test	Send test email	Organization Admin

8.10.8 Notification Preference APIs

Method	Endpoint	Description	Authorization
GET	/notification-preferences	Retrieve user preferences	Authenticated User
PUT	/notification-preferences	Update notification preferences	Authenticated User
PATCH	/notification-preferences/channels	Configure notification channels	Authenticated User

8.10.9 Communication Template APIs

Method	Endpoint	Description	Authorization
GET	/communication-templates	Retrieve templates	Organization Admin
POST	/communication-templates	Create template	Organization Admin
PUT	/communication-templates/{templateId}	Update template	Organization Admin
DELETE	/communication-templates/{templateId}	Delete template	Organization Admin

8.10.10 WebSocket APIs

The platform shall support WebSocket communication for real-time updates.

Endpoint	Description
/ ws / notifications	Real-time user notifications
/ ws / workflows	Workflow execution updates
/ ws /tasks	Task assignment updates
/ ws / approvals	Approval status updates
/ ws / system	System announcements

8.10.11 Supported Notification Channels

Channel	Supported
In-App Notification	Yes
Email	Yes
WebSocket (Real-Time)	Yes
Push Notification (Future)	Planned
SMS (Future)	Planned
Microsoft Teams (Future)	Planned
Slack (Future)	Planned
WhatsApp Business (Future)	Planned

8.10.12 Business Rules

Rule ID	Rule
NOT-BR-001	Notifications shall only be delivered to authorized recipients.
NOT-BR-002	Users may configure notification preferences and delivery channels.
NOT-BR-003	Critical security notifications shall not be disabled.
NOT-BR-004	Workflow and approval events shall automatically generate notifications.

NOT-BR-005	AI processing completion or failure shall generate notifications where configured.
NOT-BR-006	Communication templates shall support version control.
NOT-BR-007	Notification delivery failures shall be logged and retried according to policy.
NOT-BR-008	Notification history shall be retained according to the configured retention policy.
NOT-BR-009	Real-time notifications shall be delivered through WebSocket when available.
NOT-BR-010	Users shall only receive notifications related to their organization and assigned responsibilities.

8.10.13 Error Responses

HTTP Status	Error Code	Description
400	INVALID_NOTIFICATION	Invalid notification request.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	User lacks permission.
404	NOTIFICATION_NOT_FOUND	Notification does not exist.
409	TEMPLATE_CONFLICT	Communication template conflict detected.
422	DELIVERY_FAILED	Notification delivery failed.
500	COMMUNICATION_ERROR	Unexpected communication service failure.

8.10.14 Acceptance Criteria

The Notification & Communication API module shall be considered complete when:

- Users receive workflow, task, approval, and AI notifications.
- Notification preferences are configurable.
- Communication templates are reusable and manageable.
- Email delivery is supported.
- Real-time updates are available through WebSockets.
- Notification history is retained according to policy.
- Delivery failures are monitored and logged.
- Multi-tenant isolation is maintained.

8.11 Administration APIs

8.11.1 Module Overview

The Administration API provides endpoints for system configuration, platform administration, AI provider management, subscription management, audit logging, backup and recovery, health monitoring, diagnostics, and maintenance.

These APIs are primarily intended for **System Administrators** and **Organization Administrators**, depending on the operation.

Administrative operations are security-sensitive and shall be fully audited.

8.11.2 System Configuration APIs

Method	Endpoint	Description	Authorization
GET	/admin/system-config	Retrieve system configuration	System Admin
PUT	/admin/system-config	Update system configuration	System Admin
GET	/admin/system-config/security	Retrieve security configuration	System Admin
PUT	/admin/system-config/security	Update security configuration	System Admin
GET	/admin/system-config/ai	Retrieve AI configuration	System Admin
PUT	/admin/system-config/ai	Update AI configuration	System Admin

8.11.3 Organization Administration APIs

Method	Endpoint	Description	Authorization
GET	/admin/organizations	Retrieve all organizations	System Admin
GET	/admin/organizations/{organizationId}	Retrieve organization details	System Admin
PATCH	/admin/organizations/{organizationId}/status	Activate or suspend organization	System Admin
GET	/admin/organizations/{organizationId}/usage	Retrieve organization usage statistics	System Admin

8.11.4 AI Provider Administration APIs

Method	Endpoint	Description	Authorization
GET	/admin/ai/providers	Retrieve configured AI providers	System Admin

POST	/admin/ai/providers	Register AI provider	System Admin
PUT	/admin/ai/providers/{providerId}	Update provider configuration	System Admin
DELETE	/admin/ai/providers/{providerId}	Remove AI provider	System Admin
POST	/admin/ai/providers/{providerId}/health-check	Test provider connectivity	System Admin

8.11.5 Subscription & Licensing APIs

Method	Endpoint	Description	Authorization
GET	/admin/subscriptions	Retrieve subscriptions	System Admin
GET	/admin/subscriptions/{subscriptionId}	Retrieve subscription details	System Admin
PUT	/admin/subscriptions/{subscriptionId}	Update subscription	System Admin
GET	/admin/licenses	Retrieve license information	System Admin
PUT	/admin/licenses/{licenseId}	Update license	System Admin

8.11.6 Audit Management APIs

Method	Endpoint	Description	Authorization
GET	/admin/audit-logs	Retrieve audit logs	System Admin
GET	/admin/audit-logs/{logId}	Retrieve audit log details	System Admin
GET	/admin/activity-logs	Retrieve activity logs	System Admin
GET	/admin/security-events	Retrieve security events	System Admin

8.11.7 Backup & Recovery APIs

Method	Endpoint	Description	Authorization
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POST	/admin/backups	Create manual backup	System Admin
GET	/admin/backups	Retrieve backup history	System Admin
POST	/admin/backups/{backupId}/restore	Restore backup	System Admin
DELETE	/admin/backups/{backupId}	Delete backup	System Admin

8.11.8 System Monitoring APIs

Method	Endpoint	Description	Authorization
GET	/admin/health	Retrieve overall system health	System Admin
GET	/admin/metrics	Retrieve platform metrics	System Admin
GET	/admin/database/status	Retrieve database health	System Admin
GET	/admin/storage/status	Retrieve storage usage	System Admin
GET	/admin/ai/status	Retrieve AI service status	System Admin

8.11.9 Maintenance APIs

Method	Endpoint	Description	Authorization
POST	/admin/cache/clear	Clear application cache	System Admin
POST	/admin/indexes/rebuild	Rebuild search indexes	System Admin
POST	/admin/system/restart-services	Restart background services	System Admin
GET	/admin/system/version	Retrieve system version	System Admin

8.11.10 Diagnostics APIs

Method	Endpoint	Description	Authorization
GET	/admin/diagnostics	Retrieve diagnostic report	System Admin
GET	/admin/logs/errors	Retrieve application errors	System Admin
GET	/admin/logs/performance	Retrieve performance logs	System Admin
GET	/admin/logs/integration	Retrieve integration logs	System Admin

8.11.11 Business Rules

Rule ID	Rule
ADM-BR-001	Administrative APIs shall only be accessible to authorized administrators.
ADM-BR-002	All administrative actions shall generate audit log entries.
ADM-BR-003	Backup restoration shall require confirmation and appropriate authorization.
ADM-BR-004	System configuration changes shall be versioned and traceable.
ADM-BR-005	AI provider changes shall be validated before activation.
ADM-BR-006	Critical maintenance operations shall notify affected administrators.
ADM-BR-007	Health monitoring APIs shall provide real-time system status.
ADM-BR-008	System diagnostics shall not expose sensitive information to unauthorized users.
ADM-BR-009	Subscription and licensing changes shall take effect according to defined business policies.
ADM-BR-010	Administrative access shall respect the principle of least privilege.

8.11.12 Error Responses

HTTP Status	Error Code	Description
400	INVALID_ADMIN_REQUEST	Invalid administrative request.
401	UNAUTHORIZED	Authentication required.
403	FORBIDDEN	Administrative privileges required.
404	RESOURCE_NOT_FOUND	Requested administrative resource not found.
409	CONFIGURATION_CONFLICT	Configuration conflict detected.
422	VALIDATION_FAILED	Administrative validation failed.
500	ADMIN_OPERATION_FAILED	Unexpected administrative operation failure.

8.11.13 Acceptance Criteria

The Administration API module shall be considered complete when:

- System configuration can be managed securely.
- Organizations, AI providers, subscriptions, and licenses can be administered.
- Backup and recovery operations function correctly.
- Audit logs and security events are accessible to authorized administrators.

- Health monitoring and diagnostics provide accurate system information.
- Maintenance operations are controlled and audited.
- Administrative actions follow the principle of least privilege.

Chapter 9 – Security Requirements

9.1 Security Overview

9.1.1 Purpose

The Security Requirements define the policies, controls, and mechanisms implemented to protect ProcessMind AI against unauthorized access, data breaches, cyberattacks, system misuse, and operational failures.

The security architecture shall ensure the confidentiality, integrity, availability, accountability, and privacy of organizational data while supporting secure AI-assisted workflow automation.

Security controls shall be applied consistently across all system components, including:

- Web Application
- REST APIs
- AI Services
- Database
- File Storage
- Authentication Services
- Workflow Engine
- Notification Services
- Administrative Interfaces

9.1.2 Security Objectives

The platform shall achieve the following security objectives.

Objective	Description
Confidentiality	Prevent unauthorized access to organizational data.
Integrity	Prevent unauthorized modification of data.
Availability	Ensure continuous access to services.

Authentication	Verify user identities securely.
Authorization	Restrict access according to permissions.
Accountability	Record security-relevant activities through audit logging.
Privacy	Protect organizational and user information.
Resilience	Continue operating safely during failures or attacks.

9.1.3 Security Principles

The security architecture shall follow these principles:

- Zero Trust Architecture
- Least Privilege
- Defense in Depth
- Secure by Default
- Fail Secure
- Separation of Duties
- Multi-Layer Security
- Continuous Monitoring
- Secure AI Processing
- Privacy by Design

9.1.4 Security Architecture



9.1.5 Security Domains

The security architecture consists of the following domains.

Domain	Purpose
Identity Security	Authentication and identity management
Access Security	Role-Based Access Control (RBAC)
API Security	Secure API communication
Application Security	Secure application logic
Database Security	Data protection and integrity
AI Security	Secure AI interactions and prompt handling
Infrastructure Security	Server and deployment security
Network Security	Secure communications
Audit Security	Monitoring and forensic capabilities
Incident Response	Detection and response to security events

9.1.6 Protected Assets

The following assets shall be protected.

Asset	Protection Required
User Accounts	Authentication and authorization
Passwords	Secure hashing
JWT Tokens	Digital signing and expiration
SOP Documents	Encryption and access control
Workflow Definitions	Authorization and version control
AI Requests	Secure transmission
AI Responses	Validation and auditing
Organizational Data	Tenant isolation
Audit Logs	Integrity protection
System Configuration	Administrative controls
API Keys	Secure storage
Backups	Encryption and restricted access

9.1.7 Threat Categories

The platform shall protect against:

- Unauthorized access
- Credential theft
- SQL Injection
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF) (where applicable)
- Broken authentication
- Broken access control
- Prompt injection
- AI output manipulation
- Malware uploads
- Denial-of-Service (DoS)
- Brute-force attacks
- Session hijacking

- Data leakage
- Insider threats
- Supply chain vulnerabilities

9.1.8 Security Standards

The platform shall be designed in alignment with recognized security standards and best practices.

Standard	Purpose
OWASP Top 10	Web application security
OWASP ASVS	Application Security Verification Standard
NIST Cybersecurity Framework	Security governance
NIST SP 800-63	Digital identity guidelines
CWE Top 25	Common software weaknesses
CIS Benchmarks	Secure system configuration
JWT Best Practices	Secure token management

Future Enterprise Compliance (Optional):

- ISO/IEC 27001
- SOC 2 Type II
- GDPR
- HIPAA (if healthcare customers are supported)

9.1.9 Security Responsibilities

Role	Responsibilities
System Administrator	Platform security configuration
Organization Administrator	Organization-level security policies
Process Manager	Secure workflow management
Reviewer	Validate AI-generated outputs
Employee	Follow security policies

Security Administrator (Future)	Incident response and monitoring
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9.1.10 General Security Requirements

Requirement ID	Requirement
SEC-GEN-001	All communications shall use HTTPS/TLS.
SEC-GEN-002	Sensitive data shall be protected during transmission and storage.
SEC-GEN-003	Authentication shall be required for all protected resources.
SEC-GEN-004	Authorization shall follow RBAC principles.
SEC-GEN-005	All security-relevant events shall be logged.
SEC-GEN-006	Security configurations shall be centrally managed.
SEC-GEN-007	The principle of least privilege shall be enforced.
SEC-GEN-008	Security controls shall be periodically reviewed and updated.
SEC-GEN-009	Security incidents shall be monitored and investigated.
SEC-GEN-010	Tenant isolation shall be maintained across all system layers.

9.1.11 Business Rules

Rule ID	Rule
SEC-BR-001	Every authenticated user shall belong to exactly one organization unless assigned as a System Administrator.
SEC-BR-002	Security events shall generate audit log entries.
SEC-BR-003	Administrative operations shall require elevated privileges.
SEC-BR-004	AI operations shall comply with organizational security policies.
SEC-BR-005	Sensitive credentials shall never be exposed in application code or API responses.
SEC-BR-006	Security configurations shall support future compliance requirements.
SEC-BR-007	Security controls shall remain effective during system failures wherever feasible.

9.1.12 Acceptance Criteria

The security architecture shall be considered complete when:

- Security principles are consistently applied across all system components.
- Protected assets are secured according to organizational policies.
- Authentication and authorization mechanisms are enforced.
- Communications are encrypted.
- Security events are logged and monitored.
- Multi-tenant isolation is maintained.
- The platform follows recognized security best practices.
- Security responsibilities are clearly defined.

9.2 Authentication Security

9.2.1 Overview

Authentication Security defines the controls and mechanisms used to verify user identities and protect authentication credentials within ProcessMind AI.

The authentication framework shall ensure that only authorized users gain access to protected system resources while preventing credential theft, brute-force attacks, session hijacking, and unauthorized account access.

Authentication shall be implemented using secure industry-standard mechanisms and continuously monitored for suspicious activities.

9.2.2 Authentication Methods

The platform shall support the following authentication methods.

Authentication Method	Supported
Email and Password	Yes
JWT Access Tokens	Yes
JWT Refresh Tokens	Yes
Multi-Factor Authentication (Future)	Planned
Single Sign-On (SSO) (Future)	Planned
OAuth 2.0 / OpenID Connect (Future)	Planned
WebAuthn / Passkeys (Future)	Planned

9.2.3 Password Policy

User passwords shall comply with the following policy.

Requirement	Value
Minimum Length	12 characters
Maximum Length	128 characters
Uppercase Letter	Required
Lowercase Letter	Required
Numeric Character	Required
Special Character	Required
Common Password Check	Required
Password Reuse	Last 5 passwords shall not be reused
Password Expiration	Configurable (disabled by default)

Passwords shall never be stored or transmitted in plain text.

9.2.4 Credential Storage

Authentication credentials shall be protected using secure storage mechanisms.

Requirement ID	Requirement
AUTH-SEC-001	Passwords shall be hashed using BCrypt.
AUTH-SEC-002	Plain-text passwords shall never be stored.
AUTH-SEC-003	Authentication secrets shall be stored securely using environment variables or a secrets management solution.
AUTH-SEC-004	JWT signing keys shall be protected and rotated periodically.

9.2.5 JWT Token Security

The platform shall use JSON Web Tokens (JWT) for stateless authentication.

Requirement	Description
Access Token	Short-lived authentication token
Refresh Token	Used to obtain new access tokens
Digital Signature	Required
Token Expiration	Configurable
Token Revocation	Supported on logout and administrative revocation

Access tokens shall contain only the minimum claims required for authorization.

Sensitive information shall not be stored within JWT payloads.

9.2.6 Session Management

The platform shall manage authenticated sessions securely.

Requirement ID	Requirement
AUTH-SEC-005	Sessions shall expire automatically after the configured inactivity period.
AUTH-SEC-006	Logout shall invalidate refresh tokens.
AUTH-SEC-007	Multiple concurrent sessions may be allowed based on organizational policy.
AUTH-SEC-008	Session identifiers shall be unpredictable and cryptographically secure.
AUTH-SEC-009	Administrators shall be able to revoke active sessions.

9.2.7 Account Protection

The platform shall protect accounts from unauthorized access.

Requirement ID	Requirement
AUTH-SEC-010	Failed login attempts shall be monitored.
AUTH-SEC-011	Accounts shall be temporarily locked after repeated failed login attempts.
AUTH-SEC-012	Lockout duration shall be configurable.
AUTH-SEC-013	Suspicious authentication attempts shall be logged.
AUTH-SEC-014	Brute-force protection mechanisms shall be implemented.

9.2.8 Password Reset Security

Password reset operations shall be secured.

Requirement ID	Requirement
AUTH-SEC-015	Password reset tokens shall be randomly generated.
AUTH-SEC-016	Reset tokens shall expire after a configurable period.
AUTH-SEC-017	Reset tokens shall be single-use.
AUTH-SEC-018	Password reset requests shall not reveal whether an account exists.
AUTH-SEC-019	Successful password resets shall invalidate existing refresh tokens.

9.2.9 Email Verification

The platform may require email verification before account activation.

Requirement ID	Requirement
AUTH-SEC-020	Email verification tokens shall expire after a configurable period.
AUTH-SEC-021	Verification tokens shall be single-use.
AUTH-SEC-022	Email verification events shall be logged.

9.2.10 Multi-Factor Authentication (Future)

The platform architecture shall support future MFA implementation.

Supported methods may include:

- Time-Based One-Time Passwords (TOTP)
- Email verification codes
- SMS verification codes (optional)
- Hardware security keys (WebAuthn/FIDO2)
- Authenticator applications

MFA shall be configurable at the organization level.

9.2.11 Authentication Monitoring

The authentication subsystem shall monitor:

- Successful logins
- Failed login attempts
- Password reset requests
- Account lockouts
- Session revocations
- Token refresh events
- Suspicious authentication activity

Critical events shall generate administrator alerts where configured.

9.2.12 Business Rules

Rule ID	Rule
AUTH-BR-001	Every user shall authenticate before accessing protected resources.
AUTH-BR-002	Passwords shall comply with the configured password policy.
AUTH-BR-003	Passwords shall never be stored or transmitted in plain text.
AUTH-BR-004	Refresh tokens shall be revoked upon logout or administrative action.
AUTH-BR-005	Failed authentication attempts shall be recorded in audit logs.
AUTH-BR-006	Account lockout thresholds shall be configurable.
AUTH-BR-007	Authentication credentials shall be protected according to industry best practices.

9.2.13 Exception Handling

Exception	Expected System Response
Invalid credentials	Reject authentication and increment failed-attempt counter.
Expired JWT access token	Return an authentication error and require token refresh.
Invalid refresh token	Reject token renewal and require re-authentication.

Locked account	Deny access until lockout conditions are satisfied or administrative action is taken.
Expired password reset token	Reject the reset request and require a new reset token.
Compromised session detected	Revoke affected sessions, log the incident, and notify administrators where appropriate.

9.2.14 Acceptance Criteria

The authentication security requirements shall be considered complete when:

- Passwords comply with the configured security policy.
- Credentials are securely stored using strong hashing algorithms.
- JWT authentication is implemented securely.
- Sessions are managed and revoked appropriately.
- Brute-force protection and account lockout mechanisms operate correctly.
- Password reset and email verification processes are secure.
- Authentication events are monitored and audited.
- The architecture supports future Multi-Factor Authentication (MFA).

9.3 Authorization Security

9.3.1 Overview

Authorization Security defines the mechanisms used to control access to system resources after successful authentication.

The authorization framework shall ensure that authenticated users can access only the resources and operations explicitly permitted by their assigned roles, permissions, organization membership, and resource ownership.

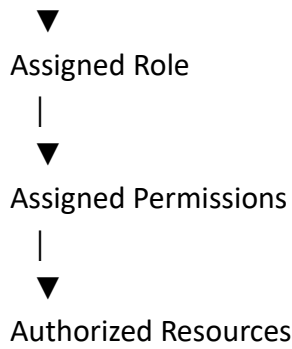
The platform shall implement **Role-Based Access Control (RBAC)** with support for future **Attribute-Based Access Control (ABAC)** extensions.

9.3.2 Authorization Model

The platform shall implement Role-Based Access Control (RBAC).

User

|



Future versions may extend this model using:

- Attribute-Based Access Control (ABAC)
- Policy-Based Access Control (PBAC)
- Context-Aware Authorization

9.3.3 Default System Roles

Role	Description
System Administrator	Full platform administration
Organization Administrator	Organization management
Process Manager	SOP and workflow management
Reviewer	Human review of AI outputs
Approver	Workflow approval authority
Employee	Standard workflow execution
Read-Only Auditor	Read-only access to reports and audit data

Organizations may define additional custom roles where permitted.

9.3.4 Permission Categories

Permissions shall be grouped into logical categories.

Category	Examples
User Management	Create User, Update User, Delete User
Organization Management	Manage Organization, Departments
SOP Management	Upload, Edit, Archive SOP
Workflow Management	Create, Deploy, Archive Workflow
Workflow Execution	Execute, Pause, Resume Workflow
AI Management	Generate Workflow, Review AI Output
Reporting	View Reports, Export Reports
Administration	Configure System, Manage AI Providers
Security	View Audit Logs, Manage Security Policies

9.3.5 Authorization Requirements

Requirement ID	Requirement
AUTHZ-SEC-001	Every protected operation shall require authorization.
AUTHZ-SEC-002	Authorization decisions shall be based on assigned roles and permissions.
AUTHZ-SEC-003	Unauthorized requests shall be rejected.
AUTHZ-SEC-004	Administrative operations shall require elevated privileges.
AUTHZ-SEC-005	Authorization shall be enforced at the API and business service layers.

9.3.6 Least Privilege

The platform shall implement the Principle of Least Privilege.

Requirement ID	Requirement
AUTHZ-SEC-006	Users shall receive only the permissions necessary for their responsibilities.
AUTHZ-SEC-007	Administrative privileges shall be granted only to authorized users.
AUTHZ-SEC-008	Default user accounts shall not receive elevated permissions.

9.3.7 Multi-Tenant Authorization

Tenant isolation shall be enforced.

Requirement ID	Requirement
AUTHZ-SEC-009	Users shall only access resources belonging to their organization.
AUTHZ-SEC-010	Cross-tenant access shall be prohibited except for System Administrators.
AUTHZ-SEC-011	Every organization-owned resource shall validate tenant ownership before access.
AUTHZ-SEC-012	Tenant isolation shall be enforced consistently across APIs, services, and data access layers.

9.3.8 Resource Ownership

Certain resources shall enforce ownership rules.

Resource	Ownership Rule
User Profile	User may update own profile
Workflow	Owner or authorized manager may modify
SOP	Authorized owner or Process Manager may modify
Reports	Accessible according to organization permissions
Notifications	Visible only to intended recipients

9.3.9 Administrative Separation

Administrative responsibilities shall be separated to reduce risk.

Administrative Role	Scope
System Administrator	Platform-wide administration
Organization Administrator	Organization-specific administration
Process Manager	Workflow and SOP administration
Security Administrator (Future)	Security policies and incident management

No single administrative role should receive unnecessary permissions beyond its defined scope.

9.3.10 Fine-Grained Authorization

The platform shall support authorization based on:

- User role
- Organization
- Department
- Resource ownership
- Workflow state
- Approval stage
- Resource sensitivity

Future releases may include contextual authorization based on device, network, and time.

9.3.11 Authorization Auditing

The following events shall be recorded.

- Permission changes
- Role assignments
- Administrative actions
- Access denials
- Privileged operations
- Authorization failures
- Resource access by privileged users

9.3.12 Business Rules

Rule ID	Rule
AUTHZ-BR-001	Every protected resource shall enforce authorization checks.
AUTHZ-BR-002	Users shall only access resources belonging to their organization.
AUTHZ-BR-003	Roles shall determine default permissions.
AUTHZ-BR-004	Permission assignments shall be auditable.
AUTHZ-BR-005	Administrative operations shall require elevated privileges.

AUTHZ-BR-006	Authorization failures shall be logged.
AUTHZ-BR-007	Resource ownership shall be validated before modification or deletion.
AUTHZ-BR-008	Custom roles shall not exceed the maximum permission scope defined by organizational policy.

9.3.13 Exception Handling

Exception	Expected System Response
Missing permission	Reject the request with an authorization error.
Cross-tenant access attempt	Deny access, log the event, and notify administrators where appropriate.
Invalid role assignment	Reject the operation and return a validation error.
Unauthorized administrative action	Deny the operation and create an audit record.
Resource ownership violation	Prevent the operation and return an authorization error.
Permission evaluation failure	Deny access by default (fail secure) and log the incident.

9.3.14 Acceptance Criteria

The authorization security requirements shall be considered complete when:

- Role-Based Access Control (RBAC) is consistently enforced.
- Users can only access authorized resources.
- Multi-tenant isolation prevents cross-organization access.
- Administrative privileges are separated appropriately.
- Authorization failures are logged and auditable.
- Resource ownership rules are enforced.
- The system follows the Principle of Least Privilege.
- The architecture supports future fine-grained authorization models.

9.4 Data Protection Security

9.4.1 Overview

Data Protection Security defines the policies, controls, and mechanisms used to protect organizational data throughout its lifecycle—from creation and storage to transmission, archival, backup, and secure deletion.

The platform shall ensure that sensitive information remains confidential, accurate, and available only to authorized users.

Data protection controls shall apply consistently across:

- Web Application
- REST APIs
- AI Processing Services
- PostgreSQL Database
- File Storage
- Backups
- Analytics
- Audit Logs

9.4.2 Data Classification

All information managed by the platform shall be classified according to its sensitivity.

Classification	Description	Examples
Public	Information intended for unrestricted access	Public documentation, product information
Internal	Information used within an organization	Workflow metadata, department lists
Confidential	Sensitive organizational information	SOPs, workflow definitions, reports
Restricted	Highly sensitive information	Password hashes, API keys, encryption secrets, audit logs

Security controls shall increase according to the classification level.

9.4.3 Encryption in Transit

All communications shall be encrypted.

Requirement ID	Requirement
DATA-SEC-001	All HTTP communication shall use HTTPS (TLS 1.2 or higher; TLS 1.3 preferred).
DATA-SEC-002	Communication between application services shall use encrypted channels where applicable.
DATA-SEC-003	Database connections shall use TLS in production environments.
DATA-SEC-004	AI provider communications shall use secure HTTPS connections.

9.4.4 Encryption at Rest

Sensitive information shall be protected while stored.

Requirement ID	Requirement
DATA-SEC-005	Database storage shall support encryption at rest.
DATA-SEC-006	Uploaded SOP documents shall be stored in encrypted storage.
DATA-SEC-007	Backup files shall be encrypted before storage.
DATA-SEC-008	Encryption keys shall not be stored with encrypted data.

9.4.5 Key Management

Cryptographic keys shall be managed securely.

Requirement ID	Requirement
DATA-SEC-009	Encryption keys shall be stored using a secure secrets management solution or equivalent.
DATA-SEC-010	Keys shall be rotated periodically according to organizational policy.
DATA-SEC-011	Access to encryption keys shall be restricted to authorized components and administrators.

DATA-SEC-012	Key usage shall be logged where appropriate.
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9.4.6 Sensitive Data Handling

Sensitive information shall receive additional protection.

Data Type	Protection
Passwords	BCrypt hashing
JWT Signing Keys	Secure secret storage
API Keys	Encrypted or managed through a secrets manager
Refresh Tokens	Secure storage and revocation support
SOP Documents	Encryption and access control
AI Requests	Secure transmission
AI Responses	Organization isolation and audit logging
Audit Logs	Integrity protection and restricted access

Sensitive information shall never be exposed in logs, error messages, or API responses.

9.4.7 File Protection

Uploaded documents shall be protected.

Requirement ID	Requirement
DATA-SEC-013	Uploaded files shall undergo malware scanning before storage.
DATA-SEC-014	Supported file types shall be validated before processing.
DATA-SEC-015	File size limits shall be enforced.
DATA-SEC-016	Access to uploaded files shall require authorization.

9.4.8 Backup Protection

Backups shall be secured throughout their lifecycle.

Requirement ID	Requirement
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DATA-SEC-017	Backup files shall be encrypted.
DATA-SEC-018	Backup access shall be restricted to authorized administrators.
DATA-SEC-019	Backup restoration shall be audited.
DATA-SEC-020	Backup integrity shall be verified before restoration.

9.4.9 Data Masking

Sensitive information shall be masked where full disclosure is unnecessary.

Examples include:

- Partial email addresses
- Partial phone numbers
- API keys
- Access tokens
- Secret configuration values

Only authorized users may view complete sensitive information where required.

9.4.10 Secure Deletion

The platform shall securely delete information when permitted by retention policies.

Requirement ID	Requirement
DATA-SEC-021	Deleted data shall comply with configured retention policies.
DATA-SEC-022	Secure deletion shall be logged.
DATA-SEC-023	Data subject to legal or compliance requirements shall not be deleted prematurely.
DATA-SEC-024	Temporary processing data shall be removed after use according to policy.

9.4.11 Data Integrity

The platform shall protect against unauthorized modification.

Requirement ID	Requirement
DATA-SEC-025	Referential integrity shall be enforced.
DATA-SEC-026	Database transactions shall comply with ACID principles.

DATA-SEC-027	Audit records shall be immutable.
DATA-SEC-028	Workflow versions shall remain immutable after deployment.

9.4.12 Business Rules

Rule ID	Rule
DATA-BR-001	Sensitive organizational data shall be protected according to its classification.
DATA-BR-002	Encryption shall be used for sensitive data in transit and at rest where appropriate.
DATA-BR-003	Encryption keys shall be managed separately from encrypted data.
DATA-BR-004	Uploaded files shall be validated before storage and processing.
DATA-BR-005	Backup operations shall preserve confidentiality and integrity.
DATA-BR-006	Secure deletion shall follow retention and compliance policies.
DATA-BR-007	Sensitive information shall not be exposed through logs or API responses.
DATA-BR-008	Tenant isolation shall be maintained throughout the data lifecycle.

9.4.13 Exception Handling

Exception	Expected System Response
Encryption failure	Reject the operation where encryption is required, log the incident, and notify administrators.
Invalid file upload	Reject the upload and return a validation error.
Malware detected	Quarantine or reject the file and record the event.
Backup integrity verification failure	Prevent restoration and require a verified backup.
Unauthorized access to sensitive data	Deny access, generate an audit log entry, and notify administrators where appropriate.

Key management failure	Prevent cryptographic operations that depend on the unavailable key material and raise an administrative alert.
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9.4.14 Acceptance Criteria

The data protection security requirements shall be considered complete when:

- Sensitive data is classified and protected appropriately.
- Encryption is applied during transmission and storage where required.
- Encryption keys are securely managed.
- Uploaded files are validated and protected.
- Backup data is encrypted and recoverable.
- Sensitive information is masked when appropriate.
- Secure deletion complies with retention policies.
- Data integrity controls protect against unauthorized modification.

9.5 Network & Infrastructure Security

9.5.1 Overview

Network & Infrastructure Security defines the controls, configurations, and operational practices used to protect the infrastructure hosting ProcessMind AI.

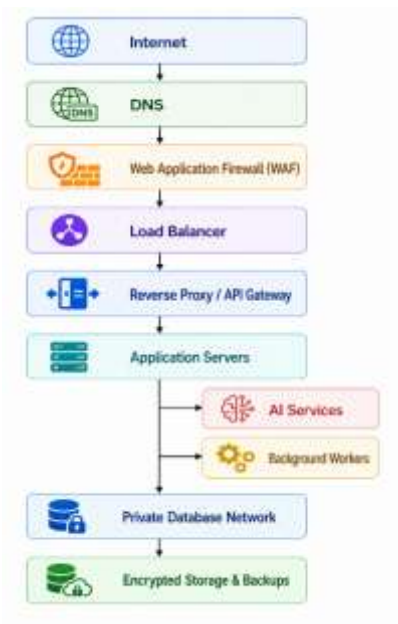
The infrastructure shall be designed using a **defense-in-depth** strategy, ensuring that failures in one security layer do not compromise the entire platform.

Security controls shall protect:

- Client communications
- Application servers
- API services
- AI integration services
- PostgreSQL database
- File storage
- Background workers
- Monitoring services
- Administrative interfaces

9.5.2 Network Architecture

The production environment shall follow a layered architecture.



The database and internal services shall not be directly accessible from the public internet.

9.5.3 Secure Communication

All network communications shall be encrypted.

Requirement ID	Requirement
NET-SEC-001	HTTPS shall be enforced for all client-server communication.
NET-SEC-002	TLS 1.2 or higher shall be required (TLS 1.3 preferred).
NET-SEC-003	HTTP requests shall be redirected to HTTPS.
NET-SEC-004	Internal service communication shall use encrypted channels where applicable.

9.5.4 Firewall Requirements

The infrastructure shall implement firewall protection.

Requirement ID	Requirement
NET-SEC-005	Only required ports shall be exposed publicly.
NET-SEC-006	Database ports shall not be publicly accessible.
NET-SEC-007	Administrative interfaces shall be restricted.

NET-SEC-008	Firewall rules shall be reviewed periodically.
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9.5.5 Reverse Proxy & API Gateway Security

The reverse proxy or API gateway shall provide:

- HTTPS termination
- Request routing
- Rate limiting
- Request size limits
- Security header injection
- API authentication forwarding
- Request logging
- WebSocket support
- Protection against malformed requests

9.5.6 Infrastructure Isolation

Infrastructure components shall be isolated.

Component	Isolation Requirement
Database	Private network only
File Storage	Restricted access
AI Services	Controlled API access
Background Workers	Internal network
Monitoring Services	Administrative access only
Backup Storage	Separate protected storage

9.5.7 Secrets Management

Sensitive configuration values shall be protected.

Requirement ID	Requirement
NET-SEC-009	Secrets shall not be stored in source code repositories.

NET-SEC-010	Environment variables or a dedicated secrets manager shall be used.
NET-SEC-011	API keys shall be rotated periodically.
NET-SEC-012	Access to secrets shall follow the principle of least privilege.

9.5.8 Container & Runtime Security

If deployed using containers, the platform shall follow secure container practices.

Requirement ID	Requirement
NET-SEC-013	Containers shall run with non-root users where feasible.
NET-SEC-014	Container images shall be scanned for known vulnerabilities.
NET-SEC-015	Unnecessary services and packages shall be removed from container images.
NET-SEC-016	Container runtime permissions shall be minimized.

9.5.9 Availability & DDoS Protection

The infrastructure shall support high availability and resilience.

Security measures include:

- Load balancing
- Automatic health checks
- Horizontal scaling
- Rate limiting
- Traffic filtering
- DDoS mitigation (through the hosting provider or CDN)
- Automatic service recovery where supported

9.5.10 Infrastructure Monitoring

Infrastructure monitoring shall include:

- CPU utilization
- Memory utilization
- Disk usage
- Database availability
- API availability
- AI provider connectivity
- Network latency
- SSL/TLS certificate status
- Error rates

Critical failures shall generate administrator alerts.

9.5.11 Logging & Monitoring

Infrastructure events shall be logged.

Examples include:

- Service startup and shutdown
- Configuration changes
- Deployment events
- Firewall events
- Reverse proxy events
- Infrastructure errors
- Security events

Logs shall be protected from unauthorized modification.

9.5.12 Business Rules

Rule ID	Rule
NET-BR-001	All production traffic shall use HTTPS.
NET-BR-002	Internal infrastructure components shall not be directly accessible from the public internet unless explicitly required.
NET-BR-003	Secrets shall be stored securely and never committed to version control.

NET-BR-004	Firewall configurations shall follow the principle of least privilege.
NET-BR-005	Infrastructure changes shall be documented and auditable.
NET-BR-006	Monitoring systems shall continuously assess infrastructure health.
NET-BR-007	Security events shall trigger alerts according to configured policies.
NET-BR-008	Infrastructure shall support scalable deployment without reducing security controls.

9.5.13 Exception Handling

Exception	Expected System Response
TLS certificate expired	Reject insecure connections where appropriate, generate alerts, and require certificate renewal.
Firewall misconfiguration	Log the issue, notify administrators, and prevent deployment where automated validation is available.
Service unavailable	Redirect traffic to healthy instances where applicable and notify administrators.
Secret retrieval failure	Prevent dependent services from starting or performing sensitive operations until secrets are available.
DDoS attack detected	Apply traffic mitigation controls and continue serving legitimate users where possible.
Infrastructure health check failure	Mark the affected service as unhealthy and initiate configured recovery procedures.

9.5.14 Acceptance Criteria

The network and infrastructure security requirements shall be considered complete when:

- All external communications use HTTPS.
- Network segmentation protects internal services.
- Firewall rules restrict unnecessary access.
- Secrets are securely managed.
- Container and runtime environments follow secure configurations.
- Infrastructure monitoring detects failures and security events.
- High-availability mechanisms support continuous operation.

- Administrative interfaces are protected from unauthorized access.

9.6 Application Security

9.6.1 Overview

Application Security defines the controls, secure development practices, and runtime protections implemented to safeguard ProcessMind AI against application-layer attacks.

The platform shall follow **Secure Software Development Lifecycle (SSDLC)** principles and implement security controls throughout development, testing, deployment, and maintenance.

Application security requirements apply to:

- Next.js Web Application
- Spring Boot Backend
- REST APIs
- AI Processing Services
- File Upload Services
- Background Processing Services
- Administrative Interfaces

9.6.2 Secure Development Principles

The application shall follow these principles:

- Secure by Design
- Secure by Default
- Defense in Depth
- Least Privilege
- Input Validation
- Output Encoding
- Fail Secure
- Separation of Duties
- Secure Error Handling
- Security Logging

9.6.3 Input Validation

All externally supplied input shall be validated before processing.

Requirement ID	Requirement
APP-SEC-001	All request parameters shall be validated on the server.
APP-SEC-002	Client-side validation shall improve usability but shall not replace server-side validation.
APP-SEC-003	Input length, type, format, and range shall be validated.
APP-SEC-004	Invalid input shall be rejected with standardized error responses.
APP-SEC-005	Unexpected fields shall be ignored or rejected according to API specifications.

9.6.4 Output Encoding

Application responses shall safely render user-supplied content.

Requirement ID	Requirement
APP-SEC-006	Dynamic content displayed in the user interface shall be properly encoded.
APP-SEC-007	User-generated content shall not be rendered as executable code.
APP-SEC-008	HTML injection shall be prevented.

9.6.5 SQL Injection Protection

The platform shall protect against SQL injection attacks.

Requirement ID	Requirement
APP-SEC-009	Parameterized queries or ORM mechanisms shall be used for database access.
APP-SEC-010	Dynamic SQL construction using untrusted input shall be prohibited.
APP-SEC-011	Database errors shall not expose implementation details.

9.6.6 Cross-Site Scripting (XSS) Protection

The application shall prevent execution of malicious scripts.

Requirement ID	Requirement
APP-SEC-012	User input displayed in web pages shall be encoded appropriately.
APP-SEC-013	Rich-text content shall be sanitized before rendering.
APP-SEC-014	A Content Security Policy (CSP) shall be implemented where feasible.

9.6.7 Cross-Site Request Forgery (CSRF)

For browser-based interactions, the platform shall mitigate CSRF risks.

Requirement ID	Requirement
APP-SEC-015	CSRF protection shall be implemented for cookie-based authenticated requests.
APP-SEC-016	Stateless JWT APIs shall be designed to minimize CSRF exposure.
APP-SEC-017	State-changing operations shall require appropriate authentication and authorization.

9.6.8 File Upload Security

The platform shall securely process uploaded documents.

Requirement ID	Requirement
APP-SEC-018	Uploaded files shall be validated against allowed file types.
APP-SEC-019	File size limits shall be enforced.
APP-SEC-020	Uploaded files shall undergo malware scanning before processing.
APP-SEC-021	File names shall be sanitized before storage.
APP-SEC-022	Uploaded files shall not be executed by the application server.

9.6.9 API Security

Application APIs shall implement appropriate protections.

Requirement ID	Requirement
APP-SEC-023	All protected APIs shall require authentication.
APP-SEC-024	Authorization shall be evaluated before business logic execution.
APP-SEC-025	Rate limiting shall protect against abuse.

APP-SEC-026	API responses shall not expose sensitive internal information.
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9.6.10 Dependency Management

Third-party software dependencies shall be managed securely.

Requirement ID	Requirement
APP-SEC-027	Dependencies shall originate from trusted sources.
APP-SEC-028	Known vulnerable dependencies shall be updated or replaced promptly.
APP-SEC-029	Dependency vulnerability scanning shall be integrated into the development process.

9.6.11 Secure Error Handling

The platform shall provide secure error responses.

Requirement ID	Requirement
APP-SEC-030	Internal implementation details shall not be exposed to users.
APP-SEC-031	Detailed stack traces shall only be available in development environments.
APP-SEC-032	Errors shall be logged securely for troubleshooting and auditing.

9.6.12 Secure Coding Practices

The development team shall follow secure coding practices.

These include:

- Code reviews.
- Static code analysis.
- Security testing.
- Principle of least privilege.
- Secure session handling.
- Avoidance of hardcoded secrets.
- Proper exception handling.

9.6.13 Vulnerability Management

The platform shall support continuous vulnerability management.

Activities include:

- Dependency vulnerability scanning.
- Static Application Security Testing (SAST).
- Dynamic Application Security Testing (DAST).
- Penetration testing.
- Security patch management.
- Remediation tracking.

9.6.14 Business Rules

Rule ID	Rule
APP-BR-001	All externally supplied input shall be validated before processing.
APP-BR-002	Database access shall use parameterized queries or ORM frameworks.
APP-BR-003	User-generated content shall be encoded or sanitized before rendering.
APP-BR-004	File uploads shall comply with validation and malware scanning policies.
APP-BR-005	Application APIs shall enforce authentication and authorization.
APP-BR-006	Sensitive implementation details shall not be exposed in error messages.
APP-BR-007	Vulnerability management shall be performed continuously throughout the software lifecycle.
APP-BR-008	Secure coding practices shall be followed across all application components.

9.6.15 Exception Handling

Exception	Expected System Response
Invalid input	Reject the request with a standardized validation error.

SQL injection attempt	Reject the request, log the event, and continue normal operation.
XSS payload detected	Sanitize or reject the input according to application policy.
Malware detected in uploaded file	Reject or quarantine the file and notify administrators where appropriate.
Dependency vulnerability identified	Assess severity, apply mitigation, and update the affected dependency as soon as practical.
Unexpected application exception	Return a generic error response, log diagnostic information securely, and preserve application stability.

9.6.16 Acceptance Criteria

The application security requirements shall be considered complete when:

- Input validation protects all externally supplied data.
- Output encoding and sanitization prevent script injection.
- Database interactions are protected from SQL injection.
- File uploads are validated and scanned.
- APIs enforce authentication and authorization.
- Dependencies are monitored for vulnerabilities.
- Error handling prevents disclosure of sensitive information.
- Secure development practices are consistently applied.

9.7 AI Security

9.7.1 Overview

AI Security defines the policies, controls, and governance mechanisms used to securely integrate Artificial Intelligence into ProcessMind AI.

The objective is to ensure that AI-assisted business process automation remains secure, reliable, explainable, auditable, and aligned with organizational policies.

AI security controls shall apply to:

- SOP Analysis
- Workflow Generation
- AI Reasoning
- AI Validation
- AI Optimization
- AI Chat
- AI Agents
- Prompt Management
- AI Provider Integrations

9.7.2 AI Security Objectives

The AI subsystem shall achieve the following objectives.

Objective	Description
Confidentiality	Protect organizational information processed by AI services.
Integrity	Prevent unauthorized modification of AI inputs and outputs.
Availability	Ensure AI services remain available and resilient.
Explainability	Provide reasoning and confidence information where applicable.
Human Oversight	Enable human review of important AI decisions.
Auditability	Maintain complete records of AI interactions.
Responsible AI	Support fair, transparent, and accountable AI usage.

9.7.3 AI Request Security

AI requests shall be protected before transmission.

Requirement ID	Requirement
AI-SEC-001	AI requests shall be transmitted over encrypted channels.
AI-SEC-002	Requests shall be associated with an authenticated user and organization.
AI-SEC-003	Requests shall be validated before being sent to AI providers.
AI-SEC-004	Sensitive information shall be minimized before transmission where feasible.

9.7.4 Prompt Security

The platform shall protect prompt construction and execution.

Requirement ID	Requirement
AI-SEC-005	Prompt templates shall be version-controlled.
AI-SEC-006	Prompt templates shall only be modified by authorized administrators.
AI-SEC-007	Dynamic prompt variables shall be validated before insertion.
AI-SEC-008	Prompt execution shall be logged.

9.7.5 Prompt Injection Protection

The platform shall implement safeguards against prompt injection attacks.

Requirement ID	Requirement
AI-SEC-009	User-supplied content shall be clearly separated from system instructions.
AI-SEC-010	AI outputs shall not automatically override system policies or business rules.
AI-SEC-011	AI instructions requesting privileged operations shall require authorization outside the model response.
AI-SEC-012	Potential prompt injection attempts shall be detected, logged, and flagged for review where appropriate.

9.7.6 AI Output Validation

AI-generated outputs shall be validated before operational use.

Requirement ID	Requirement
AI-SEC-013	AI-generated workflows shall undergo validation before deployment.
AI-SEC-014	Confidence scores shall accompany AI-generated outputs where supported.
AI-SEC-015	AI-generated business rules shall be reviewed before activation.
AI-SEC-016	Validation failures shall prevent automatic deployment.

9.7.7 Human-in-the-Loop (HITL)

Human oversight shall remain part of critical AI-assisted decisions.

Requirement ID	Requirement
AI-SEC-017	Low-confidence AI outputs shall require human review.
AI-SEC-018	High-risk workflow modifications shall require approval before activation.
AI-SEC-019	Human reviewers shall be able to approve, reject, or modify AI-generated outputs.
AI-SEC-020	Human decisions shall be recorded in audit logs.

9.7.8 AI Provider Security

External AI providers shall be integrated securely.

Requirement ID	Requirement
AI-SEC-021	API keys shall be securely managed.
AI-SEC-022	Provider communications shall use encrypted connections.
AI-SEC-023	AI provider availability shall be monitored.
AI-SEC-024	Provider failures shall trigger retry or fallback mechanisms where configured.

9.7.9 AI Governance

The platform shall support responsible AI governance.

Governance includes:

- AI model version tracking.
- Prompt version management.
- AI provider version tracking.
- Human review records.
- Decision traceability.
- Policy compliance.

9.7.10 AI Audit Logging

The following AI events shall be recorded.

- AI request submission
- Prompt template version used
- AI provider and model used
- Response time
- Confidence score
- Validation results
- Human review decisions
- Workflow deployment approval
- AI failures

Audit records shall be protected from unauthorized modification.

9.7.11 AI Abuse Prevention

The platform shall implement safeguards against misuse.

Requirement ID	Requirement
AI-SEC-025	AI APIs shall enforce authentication and authorization.

AI-SEC-026	AI requests shall be rate-limited.
AI-SEC-027	Excessive or suspicious AI usage shall be monitored.
AI-SEC-028	Organizations shall only access their own AI processing data.

9.7.12 Responsible AI Principles

The platform shall promote responsible AI by:

- Keeping humans responsible for final business decisions.
- Providing transparency into AI-generated recommendations.
- Supporting explainability where technically feasible.
- Preventing AI from bypassing security or authorization controls.
- Allowing organizations to review and validate AI outputs before operational deployment.

9.7.13 Business Rules

Rule ID	Rule
AI-BR-001	AI shall assist decision-making but shall not bypass mandatory approval workflows.
AI-BR-002	AI-generated workflows shall require validation before production deployment.
AI-BR-003	AI processing shall respect tenant isolation.
AI-BR-004	Prompt templates shall be managed under version control.
AI-BR-005	Prompt injection protections shall be applied to AI interactions.
AI-BR-006	AI provider credentials shall be securely managed.
AI-BR-007	Human review shall remain available for configurable high-risk operations.
AI-BR-008	AI activities shall be fully auditable.

9.7.14 Exception Handling

Exception	Expected System Response
AI provider unavailable	Retry according to policy or fail over to an alternative provider if configured.
Prompt injection attempt detected	Reject or isolate the request, log the event, and notify administrators where appropriate.

AI validation failure	Prevent deployment or execution of the generated output until resolved.
Low-confidence AI output	Route the result to Human-in-the-Loop review.
AI provider timeout	Cancel the request after the configured timeout and return an appropriate error response.
AI model version mismatch	Log the event, maintain compatibility where possible, and require administrative review if necessary.

9.7.15 Acceptance Criteria

The AI security requirements shall be considered complete when:

- AI communications are encrypted.
- Prompt templates are securely managed.
- Prompt injection protections are implemented.
- AI-generated outputs undergo validation.
- Human review is supported for configurable scenarios.
- AI provider integrations are securely managed.
- AI activities are fully audited.
- Responsible AI principles are consistently applied.

9.8 Audit Logging & Monitoring Security

9.8.1 Overview

Audit Logging & Monitoring Security defines the policies, controls, and mechanisms used to record, monitor, protect, and analyze security-related and operational events within ProcessMind AI.

The platform shall maintain comprehensive audit trails to support security investigations, compliance requirements, operational monitoring, and forensic analysis.

Audit logging shall apply to all major platform components, including:

- Authentication Services
- Authorization Services
- Workflow Management
- Workflow Execution
- AI Processing
- Administrative Operations

- Database Operations
- File Management
- REST APIs

9.8.2 Audit Objectives

The audit subsystem shall achieve the following objectives.

Objective	Description
Accountability	Identify who performed an action.
Traceability	Reconstruct system events accurately.
Integrity	Protect audit logs from unauthorized modification.
Compliance	Support organizational and regulatory requirements.
Security Monitoring	Detect suspicious activities.
Incident Investigation	Support forensic analysis after security incidents.

9.8.3 Events to be Logged

The following categories of events shall be audited.

Category	Examples
Authentication	Login, logout, failed login, password reset
Authorization	Permission changes, access denials
User Management	User creation, updates, deletion
Organization Management	Organization creation, suspension
SOP Management	Upload, update, archive
Workflow Management	Creation, deployment, archival
Workflow Execution	Start, completion, cancellation
AI Processing	Prompt execution, AI responses, validation
Administration	Configuration changes, backups, restores

Security	Account lockout, suspicious activity
System	Service startup, shutdown, deployment

9.8.4 Audit Log Contents

Each audit record shall contain the following information.

Field	Description
Event ID	Unique identifier
Timestamp	Date and time (UTC)
User ID	Authenticated user
Organization ID	Associated organization
Event Type	Category of event
Resource Type	Resource affected
Resource ID	Unique resource identifier
Action	Operation performed
Result	Success or failure
Client IP Address	Originating IP address
User Agent	Client application details
Additional Metadata	Optional contextual information

9.8.5 Log Integrity

Audit logs shall be protected against tampering.

Requirement ID	Requirement
AUD-SEC-001	Audit logs shall be append-only.
AUD-SEC-002	Audit records shall not be modified or deleted by standard users.
AUD-SEC-003	Administrative access to audit logs shall be restricted.
AUD-SEC-004	Audit logs shall be included in backup procedures.

9.8.6 Monitoring

The platform shall continuously monitor:

- Authentication failures
- Account lockouts
- Authorization failures
- AI provider failures
- Workflow execution failures
- API errors
- Infrastructure health
- Database availability
- High resource utilization
- Security events

Monitoring shall support configurable thresholds and alerts.

9.8.7 Security Alerts

The platform shall generate alerts for significant events.

Examples include:

- Multiple failed login attempts
- Suspicious authentication activity
- Privilege escalation attempts
- Cross-tenant access attempts
- AI provider outages
- Backup failures
- Database connectivity failures
- Excessive API requests
- Unusual administrative activity

Alerts may be delivered through:

- In-app notifications
- Email
- WebSocket notifications
- External monitoring platforms (future)

9.8.8 Log Retention

Audit logs shall be retained according to organizational policy.

Log Type	Default Retention
Authentication Logs	2 years
Security Logs	7 years
Audit Logs	7 years
Workflow Execution Logs	5 years
AI Processing Logs	2 years
System Logs	1 year
Error Logs	1 year

Retention periods shall be configurable where permitted.

9.8.9 Log Access Control

Access to audit logs shall follow RBAC.

Role	Access Level
System Administrator	Full access
Organization Administrator	Organization-specific logs
Security Administrator (Future)	Security logs
Auditor	Read-only access
Standard User	No direct access

9.8.10 Business Rules

Rule ID	Rule
AUD-BR-001	Security-relevant events shall be recorded automatically.
AUD-BR-002	Audit logs shall be immutable after creation.
AUD-BR-003	Audit records shall include sufficient information for investigation.
AUD-BR-004	Audit logs shall be protected from unauthorized access.
AUD-BR-005	Monitoring shall detect abnormal system behavior.

AUD-BR-006	Critical security events shall generate alerts.
AUD-BR-007	Audit log retention shall follow configured policies.
AUD-BR-008	Monitoring failures shall be reported to administrators.

9.8.11 Exception Handling

Exception	Expected System Response
Audit logging failure	Record the failure in system diagnostics, alert administrators, and continue operation where safe.
Monitoring service unavailable	Continue collecting logs locally where possible and notify administrators.
Unauthorized audit log access	Deny access, log the attempt, and generate a security alert.
Log storage nearing capacity	Generate warnings and initiate configured retention or archival procedures.
Alert delivery failure	Retry delivery according to policy and record the failure.
Time synchronization issue	Log the discrepancy and use synchronized system time where available.

9.8.12 Acceptance Criteria

The audit logging and monitoring security requirements shall be considered complete when:

- Security and operational events are recorded automatically.
- Audit logs are immutable and protected.
- Monitoring detects significant operational and security events.
- Critical events generate alerts.
- Audit log retention complies with organizational policies.
- Access to audit information follows RBAC.

- Monitoring supports incident investigation and operational analysis.

9.9 Incident Response & Security Acceptance Criteria

9.9.1 Overview

Incident Response defines the procedures, responsibilities, and controls for identifying, responding to, containing, recovering from, and documenting security incidents affecting ProcessMind AI.

The objective is to minimize operational disruption, protect organizational data, preserve evidence, and continuously improve the platform's security posture.

Incident response applies to:

- Authentication systems
- Authorization systems
- REST APIs
- AI services
- Workflow engine
- Database
- File storage
- Infrastructure
- Administrative services

9.9.2 Security Incident Classification

Security incidents shall be classified according to severity.

Severity	Description	Example
Critical	Immediate threat affecting confidentiality, integrity, or availability	Database compromise, ransomware, tenant data exposure
High	Significant security event requiring urgent action	Administrative account compromise, AI provider credential leakage
Medium	Limited security impact	Repeated failed login attempts, abnormal API usage
Low	Minor security issue	Single authentication failure, isolated validation error
Informational	Non-security operational event	Normal administrative configuration update

Incident priority shall determine escalation procedures and response timelines.

9.9.3 Incident Response Lifecycle

The platform shall support the following incident response process.



9.9.4 Incident Detection

Potential security incidents may be detected through:

- Authentication monitoring
- Audit logs
- Infrastructure monitoring
- AI monitoring
- API monitoring
- Database monitoring
- User reports
- Automated alerts
- External monitoring systems

Detection mechanisms shall generate alerts according to configured severity levels.

9.9.5 Incident Containment

Upon confirmation of an incident, the platform shall support appropriate containment actions.

Examples include:

- Disable compromised user accounts.
- Revoke JWT refresh tokens.
- Block suspicious IP addresses.
- Suspend affected API keys.
- Isolate compromised services.
- Temporarily disable affected AI providers.
- Prevent execution of affected workflows.

Containment actions shall be proportionate to the severity of the incident.

9.9.6 Recovery

Recovery procedures shall restore normal operations while preserving system integrity.

Recovery activities may include:

- Restore verified backups.
- Rotate compromised credentials.
- Redeploy affected services.
- Restore AI provider connectivity.
- Validate database integrity.
- Verify workflow consistency.
- Resume suspended workflows where appropriate.

Recovery actions shall be documented.

9.9.7 Post-Incident Review

Every significant security incident shall undergo a documented review.

The review shall include:

- Root cause analysis.
- Timeline reconstruction.
- Impact assessment.
- Corrective actions.
- Preventive actions.
- Lessons learned.
- Verification of implemented improvements.

9.9.8 Security Testing

The platform shall undergo regular security testing.

Test Type	Purpose
Unit Security Testing	Validate security-related code paths
Integration Security Testing	Verify secure interactions between components
Authentication Testing	Validate login and session controls
Authorization Testing	Verify RBAC and access controls
API Security Testing	Validate REST API protections
File Upload Testing	Verify upload restrictions and malware handling
AI Security Testing	Validate AI-specific safeguards
Penetration Testing	Simulate real-world attacks
Regression Security Testing	Ensure fixes do not introduce new vulnerabilities

9.9.9 Security Metrics

The platform shall track security-related metrics.

Examples include:

- Failed authentication rate
- Account lockout frequency
- API abuse attempts
- Authorization failures

- AI provider failure rate
- Detected prompt injection attempts
- Average incident response time
- Mean Time to Detect (MTTD)
- Mean Time to Respond (MTTR)
- Vulnerability remediation time

These metrics shall support continuous improvement.

9.9.10 Business Rules

Rule ID	Rule
SEC-IR-001	Security incidents shall be classified according to severity.
SEC-IR-002	Critical incidents shall generate immediate alerts.
SEC-IR-003	Incident response activities shall be documented.
SEC-IR-004	Recovery shall preserve data integrity and auditability.
SEC-IR-005	Significant incidents shall undergo post-incident review.
SEC-IR-006	Security testing shall be performed throughout the software lifecycle.
SEC-IR-007	Security metrics shall be monitored and reviewed periodically.
SEC-IR-008	Corrective actions shall be tracked until completion.

9.9.11 Exception Handling

Exception	Expected System Response
Monitoring service unavailable	Continue collecting available local logs, notify administrators, and restore monitoring as soon as practical.
False positive security alert	Document the investigation outcome and update detection rules if appropriate.
Backup restoration failure during recovery	Halt the recovery process, preserve available evidence, and use the most recent verified backup where possible.
Compromised administrative account	Immediately revoke active sessions, rotate credentials, and initiate administrative review.

AI provider security incident	Suspend or isolate the affected provider according to organizational policy and switch to an approved alternative if available.
Incomplete incident information	Continue investigation while preserving evidence and documenting assumptions.

9.9.12 Security Acceptance Criteria

The overall security architecture of ProcessMind AI shall be considered acceptable when:

Identity & Access Security

- Authentication is securely implemented.
- RBAC authorization is consistently enforced.
- Multi-tenant isolation prevents unauthorized cross-organization access.
- Administrative privileges follow the Principle of Least Privilege.

Data Security

- Sensitive information is encrypted during transmission and storage where required.
- Backups are protected.
- Secure deletion complies with retention policies.
- Sensitive information is not exposed through logs or APIs.

Application Security

- Input validation protects against malicious input.
- SQL injection and XSS protections are implemented.
- File upload security controls operate correctly.
- Dependency vulnerabilities are monitored and addressed.

AI Security

- Prompt security controls are implemented.
- Prompt injection protections are active.
- AI-generated outputs are validated.
- Human-in-the-Loop review supports configurable high-risk operations.

- AI activities are fully auditable.

Operational Security

- Security monitoring detects significant events.
- Audit logs are immutable.
- Critical alerts are generated appropriately.
- Incident response procedures are documented and tested.

Infrastructure Security

- HTTPS is enforced.
- Internal services are protected.
- Secrets are securely managed.
- Infrastructure monitoring is operational.

Compliance & Governance

- Security controls align with recognized industry best practices.
- Security testing is performed throughout development and deployment.
- Security incidents are documented and reviewed.
- Continuous improvement processes are established.

Chapter 10 – Non-Functional Requirements

10.1 Performance Requirements

10.1.1 Overview

Performance Requirements define the measurable operational characteristics that ProcessMind AI shall satisfy under expected workloads.

The platform shall provide responsive user interactions, efficient AI processing, scalable workflow execution, and reliable API performance while maintaining system stability.

Performance requirements apply to:

- Web Application
- REST APIs

- AI Processing Engine
- Workflow Execution Engine
- Database
- Analytics Engine
- Notification Services
- Background Processing Services

10.1.2 Performance Objectives

The platform shall achieve the following objectives.

Objective	Description
Responsiveness	Provide fast response times for user operations.
Throughput	Process large numbers of concurrent requests efficiently.
Scalability	Maintain performance as workload increases.
Stability	Continue operating under sustained load.
Resource Efficiency	Optimize CPU, memory, storage, and network utilization.

10.1.3 User Interface Performance

Requirement ID	Requirement
PERF-001	Standard page navigation shall complete within 2 seconds under normal operating conditions.
PERF-002	Dashboard loading shall complete within 3 seconds for typical datasets.
PERF-003	Search results shall begin displaying within 2 seconds for indexed data.
PERF-004	UI interactions shall remain responsive during background processing.

10.1.4 REST API Performance

Requirement ID	Requirement
PERF-005	Standard CRUD API requests shall complete within 500 ms under normal load.
PERF-006	Authentication requests shall complete within 1 second .
PERF-007	API responses shall support compression where appropriate.
PERF-008	Long-running operations shall use asynchronous processing where applicable.

10.1.5 AI Processing Performance

Because AI response times vary by provider and workload, targets should be defined as operational goals rather than hard guarantees.

Requirement ID	Requirement
PERF-009	AI analysis requests shall provide progress feedback for long-running operations.
PERF-010	AI workflow generation should complete within acceptable operational targets based on document size and provider performance.
PERF-011	AI validation shall execute asynchronously for computationally intensive tasks where appropriate.
PERF-012	AI provider latency shall be monitored continuously.

10.1.6 Workflow Execution Performance

Requirement ID	Requirement
PERF-013	Workflow instance creation shall complete within 2 seconds under normal conditions.
PERF-014	Task assignment shall occur within 1 second after workflow progression.
PERF-015	Workflow state changes shall be persisted reliably before confirmation is returned.
PERF-016	Concurrent workflow execution shall be supported without degrading system stability.

10.1.7 Database Performance

Requirement ID	Requirement
PERF-017	Frequently executed queries shall be optimized using appropriate indexing.
PERF-018	Database connection pooling shall be implemented.
PERF-019	Long-running queries shall be monitored and optimized.
PERF-020	Database transactions shall maintain ACID properties while minimizing lock contention.

10.1.8 Scalability Targets

The architecture shall support horizontal scaling.

Component	Scalability Requirement
Web Servers	Horizontal scaling supported
API Servers	Horizontal scaling supported
Background Workers	Horizontal scaling supported
Notification Services	Horizontal scaling supported
AI Processing	Multiple provider instances supported

The platform shall be designed so that capacity can increase without requiring major architectural changes.

10.1.9 Resource Utilization

The platform shall monitor:

- CPU utilization
- Memory utilization
- Disk usage
- Network utilization

- Database connection usage
- Queue length
- AI provider response time

Thresholds shall be configurable.

10.1.10 Performance Monitoring

The following metrics shall be collected continuously.

Metric

1. API response time
2. Page load time
3. Workflow execution time
4. AI processing duration
5. Database query latency
6. Error rate
7. Concurrent users
8. Throughput (requests/second)
9. Queue processing time
10. Resource utilization

10.1.11 Business Rules

Rule ID	Rule
PERF-BR-001	Performance targets shall be monitored continuously.

PERF-BR-002	Long-running operations shall not block unrelated user interactions.
PERF-BR-003	Performance monitoring data shall support capacity planning.
PERF-BR-004	Performance degradation beyond configured thresholds shall generate alerts.
PERF-BR-005	Resource utilization shall remain within acceptable operational limits under expected workloads.
PERF-BR-006	Performance optimizations shall not compromise security or data integrity.

10.1.12 Exception Handling

Exception	Expected System Response
High API latency	Record performance metrics, notify administrators if thresholds are exceeded, and apply scaling policies where configured.
AI provider slowdown	Inform users of processing status, continue monitoring, and switch providers if failover is configured.
Database performance degradation	Detect long-running queries, log the issue, and initiate performance diagnostics.
Queue backlog	Increase worker capacity where possible or notify administrators.
Resource exhaustion	Apply configured scaling or throttling policies while preserving system stability.

10.1.13 Acceptance Criteria

The performance requirements shall be considered complete when:

- User interactions remain responsive under expected workloads.
- REST APIs satisfy defined response-time objectives under normal operating conditions.
- AI processing provides progress visibility and reliable completion.
- Workflow execution performs consistently.
- Database operations remain efficient.
- Performance metrics are continuously monitored.
- Scalability mechanisms support increasing workloads without major architectural changes.

10.2 Reliability Requirements

10.2.1 Overview

Reliability Requirements define the characteristics and mechanisms that enable ProcessMind AI to operate consistently and correctly over time, despite hardware failures, software defects, network interruptions, or third-party service outages.

The platform shall be designed to minimize service interruptions, prevent data corruption, and recover gracefully from failures while maintaining data integrity and user confidence.

Reliability requirements apply to:

- Web Application
- REST APIs
- Workflow Engine
- AI Processing Engine
- Database
- Notification Services
- Background Workers
- File Storage

10.2.2 Reliability Objectives

The platform shall achieve the following objectives.

Objective	Description
Fault Tolerance	Continue operating despite component failures where feasible.
Data Consistency	Preserve accurate and consistent business data.
Service Resilience	Recover gracefully from failures.
Error Recovery	Detect and recover from recoverable failures.
Business Continuity	Minimize disruption to business operations.

10.2.3 Fault Tolerance

The platform shall tolerate recoverable failures.

Requirement ID	Requirement
REL-001	Failure of one application instance shall not stop the entire service.
REL-002	Background worker failures shall not affect active user sessions.
REL-003	Temporary AI provider failures shall not corrupt workflow data.
REL-004	Recoverable failures shall be detected automatically where possible.

10.2.4 Error Recovery

The system shall recover from transient failures whenever appropriate.

Requirement ID	Requirement
REL-005	Transient failures may be retried using configurable retry policies.
REL-006	Failed background jobs shall support controlled retry mechanisms.
REL-007	Recoverable API failures shall return standardized error responses.
REL-008	Recovery attempts shall be logged for auditing and diagnostics.

10.2.5 Data Consistency

The platform shall maintain consistent business data.

Requirement ID	Requirement
REL-009	Database transactions shall satisfy ACID properties.
REL-010	Partial workflow updates shall not leave inconsistent system states.
REL-011	Workflow version history shall remain consistent after recovery.
REL-012	Backup restoration shall preserve referential integrity.

10.2.6 AI Service Reliability

Because AI services depend on external providers, the platform shall implement resilience mechanisms.

Requirement ID	Requirement
REL-013	AI provider failures shall be detected automatically.
REL-014	Retry policies shall be configurable for transient AI failures.
REL-015	AI processing failures shall not affect unrelated workflows.
REL-016	Users shall receive meaningful status updates during AI processing interruptions.

10.2.7 Graceful Degradation

When non-critical services become unavailable, the platform shall continue providing core functionality where possible.

Examples include:

Service Failure	Expected Behavior
AI provider unavailable	Continue allowing manual workflow creation and editing.
Email service unavailable	Queue notifications for later delivery and continue workflow execution.
Analytics unavailable	Continue workflow operations while temporarily disabling analytics.
Report generation unavailable	Delay report generation without affecting operational workflows.

10.2.8 Service Resilience

The platform shall support resilient service operation.

Requirement ID	Requirement
REL-017	Health checks shall monitor critical services.
REL-018	Failed services shall be restartable without affecting stored data.
REL-019	Critical failures shall generate administrator alerts.
REL-020	Recovery procedures shall be documented and periodically tested.

10.2.9 Reliability Monitoring

The following reliability indicators shall be monitored:

- Service availability
- API success rate
- Workflow execution success rate
- AI provider availability
- Background job success rate
- Database availability
- Queue processing health
- Error frequency
- Retry frequency

Monitoring thresholds shall be configurable.

10.2.10 Business Rules

Rule ID	Rule
REL-BR-001	Recoverable failures shall be retried according to configured policies.
REL-BR-002	Data consistency shall be preserved during failure recovery.
REL-BR-003	Workflow execution shall remain isolated from unrelated service failures.
REL-BR-004	AI provider failures shall not prevent manual workflow management.
REL-BR-005	Recovery activities shall be logged.
REL-BR-006	Reliability metrics shall be continuously monitored.
REL-BR-007	Critical failures shall trigger operational alerts.
REL-BR-008	Recovery procedures shall be validated periodically.

10.2.11 Exception Handling

Exception	Expected System Response
AI provider outage	Retry according to policy or switch to an alternative provider if configured.
Database connection interruption	Retry connections where appropriate and prevent partial transactions.
Background worker failure	Reassign pending jobs and restart the worker where possible.
Notification delivery failure	Queue notifications for later retry without interrupting workflow execution.
Workflow execution error	Roll back incomplete operations where appropriate and preserve execution history.
File storage temporary outage	Delay affected operations, notify users, and resume automatically when storage becomes available.

10.2.12 Acceptance Criteria

The reliability requirements shall be considered complete when:

- The platform detects and handles recoverable failures appropriately.
- Data consistency is maintained during failures and recovery.
- AI provider interruptions do not compromise unrelated business operations.
- Core functionality continues during non-critical service outages.
- Reliability metrics are continuously monitored.
- Recovery procedures are documented and tested.
- Operational alerts are generated for critical failures.

10.3 Availability Requirements

10.3.1 Overview

Availability Requirements define the characteristics and mechanisms that ensure ProcessMind AI remains accessible and operational for authorized users with minimal service interruption.

The platform shall support high availability through redundancy, monitoring, automated recovery, and disaster recovery planning.

Availability requirements apply to:

- Web Application
- REST APIs
- AI Processing Services
- Workflow Engine
- Database
- Notification Services
- Background Workers
- Storage Services

10.3.2 Availability Objectives

The platform shall achieve the following objectives.

Objective	Description
High Availability	Minimize unplanned service interruptions.
Business Continuity	Maintain critical operations during component failures.
Rapid Recovery	Restore services quickly after failures.
Fault Isolation	Prevent failures from affecting unrelated services.
Operational Resilience	Support continuous business operations.

10.3.3 Service Availability Targets

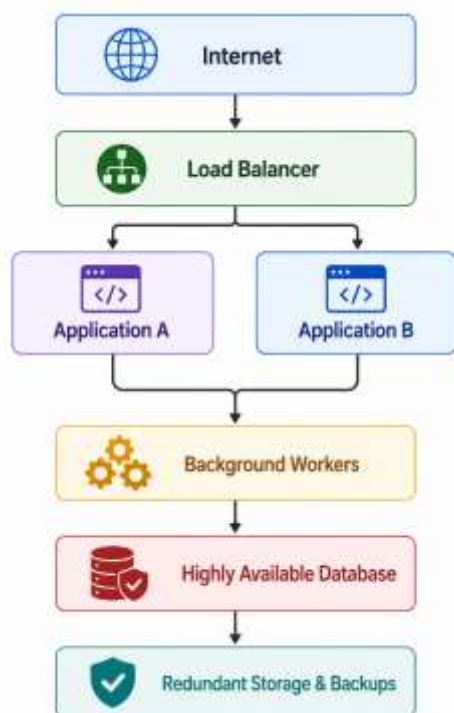
The following service availability targets are defined.

Service	Target Availability
Web Application	≥ 99.9% monthly uptime
REST APIs	≥ 99.9% monthly uptime
Authentication Services	≥ 99.95% monthly uptime
Workflow Engine	≥ 99.9% monthly uptime
Database Services	≥ 99.95% monthly uptime
AI Processing Services	Best effort, subject to external AI provider availability
Notification Services	≥ 99.5% monthly uptime

Availability targets exclude scheduled maintenance windows.

10.3.4 High Availability Architecture

The production deployment shall support redundant components where appropriate



This architecture enables continued service if a single application instance becomes unavailable.

10.3.5 Planned Maintenance

Planned maintenance activities shall minimize service disruption.

Requirement ID	Requirement
AVAIL-001	Scheduled maintenance shall be announced in advance where applicable.
AVAIL-002	Maintenance activities shall be planned to minimize user impact.
AVAIL-003	Maintenance windows shall be documented.
AVAIL-004	Critical security patches may be applied outside normal maintenance windows if necessary.

10.3.6 Failover

The platform shall support failover for critical services.

Requirement ID	Requirement
AVAIL-005	Application instances shall support automatic restart where possible.
AVAIL-006	Load balancers shall route traffic away from unhealthy instances.
AVAIL-007	AI provider failover may be supported if multiple providers are configured.
AVAIL-008	Background processing shall resume automatically after recoverable failures.

10.3.7 Backup & Disaster Recovery

The platform shall support disaster recovery procedures.

Requirement ID	Requirement
AVAIL-009	Regular backups shall be performed according to the backup policy.
AVAIL-010	Backup restoration procedures shall be tested periodically.
AVAIL-011	Disaster recovery procedures shall be documented.
AVAIL-012	Recovery operations shall preserve data integrity.

10.3.8 Recovery Objectives

The platform shall define operational recovery objectives.

Objective	Target
Recovery Time Objective (RTO)	≤ 4 hours

Recovery Point Objective (RPO)	≤ 15 minutes
Critical Security Incident Response	Immediate containment, followed by recovery according to incident response procedures

These values may be refined according to deployment architecture and customer service-level agreements.

10.3.9 Availability Monitoring

The platform shall continuously monitor:

- Service uptime
- API availability
- Database availability
- AI provider connectivity
- Background worker status
- Storage availability
- Backup success
- Health check status
- SSL/TLS certificate validity

Monitoring failures shall generate alerts.

10.3.10 Business Rules

Rule ID	Rule
AVAIL-BR-001	Critical services shall be monitored continuously.
AVAIL-BR-002	Unhealthy services shall be removed from active traffic where supported.
AVAIL-BR-003	Backup procedures shall support disaster recovery.
AVAIL-BR-004	Availability metrics shall be reviewed regularly.
AVAIL-BR-005	Planned maintenance shall be documented.

AVAIL-BR-006	Recovery procedures shall preserve system integrity.
AVAIL-BR-007	Critical failures shall trigger operational alerts.
AVAIL-BR-008	Disaster recovery plans shall be periodically validated.

10.3.11 Exception Handling

Exception	Expected System Response
Application instance failure	Remove the instance from service, restart it if possible, and continue serving requests through healthy instances.
Database failover event	Redirect traffic to the standby database where configured and preserve transactional integrity.
AI provider outage	Notify users, retry according to policy, or switch to an alternative provider if configured.
Storage service interruption	Delay affected operations until storage is restored while preserving queued work where possible.
Backup failure	Generate an alert, investigate the cause, and perform a successful backup as soon as practical.
Load balancer failure	Activate redundancy where available and notify administrators immediately.

10.3.12 Acceptance Criteria

The availability requirements shall be considered complete when:

- Critical services satisfy defined availability targets.
- High-availability architecture supports continued operation during component failures.
- Failover mechanisms function correctly.
- Backup and disaster recovery procedures are documented and tested.

- Recovery objectives are measurable and achievable.
- Availability monitoring provides timely alerts.
- Planned maintenance minimizes disruption to users.

10.4 Scalability Requirements

10.4.1 Overview

Scalability Requirements define the ability of ProcessMind AI to efficiently handle increasing workloads, user growth, organizational expansion, and larger data volumes while maintaining acceptable performance and reliability.

The platform shall support both horizontal and vertical scaling and shall be designed using cloud-native architectural principles.

Scalability requirements apply to:

- Web Application
- REST APIs
- Workflow Engine
- AI Processing Services
- Database
- File Storage
- Notification Services
- Background Workers

10.4.2 Scalability Objectives

The platform shall achieve the following objectives.

Objective	Description
User Growth	Support increasing numbers of users and organizations.
Workload Growth	Handle increased workflow executions and AI requests.
Data Growth	Efficiently manage expanding databases and file storage.
Resource Elasticity	Increase or decrease computing resources based on demand.
Cost Efficiency	Scale resources without unnecessary operational costs.

10.4.3 Horizontal Scalability

The platform shall support horizontal scaling.

Requirement ID	Requirement
SCALE-001	Multiple application server instances shall be supported.
SCALE-002	REST API services shall operate behind a load balancer.
SCALE-003	Background worker instances shall scale independently.
SCALE-004	Stateless application services shall enable efficient horizontal scaling.

10.4.4 Vertical Scalability

The platform shall support vertical scaling where appropriate.

Requirement ID	Requirement
SCALE-005	Application servers shall support increased CPU and memory allocation.
SCALE-006	Database resources shall be expandable according to workload requirements.
SCALE-007	AI processing resources shall be configurable based on processing demand.

10.4.5 Concurrent User Support

The architecture shall support concurrent users.

Requirement ID	Requirement
SCALE-008	The platform shall support multiple concurrent authenticated users.
SCALE-009	Concurrent workflow execution shall not compromise data consistency.
SCALE-010	User sessions shall remain isolated and secure.
SCALE-011	System responsiveness shall be maintained under expected concurrent workloads.

Note: Exact user capacity depends on deployment infrastructure and hardware resources.

10.4.6 AI Workload Scalability

AI processing shall support increasing workloads.

Requirement ID	Requirement
SCALE-012	AI requests shall be processed asynchronously where appropriate.
SCALE-013	Multiple AI requests may be processed concurrently subject to configured resource limits.
SCALE-014	AI provider utilization shall be monitored to support capacity planning.
SCALE-015	AI processing queues shall support scalable background execution.

10.4.7 Database Scalability

The database architecture shall support growth.

Requirement ID	Requirement
SCALE-016	Database indexing shall support efficient query execution.
SCALE-017	Connection pooling shall optimize database utilization.
SCALE-018	Database maintenance operations shall support long-term growth.

SCALE-019	Database storage shall be expandable without application redesign.
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10.4.8 Storage Scalability

The platform shall support scalable storage.

Requirement ID	Requirement
SCALE-020	SOP document storage shall support organizational growth.
SCALE-021	Backup storage shall expand according to retention policies.
SCALE-022	Uploaded files shall be managed independently from application servers.
SCALE-023	Storage capacity shall be monitored continuously.

10.4.9 Queue Scalability

Background processing shall scale independently.

Requirement ID	Requirement
SCALE-024	AI processing queues shall support multiple workers.
SCALE-025	Notification queues shall support concurrent processing.
SCALE-026	Report generation queues shall support asynchronous execution.
SCALE-027	Queue backlogs shall be monitored and managed.

10.4.10 Auto-Scaling

The architecture shall support automatic scaling where infrastructure permits.

Requirement ID	Requirement
SCALE-028	Application instances may scale automatically based on CPU, memory, or request load.
SCALE-029	Background workers may scale based on queue length.

SCALE-030	Scaling events shall be logged for operational analysis.
SCALE-031	Auto-scaling policies shall be configurable.

10.4.11 Capacity Planning

The platform shall support ongoing capacity planning.

Monitoring shall include:

- Number of organizations
- Active users
- Concurrent sessions
- Workflow executions
- AI request volume
- Storage utilization
- Database growth
- API throughput
- Queue depth
- Network utilization

Capacity reports shall assist administrators in forecasting future infrastructure requirements.

10.4.12 Business Rules

Rule ID	Rule
SCALE-BR-001	The platform shall support horizontal scaling of stateless services.
SCALE-BR-002	Data consistency shall be maintained during scaling operations.
SCALE-BR-003	Auto-scaling shall not compromise security controls.
SCALE-BR-004	AI processing shall remain isolated from unrelated workloads.
SCALE-BR-005	Queue processing shall scale independently of user-facing services.
SCALE-BR-006	Capacity planning metrics shall be collected continuously.
SCALE-BR-007	Scaling operations shall be transparent to end users wherever possible.
SCALE-BR-008	Storage expansion shall not require application redesign.

10.4.13 Exception Handling

Exception	Expected System Response
Sudden increase in user traffic	Apply auto-scaling policies where available and continue monitoring system health.
AI processing queue overload	Scale background workers where possible, prioritize queued jobs, and inform users of expected delays.
Database resource exhaustion	Generate alerts, optimize resource usage, and expand capacity according to operational procedures.
Storage nearing capacity	Notify administrators and initiate storage expansion according to policy.
Auto-scaling failure	Continue operating with available resources, record the event, and notify administrators.
Queue processing bottleneck	Redistribute work across available workers and monitor queue recovery.

10.4.14 Acceptance Criteria

The scalability requirements shall be considered complete when:

- Stateless application components support horizontal scaling.
- Vertical scaling can increase available computing resources without application redesign.
- The platform maintains acceptable performance under increasing workloads.
- AI processing, background workers, and notification services scale independently.
- Database and storage architectures support long-term growth.
- Auto-scaling policies operate correctly where supported.
- Capacity planning metrics are continuously collected and reviewed.

10.5 Maintainability Requirements

10.5.1 Overview

Maintainability Requirements define the characteristics that enable ProcessMind AI to be efficiently maintained, enhanced, tested, and supported throughout its lifecycle.

The platform shall adopt modular software architecture, coding standards, automated testing, documentation, and continuous integration practices to facilitate long-term evolution.

Maintainability applies to:

- Frontend Application
- Backend Services
- REST APIs
- AI Processing Components
- Workflow Engine
- Database
- Infrastructure Configuration
- CI/CD Pipeline

10.5.2 Maintainability Objectives

The platform shall achieve the following objectives.

Objective	Description
Modularity	Separate functionality into independent components.
Readability	Maintain clear and understandable source code.
Extensibility	Support future features with minimal impact.
Testability	Enable efficient automated and manual testing.
Reusability	Promote reuse of software components.

10.5.3 Modular Architecture

The platform shall follow a modular architecture.

Requirement ID	Requirement
MAINT-001	System functionality shall be organized into independent modules.
MAINT-002	Modules shall communicate through well-defined interfaces.
MAINT-003	Changes within one module should minimize impacts on unrelated modules.
MAINT-004	Business logic shall remain independent of presentation concerns.

10.5.4 Coding Standards

The development team shall follow consistent coding standards.

Requirement ID	Requirement
MAINT-005	Source code shall follow agreed project coding conventions.
MAINT-006	Source code shall use meaningful identifiers and consistent naming.
MAINT-007	Complex business logic shall include appropriate documentation.
MAINT-008	Deprecated functionality shall be identified before removal.

10.5.5 Documentation

The platform shall maintain comprehensive documentation.

Required documentation includes:

- Software Requirements Specification (SRS)
- Software Architecture Document (SAD)
- API Documentation (OpenAPI/Swagger)
- Database Documentation
- Deployment Guide
- Administrator Guide
- User Manual
- Developer Guide
- Release Notes
- Change Log

Documentation shall be updated as part of the software development lifecycle.

10.5.6 Logging

Application logging shall support maintenance and troubleshooting.

Requirement ID	Requirement
MAINT-009	Operational logs shall support diagnostics.
MAINT-010	Error logs shall provide sufficient technical information for troubleshooting without exposing sensitive data.
MAINT-011	Log formats shall remain consistent across services.

MAINT-012	Log verbosity shall be configurable.
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10.5.7 Configuration Management

Application configuration shall be centrally managed.

Requirement ID	Requirement
MAINT-013	Environment-specific configuration shall be externalized from source code.
MAINT-014	Configuration changes shall be documented.
MAINT-015	Sensitive configuration values shall be managed securely.
MAINT-016	Default configurations shall support secure deployment.

10.5.8 Testing

The platform shall support comprehensive testing.

Requirement ID	Requirement
MAINT-017	Unit testing shall be performed for core business logic.
MAINT-018	Integration testing shall verify interactions between components.
MAINT-019	End-to-end testing shall validate major business workflows.
MAINT-020	Regression testing shall be performed before major releases.
MAINT-021	Automated testing shall be integrated into the development workflow.

10.5.9 Version Control

The project shall use version control.

Requirement ID	Requirement
MAINT-022	Source code shall be managed using Git.
MAINT-023	Releases shall be versioned according to an agreed versioning.
MAINT-024	Source code changes shall be traceable through commit history.

MAINT-025	Release artifacts shall be reproducible.
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10.5.10 CI/CD Support

The development process shall support continuous integration and deployment.

Requirement ID	Requirement
MAINT-026	Automated builds shall validate code changes.
MAINT-027	Automated tests shall execute during the CI pipeline.
MAINT-028	Deployment processes shall be repeatable and documented.
MAINT-029	Build failures shall prevent deployment to production.

10.5.11 Technical Debt Management

The development team shall actively manage technical debt.

Activities include:

- Regular code refactoring.
- Removal of obsolete components.
- Dependency updates.
- Architecture reviews.
- Performance optimization.
- Security improvements.

Technical debt shall be tracked and prioritized during project planning.

10.5.12 Business Rules

Rule ID	Rule
MAINT-BR-001	Software modules shall remain loosely coupled and highly cohesive.
MAINT-BR-002	Documentation shall be maintained alongside software changes.

MAINT-BR-003	Source code shall follow established coding standards.
MAINT-BR-004	Automated testing shall support software quality.
MAINT-BR-005	Configuration shall be externalized from application code.
MAINT-BR-006	Releases shall be version-controlled and traceable.
MAINT-BR-007	Technical debt shall be monitored and managed continuously.
MAINT-BR-008	CI/CD pipelines shall enforce quality gates before deployment.

10.5.13 Exception Handling

Exception	Expected System Response
Build failure	Stop the CI/CD pipeline, report the failure, and prevent deployment.
Automated test failure	Block deployment until the failure is resolved or an approved exception process is followed.
Configuration error	Prevent application startup where the configuration is invalid and log diagnostic information.
Documentation outdated	Flag the discrepancy during review and update documentation before release where required.
Dependency incompatibility	Prevent deployment until compatibility is restored or the dependency is replaced.
Logging configuration failure	Fall back to a safe default logging configuration and notify administrators.

10.5.14 Acceptance Criteria

The maintainability requirements shall be considered complete when:

- The application follows a modular architecture.
- Coding standards are consistently applied.
- Documentation is complete and maintained.
- Logging supports diagnostics and troubleshooting.
- Configuration is externalized and securely managed.
- Automated testing is integrated into development.
- Version control and release management are implemented.
- CI/CD pipelines enforce quality controls before deployment

10.6 Usability Requirements

10.6.1 Overview

Usability Requirements define the characteristics that ensure ProcessMind AI is easy to learn, efficient to use, accessible, and provides a consistent user experience across supported devices and user roles.

The user interface shall support users with varying levels of technical expertise while enabling efficient completion of business tasks.

Usability requirements apply to:

- Web Application
- Dashboard
- Workflow Designer
- AI Assistant
- Administration Console
- Reporting Interface
- Mobile Web Browsers

10.6.2 Usability Objectives

The platform shall achieve the following objectives.

Objective	Description
Learnability	New users should learn the system quickly.
Efficiency	Frequent tasks should require minimal effort.
Consistency	Similar actions should behave consistently throughout the system.
Accessibility	The platform should be usable by people with diverse abilities.
User Satisfaction	Users should have a positive experience while using the platform.

10.6.3 User Interface Design Principles

The user interface shall follow these principles:

- Consistent navigation
- Responsive design
- Clear visual hierarchy
- Minimal unnecessary complexity
- Predictable interactions
- Immediate feedback for user actions
- Error prevention where possible
- Accessible color contrast
- Readable typography
- Mobile-friendly layouts

10.6.4 Navigation

Requirement ID	Requirement
USE-001	Navigation shall remain consistent throughout the platform.
USE-002	Users shall be able to access major functions within a minimal number of navigation steps.
USE-003	Breadcrumbs or equivalent navigation aids shall be provided where appropriate.
USE-004	Frequently used features shall be easily accessible.

10.6.5 User Experience

Requirement ID	Requirement
USE-005	The system shall provide immediate visual feedback after user actions.
USE-006	Long-running operations shall display progress indicators.
USE-007	Confirmation dialogs shall be presented before irreversible operations.
USE-008	The interface shall minimize unnecessary user input by using defaults and reusable selections where appropriate.

10.6.6 Responsive Design

The platform shall support responsive layouts.

Requirement ID	Requirement
USE-009	The user interface shall adapt to common desktop screen resolutions.
USE-010	The application shall remain usable on tablets.
USE-011	Core functionality shall be accessible on modern mobile web browsers where supported.
USE-012	Layouts shall adjust without requiring horizontal scrolling under normal conditions.

10.6.7 Accessibility

The platform shall support accessible design practices.

Requirement ID	Requirement
USE-013	User interface components shall be keyboard accessible where practical.
USE-014	Images conveying important information shall include descriptive alternative text where applicable.
USE-015	Color shall not be the sole means of conveying important information.
USE-016	Text shall maintain sufficient contrast against its background.
USE-017	Form controls shall have clear labels and validation messages.

The platform should be designed with consideration for the principles of **WCAG 2.1 Level AA** where feasible.

10.6.8 Error Messages

Requirement ID	Requirement
USE-018	Error messages shall clearly describe the problem.
USE-019	Error messages shall provide guidance for resolving the issue where appropriate.
USE-020	Error messages shall not expose sensitive technical details.
USE-021	Validation errors shall identify the affected input field whenever possible.

10.6.9 Help & Documentation

The platform shall provide user assistance.

Examples include:

- User Guide
- Administrator Guide
- Context-sensitive help
- Tooltips
- Frequently Asked Questions (FAQ)
- Guided onboarding for new users (future enhancement)

10.6.10 Internationalization & Localization

The platform architecture shall support future internationalization.

Requirement ID	Requirement
USE-022	User-visible text shall support localization.
USE-023	Date and time formats shall be configurable according to user or organization preferences.
USE-024	Number and currency formatting shall support regional settings where applicable.

10.6.11 User Satisfaction Metrics

The organization may monitor:

- Task completion time
- User error rate
- Navigation efficiency
- Help request frequency
- Feature adoption
- User feedback
- Customer satisfaction surveys

These metrics shall support continuous usability improvements.

10.6.12 Business Rules

Rule ID	Rule
USE-BR-001	User interfaces shall maintain a consistent design language.
USE-BR-002	Critical user actions shall require confirmation where appropriate.
USE-BR-003	Long-running operations shall provide progress feedback.
USE-BR-004	Error messages shall assist users without exposing sensitive system information.
USE-BR-005	Core functionality shall remain usable across supported devices.
USE-BR-006	Accessibility shall be considered during interface design.
USE-BR-007	User assistance resources shall be available for common tasks.
USE-BR-008	Localization support shall be considered for future releases.

10.6.13 Exception Handling

Exception	Expected System Response
Invalid user input	Highlight the affected fields, display clear validation messages, and preserve entered data where appropriate.
Long-running operation	Display progress indicators or status updates until completion or failure.
Network interruption	Inform the user of the connectivity issue and allow retry where appropriate.
Unsupported browser feature	Notify the user of limited functionality and recommend a supported browser if necessary.
Accessibility limitation detected	Provide an accessible alternative where feasible or document the limitation.
Missing help content	Direct users to the general documentation or support resources until the specific content is available.

10.6.14 Acceptance Criteria

The usability requirements shall be considered complete when:

- Navigation is consistent and intuitive.
- The interface provides clear feedback for user actions.

- Responsive layouts function correctly across supported devices.
- Accessibility considerations are incorporated into the design.
- Error messages are informative and user-friendly.
- Help resources are available for common tasks.
- The architecture supports future localization and internationalization.
- User satisfaction metrics can be collected and analyzed.

10.7 Compatibility Requirements

10.7.1 Overview

Compatibility Requirements define the environments, platforms, technologies, and external systems that ProcessMind AI shall support.

The platform shall maintain consistent functionality and performance across supported browsers, operating systems, cloud infrastructure, databases, AI providers, and third-party integrations.

Compatibility requirements apply to:

- Web Application
- Backend Services
- REST APIs
- AI Processing
- Database
- Cloud Infrastructure
- Third-Party Integrations

10.7.2 Compatibility Objectives

The platform shall achieve the following objectives.

Objective	Description
Cross-Platform Support	Operate consistently across supported operating systems and browsers.
Technology Interoperability	Integrate with external services using standard protocols.
API Compatibility	Provide stable and versioned APIs.
Cloud Compatibility	Support deployment on modern cloud platforms.

Future Extensibility	Allow integration with additional technologies in future releases.
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10.7.3 Supported Web Browsers

The platform shall support current stable versions of the following browsers.

Browser	Support Level
Google Chrome	Full Support
Microsoft Edge	Full Support
Mozilla Firefox	Full Support
Safari	Full Support
Opera	Best Effort

Older browser versions may provide reduced functionality.

10.7.4 Supported Operating Systems

The web application shall support modern operating systems capable of running supported browsers.

Operating System	Support Level
Windows 10/11	Full Support
macOS	Full Support
Linux (Desktop)	Full Support
Android	Supported through mobile web browsers
iOS	Supported through mobile web browsers

10.7.5 Device Compatibility

The platform shall support the following device categories.

Device	Support Level
Desktop Computers	Full Support
Laptop Computers	Full Support
Tablets	Full Support

Smartphones	Core functionality supported through responsive web design
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10.7.6 Database Compatibility

The platform shall support the following database technologies.

Database	Support Level
PostgreSQL	Primary Supported Database
MySQL (Future)	Planned
Microsoft SQL Server (Future)	Planned

The application architecture shall isolate database-specific implementation details where practical.

10.7.7 Cloud Compatibility

The platform shall support deployment on modern cloud infrastructure.

Cloud Platform	Support Level
Self-Hosted VPS	Supported
Docker-Based Deployment	Fully Supported
Kubernetes	Supported
Amazon Web Services (AWS)	Supported
Microsoft Azure	Supported
Google Cloud Platform (GCP)	Supported

The deployment architecture shall remain cloud-provider independent where feasible.

10.7.8 AI Provider Compatibility

The AI integration layer shall support multiple providers.

AI Provider	Support Level
Google Gemini	Primary Supported Provider
OpenAI	Future Integration

Anthropic Claude	Future Integration
Local LLMs (Ollama, etc.)	Future Integration
Azure OpenAI	Future Integration

The platform shall abstract provider-specific implementations to simplify future integrations.

10.7.9 API Compatibility

Requirement ID	Requirement
COMP-001	REST APIs shall use standard HTTP methods.
COMP-002	JSON shall be the primary request and response format.
COMP-003	APIs shall support versioning.
COMP-004	Backward compatibility should be maintained where practical during API evolution.

10.7.10 Third-Party Integration Compatibility

The platform architecture shall support integration with external systems.

Potential integrations include:

- Email services
- Identity providers
- ERP systems
- HR systems
- CRM platforms
- Document management systems
- Business intelligence platforms
- Collaboration platforms (Microsoft Teams, Slack)

Integration shall be performed through secure REST APIs or other standardized interfaces.

10.7.11 File Format Compatibility

The platform shall support the following document formats.

File Format	Purpose
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PDF	SOP uploads
DOCX	SOP uploads
TXT	Plain text SOPs
CSV	Report export
XLSX	Report export
JSON	API data exchange

Additional formats may be supported in future releases.

10.7.12 Business Rules

Rule ID	Rule
COMP-BR-001	The platform shall support all officially supported browsers consistently.
COMP-BR-002	APIs shall follow open standards and maintain version compatibility where practical.
COMP-BR-003	AI provider implementations shall be replaceable without major architectural changes.
COMP-BR-004	Cloud deployment shall not depend on vendor-specific features unless explicitly documented.
COMP-BR-005	External integrations shall use secure, documented interfaces.
COMP-BR-006	Supported file formats shall be validated before processing.
COMP-BR-007	Compatibility requirements shall be reviewed when introducing new platform dependencies.

10.7.13 Exception Handling

Exception	Expected System Response
Unsupported browser	Inform the user that some functionality may be unavailable and recommend a supported browser.
Unsupported file format	Reject the upload and display the list of supported formats.
Incompatible API version	Return a standardized API version error and guide clients toward supported versions.

AI provider incompatibility	Disable the affected integration, log the issue, and continue operating with supported providers where available.
Cloud platform limitation	Document the limitation and provide deployment guidance where feasible.
Third-party integration failure	Isolate the failure, log diagnostic information, and continue core platform operations where possible.

10.7.14 Acceptance Criteria

The compatibility requirements shall be considered complete when:

- The platform operates consistently across supported browsers and operating systems.
- Responsive design supports desktop, tablet, and mobile web browsers.
- PostgreSQL is fully supported as the primary database.
- Deployment is supported on modern cloud infrastructure and container platforms.
- The AI integration layer supports Google Gemini and can be extended to additional providers.
- APIs are versioned and follow standard HTTP and JSON conventions.
- Third-party integrations use standardized and secure interfaces.
- Supported file formats are validated and processed correctly.

10.8 Portability Requirements

10.8.1 Overview

Portability Requirements define the ability of ProcessMind AI to be deployed, migrated, configured, and operated across different computing environments with minimal changes.

The platform shall support portable deployment, configuration, data migration, and infrastructure independence to facilitate future growth and customer-specific deployment requirements.

Portability requirements apply to:

- Web Application
- Backend Services
- Database

- AI Integration Layer
- Infrastructure Configuration
- File Storage
- Deployment Pipeline

10.8.2 Portability Objectives

The platform shall achieve the following objectives.

Objective	Description
Deployment Portability	Deploy across multiple environments with minimal changes.
Cloud Independence	Avoid unnecessary dependence on a specific cloud provider.
Configuration Portability	Allow environment-specific configuration without code modifications.
Data Portability	Support import and export of organizational data.
Migration Support	Facilitate migration between supported environments.

10.8.3 Deployment Portability

Requirement ID	Requirement
PORT-001	The platform shall support deployment using Docker containers.
PORT-002	Deployment procedures shall be documented.
PORT-003	Application components shall be deployable independently where practical.
PORT-004	Production, staging, testing, and development environments shall use consistent deployment processes.

10.8.4 Environment Portability

The platform shall support multiple environments.

Environment	Supported
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Local Development	Yes
Testing	Yes
Staging	Yes
Production	Yes
On-Premises	Supported
Cloud Deployment	Supported

Environment-specific settings shall be externalized through configuration.

10.8.5 Containerization

The application shall support containerized deployment.

Requirement ID	Requirement
PORT-005	Application services shall support Docker-based deployment.
PORT-006	Container images shall remain environment-independent where feasible.
PORT-007	Configuration shall be injected during deployment rather than embedded in container images.
PORT-008	Container images shall be version-controlled.

10.8.6 Data Portability

The platform shall support secure movement of organizational data.

Requirement ID	Requirement
PORT-009	Organizations shall be able to export authorized business data.
PORT-010	Data imports shall validate integrity before processing.
PORT-011	Data exports shall preserve referential integrity where applicable.
PORT-012	Export formats shall use open, documented standards where practical.

10.8.7 Backup Portability

The backup architecture shall support migration and recovery.

Requirement ID	Requirement
PORT-013	Backup files shall support restoration into supported deployment environments.
PORT-014	Backup procedures shall be documented.
PORT-015	Backup integrity shall be verified before restoration.
PORT-016	Backup metadata shall accompany backup data where required for restoration.

10.8.8 Configuration Portability

The platform shall separate configuration from application code.

Requirement ID	Requirement
PORT-017	Environment variables shall control environment-specific settings.
PORT-018	Sensitive configuration values shall be externalized.
PORT-019	Default configuration templates shall be provided.
PORT-020	Configuration changes shall not require application recompilation.

10.8.9 Vendor Independence

The platform architecture shall minimize vendor lock-in.

Examples include:

- Standard REST APIs
- OpenAPI specifications
- Docker containers
- PostgreSQL
- Open authentication standards
- Cloud-neutral deployment architecture

Vendor-specific implementations shall be isolated behind abstraction layers where practical.

10.8.10 Migration Support

The platform shall support migration activities.

Supported migration scenarios include:

- Development to production
- VPS to cloud infrastructure
- Cloud provider migration
- Database migration (supported databases)
- AI provider replacement
- Infrastructure upgrades

Migration procedures shall be documented and tested where practical.

10.8.11 Business Rules

Rule ID	Rule
PORT-BR-001	Deployment shall support multiple environments using the same application codebase.
PORT-BR-002	Configuration shall be externalized from application binaries.
PORT-BR-003	Data portability shall preserve integrity and security.
PORT-BR-004	Vendor-specific implementations shall be isolated where practical.
PORT-BR-005	Migration procedures shall be documented.
PORT-BR-006	Backup portability shall support disaster recovery objectives.
PORT-BR-007	Containerization shall simplify deployment and migration.

10.8.12 Exception Handling

Exception	Expected System Response
Invalid configuration during deployment	Stop deployment, report validation errors, and prevent startup until configuration is corrected.
Failed data migration	Roll back the migration where supported, preserve source data, and log detailed diagnostics.

Backup restoration incompatibility	Reject the restoration, report compatibility issues, and recommend supported recovery procedures.
Missing environment variables	Prevent application startup and identify the missing configuration values.
Unsupported deployment platform	Notify administrators of unsupported requirements and provide deployment guidance where possible.
Container image incompatibility	Reject deployment and require a compatible image version.

10.8.13 Acceptance Criteria

The portability requirements shall be considered complete when:

- The platform can be deployed consistently across supported environments.
- Docker-based deployment is supported.
- Environment-specific configuration is externalized.
- Organizational data can be securely exported and imported where authorized.
- Backup and restoration procedures support migration and disaster recovery.
- Vendor-specific dependencies are minimized.
- Migration procedures are documented and validated.

10.9 Accessibility Requirements

10.9.1 Overview

Accessibility Requirements define the characteristics that enable ProcessMind AI to be used effectively by individuals with diverse abilities and assistive technologies.

The platform shall incorporate accessibility principles into its user interface, interactions, navigation, and content presentation to provide an inclusive user experience.

Accessibility requirements apply to:

- Web Application
- Dashboard
- Workflow Designer
- AI Assistant

- Reports
- Administration Console
- Forms
- Data Tables

10.9.2 Accessibility Objectives

The platform shall achieve the following objectives.

Objective	Description
Inclusive Design	Support users with diverse abilities.
Keyboard Accessibility	Enable operation without a pointing device.
Assistive Technology Support	Support commonly used assistive technologies where applicable.
Readability	Present information clearly and legibly.
Accessible Navigation	Ensure users can navigate efficiently throughout the platform.

10.9.3 Accessibility Standard

The platform should be designed with consideration for the principles of **WCAG 2.1 Level AA** where feasible.

Accessibility shall include:

- Perceivable content
- Operable interfaces
- Understandable interactions
- Robust implementation compatible with assistive technologies

10.9.4 Keyboard Accessibility

Requirement ID	Requirement
ACC-001	All major interactive functions shall be operable using a keyboard.
ACC-002	Keyboard focus shall remain visible.

ACC-003	Keyboard navigation order shall follow a logical sequence.
ACC-004	Keyboard users shall not become trapped within interface components.

10.9.5 Screen Reader Support

Requirement ID	Requirement
ACC-005	Form controls shall have accessible labels.
ACC-006	Images conveying important information shall include descriptive alternative text.
ACC-007	Headings shall follow a logical hierarchy.
ACC-008	Dynamic content updates should notify assistive technologies where appropriate.

10.9.6 Color & Contrast

Requirement ID	Requirement
ACC-009	Color shall not be the sole means of conveying important information.
ACC-010	Text and interface elements shall maintain sufficient contrast against their backgrounds.
ACC-011	Status indicators shall include text or icons in addition to color where practical.

10.9.7 Accessible Forms

Requirement ID	Requirement
ACC-012	Required fields shall be clearly identified.
ACC-013	Validation errors shall identify affected fields.
ACC-014	Validation messages shall provide guidance for correction.
ACC-015	Form controls shall have associated labels.

10.9.8 Responsive Accessibility

The platform shall maintain accessibility across supported devices.

Requirement ID	Requirement
ACC-016	Responsive layouts shall preserve accessibility features.

ACC-017	Touch targets shall be appropriately sized on touch-enabled devices.
ACC-018	Zooming shall not prevent access to core functionality under supported conditions.

10.9.9 Accessible Content

The platform shall present content clearly.

Examples include:

- Plain language where appropriate.
- Consistent terminology.
- Clear instructions.
- Readable typography.
- Predictable layouts.
- Consistent icons.

10.9.10 Accessibility Testing

Accessibility shall be evaluated during development.

Testing activities may include:

- Keyboard-only navigation testing.
- Screen reader testing.
- Color contrast verification.
- Responsive accessibility testing.
- Manual accessibility review.
- Automated accessibility scanning.

10.9.11 Business Rules

Rule ID	Rule
ACC-BR-001	Accessibility shall be considered during interface design and implementation.
ACC-BR-002	Keyboard accessibility shall be maintained for major platform functionality.
ACC-BR-003	User interfaces shall not rely solely on color to communicate important information.
ACC-BR-004	Forms shall provide clear validation and guidance.

ACC-BR-005	Accessibility testing shall be incorporated into quality assurance activities.
ACC-BR-006	Responsive layouts shall preserve accessibility features.
ACC-BR-007	Accessibility improvements shall be considered during future feature development.

10.9.12 Exception Handling

Exception	Expected System Response
Missing alternative text	Flag the issue during quality assurance and require correction before release where applicable.
Keyboard navigation issue	Record the defect, prioritize remediation, and verify accessibility before deployment.
Insufficient color contrast	Identify the issue through design review or testing and update the interface accordingly.
Screen reader compatibility issue	Investigate the affected component, implement corrective changes, and retest with assistive technologies.
Accessibility regression	Prevent release until the regression is resolved or an approved exception process is followed.

10.9.13 Acceptance Criteria

The accessibility requirements shall be considered complete when:

- Major platform functionality is accessible using a keyboard.
- Forms include accessible labels and validation messages.
- User interfaces maintain sufficient color contrast.
- Accessibility is preserved across supported responsive layouts.
- Accessibility testing is integrated into quality assurance.
- The platform is designed with consideration for **WCAG 2.1 Level AA** principles where feasible.
- Accessibility improvements are incorporated into future platform enhancements.

10.10 Compliance Requirements

10.10.1 Overview

Compliance Requirements define the legal, regulatory, organizational, and technical obligations that ProcessMind AI shall support throughout its lifecycle.

The platform shall be designed to facilitate compliance with applicable security, privacy, operational, and software engineering best practices while allowing organizations to implement additional compliance controls according to their specific requirements.

Compliance requirements apply to:

- Web Application
- REST APIs
- AI Processing
- Database
- Workflow Engine
- File Storage
- Audit Logging
- Infrastructure
- Administrative Operations

10.10.2 Compliance Objectives

The platform shall achieve the following objectives.

Objective	Description
Regulatory Readiness	Support future regulatory compliance where applicable.
Security Compliance	Follow recognized security best practices.
Privacy Protection	Protect personal and organizational information.
Audit Readiness	Maintain sufficient evidence for audits and investigations.
Responsible AI	Promote transparent and accountable AI usage.

10.10.3 Legal & Regulatory Compliance

The platform architecture shall support compliance with applicable laws and regulations.

Potential examples include:

Regulation / Framework	Applicability
GDPR	Organizations operating in applicable jurisdictions

ISO/IEC 27001	Information security management (future organizational certification)
SOC 2	Service organization controls (future)
Local Data Protection Laws	According to deployment jurisdiction

Support for a specific regulation depends on deployment configuration, organizational processes, and legal obligations.

10.10.4 Privacy Requirements

Requirement ID	Requirement
COMP-NFR-001	Personal information shall be processed according to applicable privacy requirements.
COMP-NFR-002	Access to personal information shall follow authorization policies.
COMP-NFR-003	Audit logs shall record access to sensitive resources where appropriate.
COMP-NFR-004	Data retention and deletion shall follow organizational policies and applicable legal requirements.

10.10.5 Software Licensing Compliance

The development team shall manage software licenses appropriately.

Requirement ID	Requirement
COMP-NFR-005	Third-party software shall be used according to applicable license terms.
COMP-NFR-006	Open-source dependencies shall be reviewed periodically.
COMP-NFR-007	License compliance shall be verified during release preparation.

10.10.6 Security Compliance

The platform shall align with recognized security practices.

Examples include:

- OWASP Top 10

- OWASP ASVS
- NIST Cybersecurity Framework
- Secure coding guidelines
- Security testing practices

These frameworks provide guidance rather than mandatory certification requirements.

10.10.7 AI Governance Compliance

AI-assisted functionality shall support responsible governance.

Requirement ID	Requirement
COMP-NFR-008	AI-generated outputs shall remain subject to organizational review where configured.
COMP-NFR-009	AI processing activities shall be auditable.
COMP-NFR-010	AI provider information and model versions shall be traceable.
COMP-NFR-011	Human oversight shall remain available for configurable high-risk operations.

10.10.8 Audit Compliance

The platform shall maintain sufficient records to support audits.

Audit capabilities shall include:

- Authentication history
- Administrative actions
- Workflow execution history
- AI processing history
- Configuration changes
- Security events
- System activities

Audit information shall be protected from unauthorized modification.

10.10.9 Compliance Monitoring

The platform shall support compliance monitoring.

Monitoring activities include:

- Security policy compliance
- Audit log review
- Configuration review
- Backup verification
- Vulnerability management
- Access review
- AI governance review

Organizations may define additional monitoring procedures according to internal policies.

10.10.10 Business Rules

Rule ID	Rule
COMP-BR-001	Compliance-related activities shall be documented where applicable.
COMP-BR-002	Security controls shall support organizational compliance objectives.
COMP-BR-003	Audit records shall support compliance verification.
COMP-BR-004	AI governance shall support transparency and accountability.
COMP-BR-005	Software licensing obligations shall be respected.
COMP-BR-006	Privacy controls shall be enforced according to organizational policies and applicable legal requirements.
COMP-BR-007	Compliance monitoring shall be performed periodically.
COMP-BR-008	Compliance requirements shall be reviewed as regulations and organizational needs evolve.

10.10.11 Exception Handling

Exception	Expected System Response
Missing audit records	Record the issue, investigate the cause, and notify authorized administrators.
Privacy policy violation	Restrict the affected operation where appropriate, log the event, and initiate administrative review.
License compliance issue	Review the affected dependency, assess impact, and resolve the licensing concern before release.
AI governance policy violation	Prevent or suspend the affected AI operation until compliance requirements are satisfied.
Configuration non-compliance	Report the issue through compliance monitoring and require corrective action.
Expired security certificate or credential	Generate an alert and require renewal according to operational procedures.

10.10.12 Acceptance Criteria

The compliance requirements shall be considered complete when:

- The platform architecture supports applicable legal and regulatory requirements.
- Privacy controls protect sensitive information.
- Security controls align with recognized industry best practices.
- AI governance supports transparency, accountability, and human oversight.
- Audit records provide sufficient evidence for operational and compliance reviews.
- Software licensing obligations are managed appropriately.
- Compliance monitoring activities are documented and periodically reviewed.